Network Infrastructures

Transport Networks

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About these Notes

These notes were developed from the material used in the *Network Infrastructures* course, including lecture slides, handwritten whiteboards, and transcripts of the video lectures recorded through Zoom.

The original material was reorganized and rewritten with the assistance of generative AI in order to transform the lectures into a more coherent and self-contained set of study notes. The objective was not to produce a verbatim transcription of the lectures, but to preserve their structure, explanations, examples, and overall teaching approach while improving readability and continuity.

The figures included in the notes are largely based on diagrams originally drawn during the lectures. They have been redrawn and, where appropriate, simplified or revised to improve clarity and consistency.

Generative AI was also used to suggest additional explanations, examples, and transitions when these were considered useful for understanding the material. Such additions were reviewed by the instructor before being incorporated into the final version.

Despite this review process, the notes may still contain errors or inaccuracies. Students are therefore encouraged to report any issue they may identify.

These notes are intended as supporting material for the course and should be used together with the other teaching resources made available to students.

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Chapter 1

Transport Networks: Role and Architectural Principles

1.1 From End Users to Transport Networks

The objective of this first chapter is to understand what a transport network is and why transport networks are needed. Rather than starting from a formal definition, it is useful to begin from the network as it is seen by its users.

An end user can be many different things. It may be a single laptop or smartphone, a household containing several devices, or a much larger organization such as a university or a company. From the perspective adopted here, all of them play the same role: they are located at the edge of the communication infrastructure and need a way to exchange traffic with remote destinations.

End users normally reach the network through an *access network*. Different access technologies may be used—fixed, wireless, or satellite—but their common objective is to connect users to a provider infrastructure. This is the part of the network that is closest to the user and, for this reason, it is the part that users interact with most directly.

The access network alone, however, is not enough. Suppose that a user located in Rome wants to communicate with a server located in Milan. The access network can bring the traffic from the user to a local gateway, but another infrastructure is required to move that traffic over a much larger geographical area. This is the role of the *transport network*: it interconnects geographically separated network regions and carries traffic between them.

Transport networks are therefore often associated with the concept of a *Wide Area Network* (WAN). The term emphasizes one of their first distinguishing characteristics: while an access infrastructure is usually confined to a relatively limited geographical area, a transport infrastructure may span cities, regions, countries, or even continents.

Figure 1.1 summarizes this first view of the network. End users reach a local access infrastructure, while a wider transport network interconnects geographically separated access regions.

A useful way of thinking about this distinction is that an access network solves the "last part" of the connectivity problem from the user's point of view, whereas the transport network solves the problem of moving already aggregated traffic between distant parts of the infrastructure. The exact boundary is operator- and technology-dependent, but the two roles remain conceptually different.

1.2 Looking at an End-to-End Path

The distinction becomes more concrete if we look at the path followed by real traffic. Consider again a host in Rome communicating with a remote server in Milan. A tool such

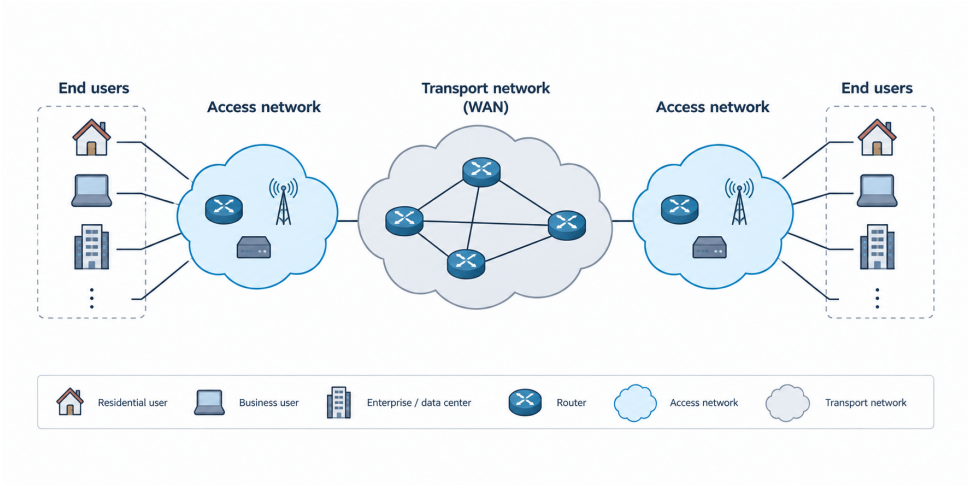


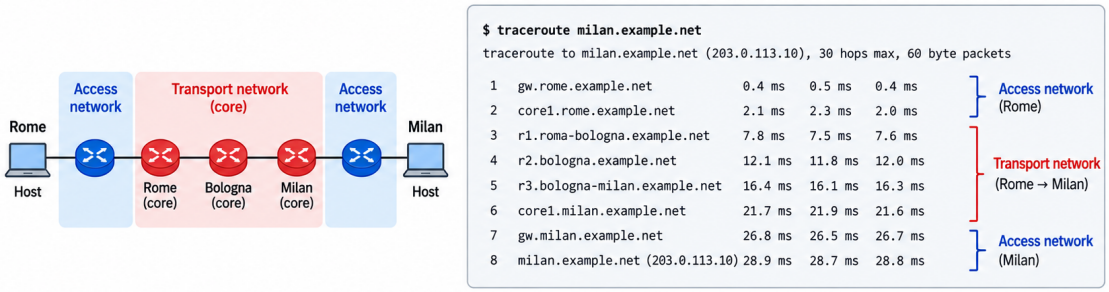
Figure 1.1: From end users and access networks to a wide-area transport network

as `traceroute` can be used to observe a sequence of Layer-3 hops encountered by packets along the path.

At the beginning of the path, the host reaches the gateway of the local access network. At the other end, the destination is reached through another access infrastructure. Between the two access portions, traffic crosses one or more provider networks that carry it over the long distance separating the two cities. These intermediate infrastructures are the transport part of the end-to-end path.

This simple example already reveals an important property of the Internet: an end-to-end communication is rarely provided by a single network. Different parts of the path may be operated by different organizations, and each organization manages its own infrastructure. A local university network, for example, may provide connectivity to its users but rely on another provider to carry traffic toward other cities; that provider may in turn exchange traffic with still other networks before the destination is reached.

This end-to-end view is represented in Figure 1.2. The figure follows the Rome–Milan example and highlights the transition from the source access network to the transport portion of the path and, finally, to the destination access network.



(a) End-to-end path between Rome and Milan

(b) Example traceroute from Rome to Milan

Figure 1.2: A conceptual traceroute from Rome to Milan.

The figure should be understood conceptually. A traceroute exposes IP-layer hops and does not necessarily reveal the complete physical path followed by the signal underneath, nor does it always identify the exact geographical location of every router. This limitation will

become particularly important later in the chapter, when logical and physical connectivity are separated explicitly.

Starting from this end-to-end view, we can now identify the main characteristics that distinguish transport networks from access networks. These characteristics will emerge progressively from the same example, allowing us to relate each of them to a concrete networking scenario rather than introducing them as an abstract classification.

1.3 Who Is the Customer?

Every network is operated by some organization, and whenever connectivity is offered as a service there is a customer–provider relationship. In an access network this relationship is familiar: the customer is typically an end user, a household, or an organization, while the access operator provides connectivity.

The situation changes when we move into the transport network. A transport provider generally does not serve individual residential users directly. Its customers are other network operators, access providers, large organizations, or even other transport providers. In other words, the customer of a transport network is often itself a network.

This difference is important. An access provider may have hundreds of thousands or millions of individual customers, each generating a relatively limited amount of traffic. A transport provider may have a much smaller number of customers, but each of those customers can inject traffic aggregated from a very large population of users.

The same end-to-end path can therefore contain several customer–provider relationships. A user may be a customer of an access network; that access network may be a customer of a transport carrier; and the carrier may in turn rely on connectivity or interconnection agreements with other networks.

This recursive customer–provider view is useful throughout the course because the word *client* will not always mean an end user. At one layer, the client of a transport service may be a router, an IP network, or another provider. What matters is the role played in the service relationship: the client requests connectivity, while the provider realizes it using its own resources.

1.4 Traffic Aggregation and Capacity

The different customer base immediately leads to another major difference: the amount of traffic carried by the network.

Consider an access network serving several end users. Each user generates one or more traffic flows. The maximum rate of an individual user's traffic is constrained by the capacity of that user's access connection. As traffic moves toward the gateway, however, flows generated by different users are aggregated. If we denote these tributary flows as $F_1, F_2, \ldots, F_N$, the traffic entering the upstream network is the aggregate of the different contributions. At a conceptual level we can write

$$F_{\text{agg}} = \sum_{i=1}^{N} F_i.$$

The important point is not the formula itself, but what it represents. The transport network does not normally carry the traffic of one residential customer in isolation. It carries traffic that has already been collected and aggregated by other networks. Consequently, the data rates handled by transport links can be much larger than those found at the edge.

The aggregation process is illustrated in Figure 1.3. Several tributary flows generated at the edge are collected by the access network and presented to the transport infrastructure as a larger aggregate flow.

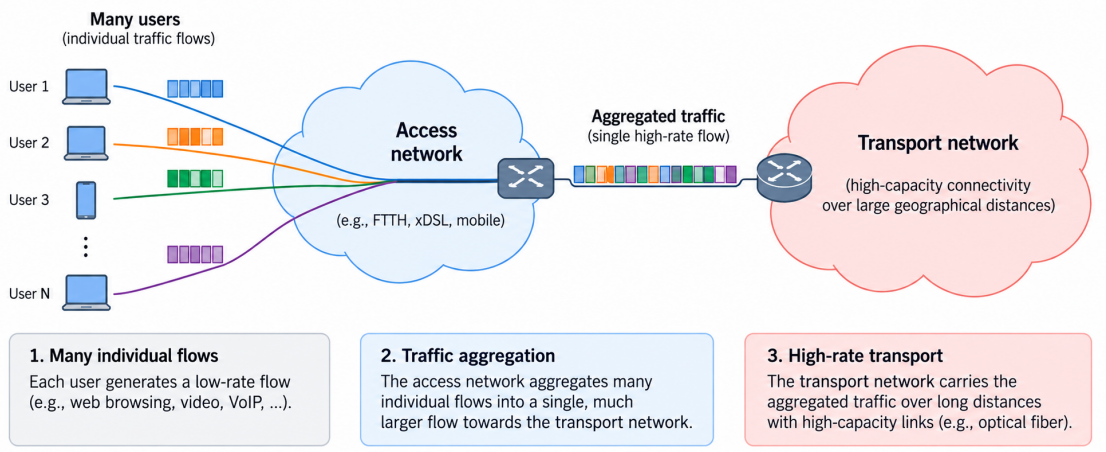


Figure 1.3: Traffic aggregation from access users toward the transport network. Multiple individual traffic flows are collected by the access network and carried toward the transport infrastructure as a higher-rate aggregate flow.

The expression above should not be interpreted as saying that every customer transmits continuously at its maximum access rate. Real traffic is bursty, and not all users are active at the same time. This statistical behavior is one of the reasons why sharing network resources is efficient. Nevertheless, as traffic is progressively aggregated, transport links must be dimensioned for much larger volumes than individual access links.

Consider, for example, an FTTH access network serving 1,000 residential users, each subscribing to a 1 Gbps access service. The sum of the nominal access rates is therefore

$$1000 \times 1 \text{ Gbps} = 1 \text{ Tbps}.$$

This does not mean that the transport network must necessarily reserve 1 Tbps for these users. Residential customers do not normally transmit continuously at their maximum access rate, and their traffic peaks do not occur at exactly the same time. The aggregation network can therefore exploit statistical multiplexing. For instance, if the users generate an average of 50 Mbps each during a busy period, the resulting aggregate traffic would be approximately

$$1000 \times 50 \text{ Mbps} = 50 \text{ Gbps}.$$

The example illustrates two important aspects of traffic aggregation: the transport network carries traffic volumes that are much larger than those of a single access connection, while the capacity required upstream can still be significantly lower than the sum of all subscribed access rates.

This is why capacity is a fundamental requirement in transport networks. Once many lower-rate flows are combined, tens or hundreds of gigabits per second can quickly become relevant. The network must therefore be designed so that both its links and its switching nodes can sustain large aggregate traffic volumes.

1.5 Topology Makes Routing Important

A third difference appears when we compare network topologies.

In an access network, the path from a user toward the aggregation point is often structurally simple. A user may have a dedicated point-to-point connection toward a central office. When the number of users becomes large, several user branches can be aggregated into a tree topology. In both cases, the route from a given user toward the aggregation point is essentially determined by the topology itself: there is normally only one possible path.

Ring topologies introduce a second possible direction. Even in that case, however, the two paths are often used in a simple protection scheme: one path acts as the primary path and the other as a backup that becomes active when a failure affects the primary one. The main purpose of the alternative path is therefore resilience rather than general path selection.

Figure 1.4 illustrates these typical access-network structures. In each case, the topology strongly constrains the path from the user toward the aggregation point, so there is little or no need for a general routing decision.

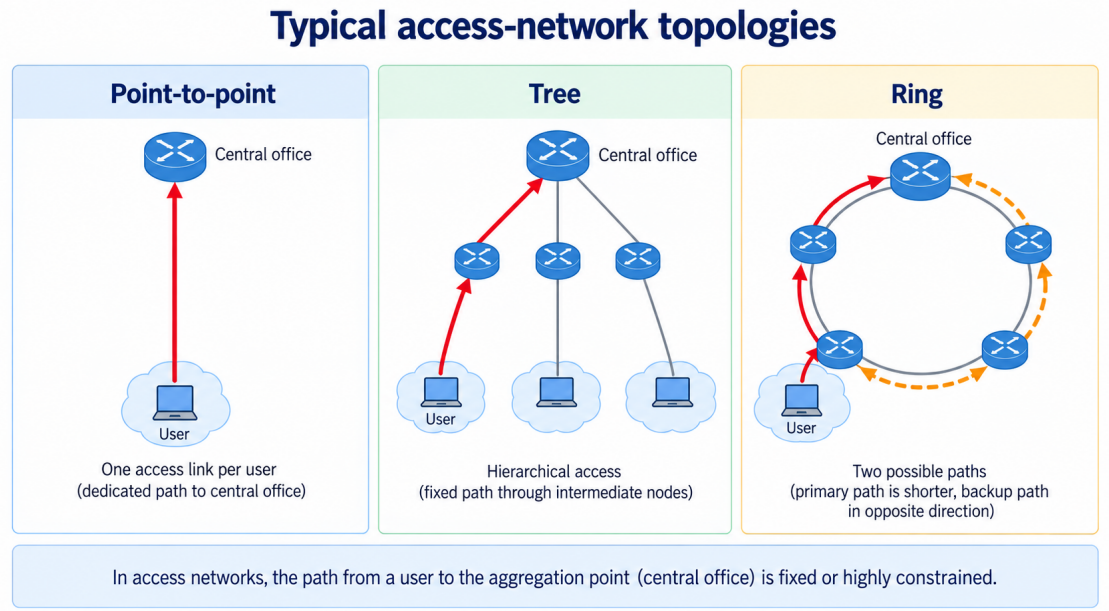


Figure 1.4: Typical access-network topologies and the corresponding path from a user to the central office.

A transport network has a different structure. Its topology is generally a *mesh*: nodes have several interconnections and there can be many possible paths between the same pair of endpoints. At this point, routing becomes a real decision problem. The network must determine which sequence of intermediate nodes should be used to carry traffic from one point to another.

This contrast is shown in Figure 1.5. Unlike the access topologies of Figure 1.4, the mesh offers several feasible routes between the same endpoints. Choosing one of these routes is therefore an explicit network function.

The availability of several paths is not merely a complication. It also gives the operator freedom. Different paths can be selected according to the state of the network and the objectives of the operator. Since transport networks carry large traffic aggregates, path selection can have a substantial impact on the utilization of network resources and on the quality of the service offered to customers.

For example, if two paths connect the same pair of transport nodes, sending every large flow through the same path may overload some links while leaving other resources underused. Alternative paths can also be valuable after a failure or when a service has particularly

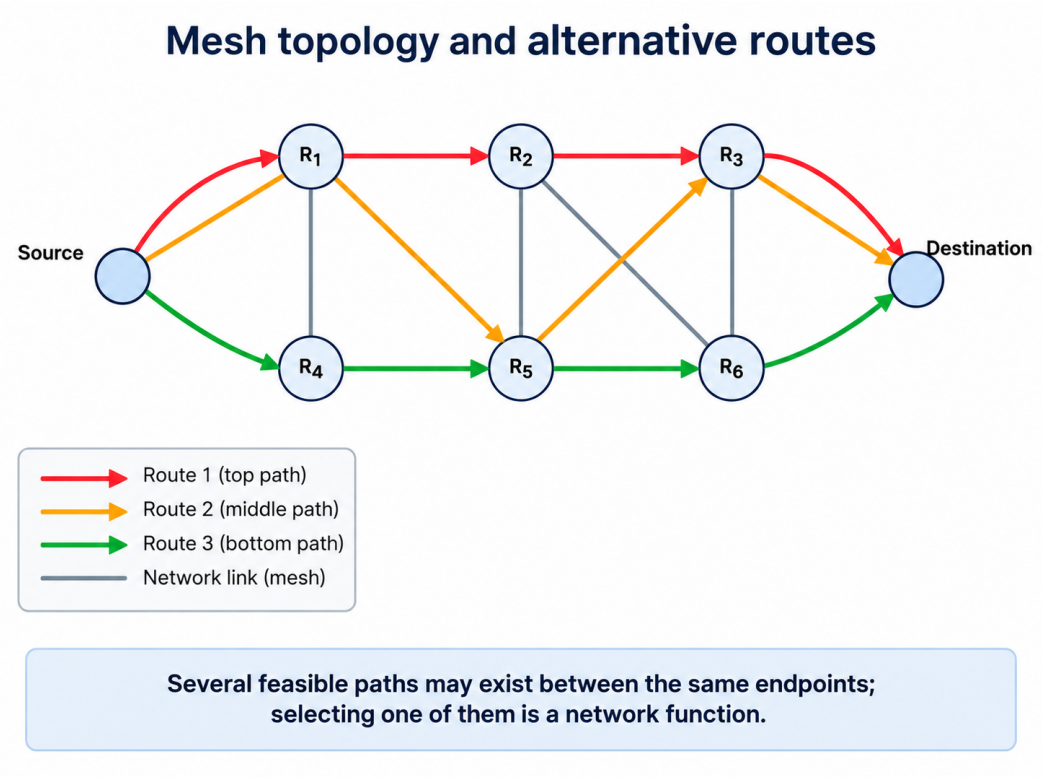


Figure 1.5: Mesh topology in a transport network.

strict delay or capacity requirements. These are early examples of the motivations behind traffic engineering and resilience mechanisms, which will be studied later in the course.

There are actually two routing problems hidden in the end-to-end example. First, an operator has to select paths *inside* its own infrastructure. This is an intradomain routing problem. Second, the complete end-to-end communication may cross several independently operated networks, so another decision concerns the sequence of networks—or Autonomous Systems—that traffic has to traverse. This is the interdomain routing problem.

The course will return to this distinction later. For the moment, the important observation is that routing, which plays only a limited role in many access topologies, becomes a central function in a mesh transport infrastructure.

1.6 Distance and the Physical Connectivity Problem

The previous discussion was mostly at the logical network level. The fourth characteristic of a transport network brings us closer to the physical infrastructure: distance.

Inside a local network, it is easy to imagine a cable directly connecting two nearby devices. A host can be connected to a local switch or access point, and buildings that are close to each other can be interconnected by relatively short physical links. The situation is very different when two routers are located in different cities.

Suppose, for example, that a router in Rome and a router in Bologna appear as directly connected from the IP point of view. Drawing a line between the two routers is perfectly reasonable in an IP topology, but that line is a *logical* representation. It says that the two interfaces belong to the same logical link; it does not explain how the physical signal actually travels between Rome and Bologna.

A first, naive idea would be to use a single physical cable from one city to the other. Apart from the obvious deployment and cost problems, a signal degrades while propagating

and cannot simply be assumed to remain usable over an arbitrary distance. Intermediate physical infrastructure is therefore required.

One possible solution is to split the long connection into shorter physical spans and place intermediate devices along the route. If attenuation is the main limitation, amplifiers can restore the signal power and make transmission over much longer distances possible.

Figure 1.6 makes the distinction explicit. The upper part shows the single logical link visible at the IP layer, while the lower part shows one possible physical realization based on several transmission spans and intermediate amplification.

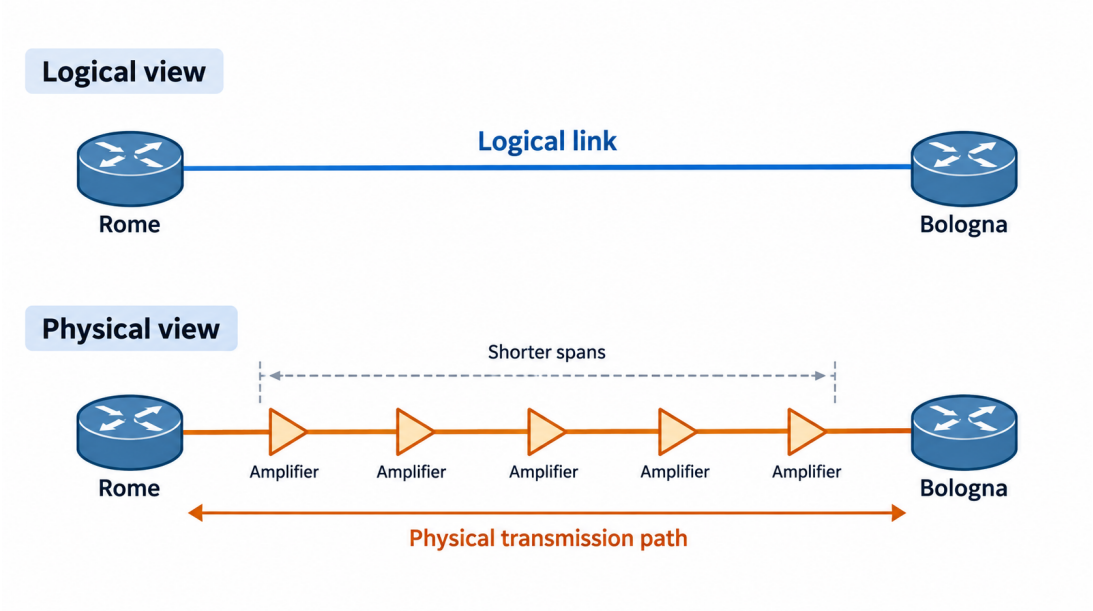


Figure 1.6: Logical and physical views of a long-distance connection. At the network layer, Rome and Bologna appear to be connected by a single logical link. The underlying physical realization may instead consist of multiple shorter transmission spans separated by intermediate amplifiers.

In optical fiber systems, attenuation is normally characterized in decibels per kilometer rather than with a free-space inverse-square law. The precise physical model is not needed here; the key networking consequence is that transmission distance is finite and the long-haul infrastructure must be engineered to compensate for propagation impairments.

A dedicated physical chain of links can solve the distance problem, but it creates a new one. Building a long-distance infrastructure is expensive. If a complete Rome–Bologna path were dedicated to only one pair of routers, all of that infrastructure would remain unavailable to other clients even when its capacity was not fully used.

This leads to the next question: if another customer in Rome also needs connectivity toward Bologna, should a second independent long-distance infrastructure be built from scratch? In general, the answer is no. The physical resources have to be shared.

1.7 The Transport Network as a Shared Lower-Layer Infrastructure

Sharing changes the way we should think about the transport provider. Imagine several routers belonging to one or more client networks: a router in Rome, another one in Florence, another in Bologna, and so on. From the Layer-3 perspective, the client may want some pairs of routers to appear directly connected. These IP links are logical abstractions. They

specify the desired connectivity but do not prescribe how the provider must realize it physically.

Underneath those routers we can place a Layer-1 network owned by a transport provider. The provider owns the long-distance physical infrastructure and offers connectivity as a service. A client that wants to connect two remote routers does not need to deploy its own nationwide physical network. Instead, it asks the Layer-1 provider to create the required connectivity between the two endpoints.

The resulting layered view is shown in Figure 1.7. The client routers see the logical connectivity they require, while the transport provider maps those connections onto a common Layer-1 infrastructure shared by multiple clients.

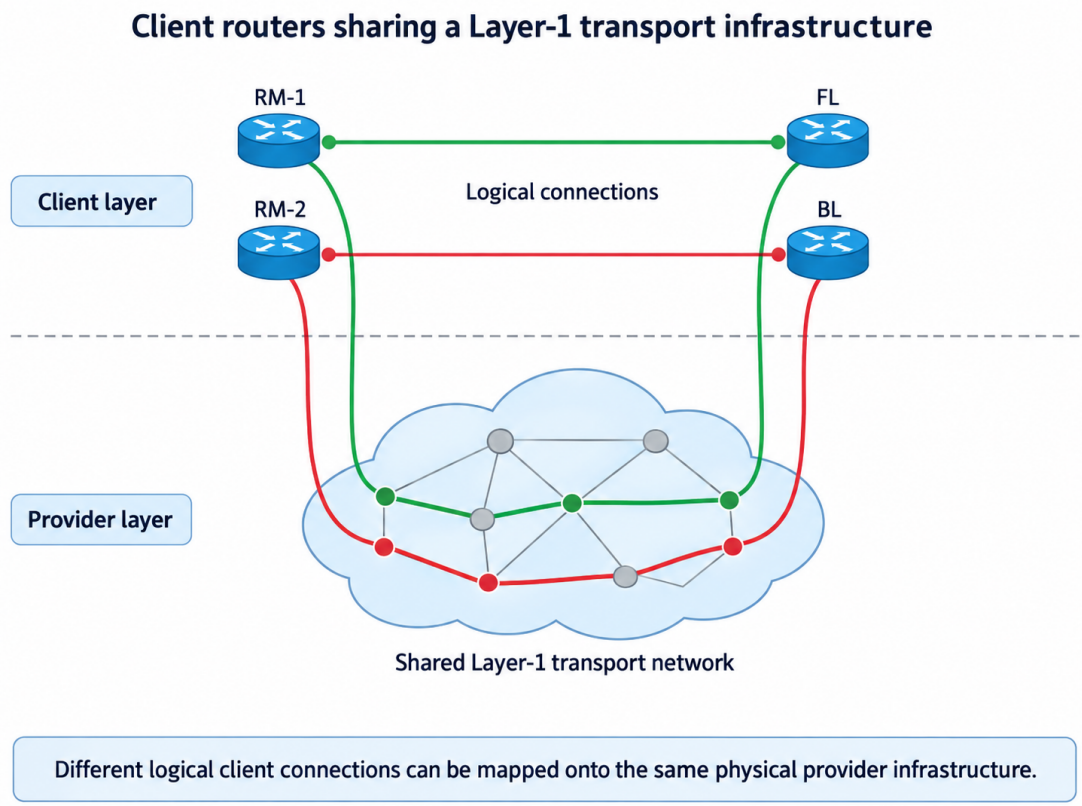


Figure 1.7: Different logical client connections mapped onto a shared Layer-1 transport infrastructure. The client layer sees independent logical links, while the provider layer realizes those connections over a common physical network.

This architecture makes the physical infrastructure reusable. The same set of long-distance links and intermediate nodes can support connections requested by many different clients. The high cost of deployment is therefore shared across multiple services instead of being associated with one dedicated end-to-end pair.

The picture also reinforces the client–provider relationship introduced earlier, now at a different layer of the stack. The client is no longer necessarily a human user. It can be a router or an entire packet network requesting lower-layer connectivity. The provider operates a physical network and exposes a connectivity service while hiding most of the internal details of that infrastructure.

This form of abstraction is one of the recurring ideas in networking. A layer can present a simple service—for example, an apparent direct link—even when implementing that service requires several internal hops, devices, and physical resources. The same idea will reappear in different forms when we discuss lightpaths, MPLS Label Switched Paths, and virtual private networks.

The logical topology seen by the client and the physical topology operated by the provider therefore do not have to be the same. A single logical link at Layer 3 may correspond to a path crossing many nodes at Layer 1. Conversely, the same Layer-1 infrastructure may simultaneously realize many logical links belonging to different clients.

At this point, one central requirement becomes unavoidable: the provider needs a mechanism that allows multiple customers and connections to share the same physical resources without losing their logical separation. The specific mechanism depends on the transport technology. In the next chapter we will see how optical networks address this problem through wavelength-based transmission and multiplexing.

1.8 From Transport Networks to Optical Networks

We can now summarize the reasoning developed so far. Starting from an end user, the access network provides the first connection to the communication infrastructure. When traffic must travel over a large geographical area, it enters transport networks operated by carriers. Compared with access networks, these infrastructures typically carry much larger aggregate traffic volumes, serve other networks as customers, use mesh topologies in which routing is a major function, and span much larger distances.

Distance makes a direct physical realization of every logical connection impractical, while the cost of long-distance infrastructure makes dedicated end-to-end facilities inefficient for most clients. A transport provider therefore builds a shared lower-layer network and uses it to offer connectivity between remote client devices.

Today, one of the fundamental technologies used to build this Layer-1 transport infrastructure is optical networking. Optical fibers provide the physical medium, while optical networking mechanisms make it possible to share fiber capacity, create end-to-end optical connections, switch them through intermediate nodes, and operate the resulting infrastructure.

This observation provides the natural starting point for Chapter 2. We will move from the generic transport-network model developed here to a concrete question: how can an optical network create a connection between two remote client devices while allowing many clients to share the same physical fiber infrastructure?

Seen in this way, the optical-network chapter is not an isolated discussion of transmission technology. It is the first concrete implementation of the architectural problem introduced here: turning an expensive, geographically distributed physical infrastructure into a reusable transport service for higher-layer networks.

1.9 Chapter Summary

A transport network is best understood by looking at the role it plays between access networks and remote destinations. It carries aggregated traffic over large geographical areas and is generally operated by a provider whose customers are other networks or large organizations rather than individual residential users. Because transport networks use mesh topologies, routing becomes a central function, both inside a provider network and across independently operated networks.

The logical links visible to packet-network devices do not directly describe the underlying physical infrastructure. Long distances, signal degradation, and deployment costs require a more complex physical realization. Rather than building a dedicated long-distance infrastructure for each pair of clients, a Layer-1 transport provider deploys a shared network and uses it to realize many logical connections. Optical networking, introduced in the next chapter, is the first technology we will study for implementing this shared transport infrastructure.

Further Reading

The following references provide additional background on the architectural principles introduced in this chapter.

- A. S. Tanenbaum, N. Feamster, and D. Wetherall, *Computer Networks*, 6th ed., Pearson, 2021.

  A broad introduction to computer networking. The chapters on network architecture, switching, routing, and wide-area networks provide useful background for understanding the distinction between access and transport infrastructures.

- L. L. Peterson and B. S. Davie, *Computer Networks: A Systems Approach*, 6th ed., Morgan Kaufmann, 2021.

  Particularly useful for reviewing the concepts of packet switching, multiplexing, routing, and the layered organization of communication networks.

Chapter 2

Optical Networks

2.1 From a Logical Link to an Optical Connection

In the previous chapter, we introduced a transport network as a shared lower-layer infrastructure that can provide connectivity between remote client devices. We can now make this idea concrete by considering two routers, R_1 and R_2, located far away from each other. From the IP point of view, we would like the two routers to appear as directly connected: their interfaces can be configured in the same IP subnet and the resulting line in the Layer-3 topology represents a logical IP link.

That logical configuration, however, is not sufficient to move any packet between the two routers. A physical transport service must exist underneath it. In the architecture considered in this chapter, this service is provided by an optical network.

The first important characteristic of the optical network is that the service considered here is connection-oriented. Before client traffic can be transported, the client asks the optical-network provider to establish a connection between two endpoints. The request identifies the endpoints and may also include requirements concerning capacity, performance, or protection. These requirements form part of the service that the provider agrees to deliver.

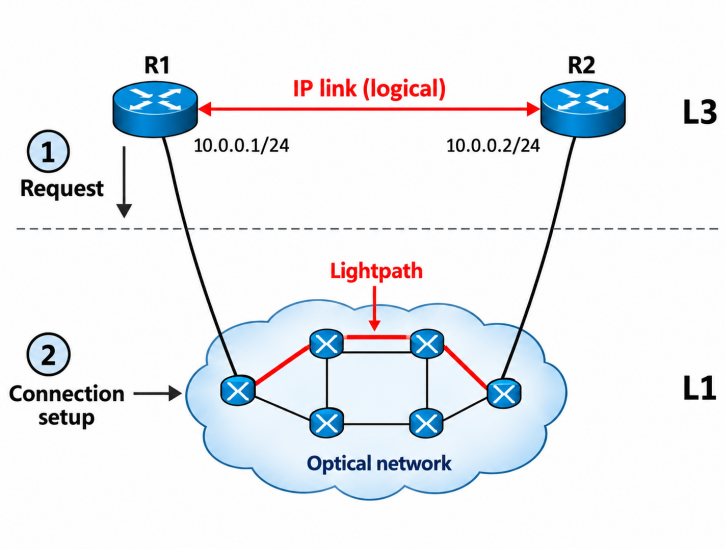


Figure 2.1: Request and establishment of an optical connection between two remote routers. At Layer 3, routers R_1 and R_2 are connected through a logical IP link. At Layer 1, this connectivity is provided by the optical transport network, which receives the connection request and establishes a lightpath across the underlying optical topology.

Figure 2.1 emphasizes the two views of the same service. At the client layer, the objective is simply to obtain connectivity between the two routers. At the provider layer, the request must be translated into a concrete path through the optical infrastructure.

The provider therefore has to solve a sequence of problems. First, it must determine a route between the ingress and egress points of the optical network. Second, because the physical links are shared by many customers, it must reserve a channel on each link of that route. Finally, the intermediate optical devices must be configured so that the signal entering on a particular port and channel is switched toward the correct outgoing port.

Once these operations have been completed, an end-to-end optical connection exists. This connection is called a *lightpath*. The lightpath is the lower-layer transport service that supports the logical link seen by the client.

The distinction between the logical IP link and the lightpath is important. The IP routers do not need to know the sequence of optical nodes traversed by the connection. From their point of view, the service simply behaves as a direct link. This is the same client–provider abstraction introduced in Chapter 1, now instantiated by a specific technology.

2.2 Sharing the Fiber: From Multiplexing to WDM

A dedicated optical fiber for every client connection would defeat one of the main goals of a transport infrastructure: sharing expensive physical resources. Several connections must therefore coexist on the same transmission medium while remaining distinguishable from one another.

This is the general problem addressed by *multiplexing*: multiple independent traffic streams share a common communication resource, while the network provides a mechanism to keep them logically separated.

More generally, there are different ways in which multiple users or traffic flows can share the same resource. A useful high-level classification distinguishes among *static multiplexing*, *statistical multiplexing*, and *turn-taking* mechanisms.

In *static multiplexing*, the available resource is partitioned in advance. Each user is assigned a portion of the resource independently of whether that user is currently transmitting. The partition can be performed along different dimensions, such as time, frequency, wavelength, or space.

In *statistical multiplexing*, resources are not permanently assigned to individual users. Instead, they are used dynamically according to the traffic that is actually present. Packet-switched networks provide the most familiar example: packets generated by different flows share the same links, and link capacity is consumed only when packets are actually transmitted.

A third possibility is to coordinate access by allowing users to transmit in turn. Examples include polling or token-based mechanisms, where a user is allowed to access the shared resource only when explicitly authorized to do so. Strictly speaking, these mechanisms are often classified as medium-access techniques rather than multiplexing schemes, but they illustrate another fundamental way of coordinating access to a shared resource.

The distinction is important because transport technologies may use different sharing principles at different layers. In this chapter, however, we are mainly interested in *static multiplexing*, and in particular in how several signals can be carried simultaneously over the same optical fiber.

2.2.1 Static Multiplexing Techniques

In static multiplexing, the available transmission resource is partitioned in advance into a number of separate channels. Each channel can then be assigned to a different user or

tributary flow.

The key idea is that the separation among users is obtained by assigning them different portions of some physical resource. Two particularly important examples are Frequency Division Multiplexing (FDM), where users are separated in the frequency domain, and Time Division Multiplexing (TDM), where they are separated in the time domain.

Frequency Division Multiplexing

Frequency Division Multiplexing (FDM) separates different users in the *frequency domain*. A useful analogy is an orchestra: different instruments can play at the same time because their sounds occupy different portions of the frequency spectrum. The listener receives the superposition of all the sounds, but the individual contributions remain distinguishable because they are associated with different spectral components. FDM applies the same basic principle to communication signals: several tributary streams are transmitted simultaneously over the same physical medium, but each one is assigned a different portion of the available spectrum.

To describe this mechanism, it is useful to introduce the concept of *bandwidth*. A transmission channel does not normally support all possible frequencies in the same way. Instead, there is a range of frequencies over which signals can be transmitted with acceptable attenuation and distortion. If the usable spectrum extends from $f_{\min}$ to $f_{\max}$, the available channel bandwidth is

$$B_{\text{ch}} = f_{\max} - f_{\min}.$$

Bandwidth is therefore measured in hertz and represents the width of the frequency interval available for transmission. In FDM, this interval is partitioned into several smaller frequency regions, or *sub-bands*, and each tributary is assigned one of them.

A tributary signal is moved into its assigned sub-band by modulating a carrier frequency. If three tributaries use carrier frequencies f_1, f_2, and f_3, their spectra are centered around these three different frequencies. The resulting signals can then be added together and transmitted simultaneously over the same physical channel. At the receiving side, frequency-selective components separate the different spectral regions and recover the individual tributaries.

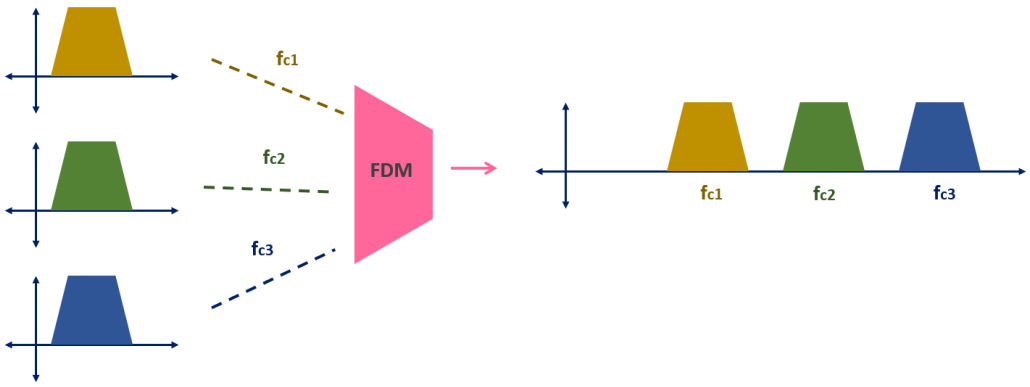


Figure 2.2: Principle of Frequency Division Multiplexing (FDM). Three tributary signals are associated with different carrier frequencies.

Figure 2.2 illustrates the basic idea. The same transmission medium carries all the tributaries at the same time, but the signals remain distinguishable because they occupy

different portions of the frequency spectrum.

The number of channels that can be multiplexed depends on the total bandwidth made available by the physical medium and on the amount of spectrum required by each tributary. The latter is related to the bitrate of the tributary: in general, transmitting information at a higher rate requires a signal that occupies a wider frequency range.

This relationship is not determined by the bitrate alone, because it also depends on the modulation format and on how efficiently the available spectrum is used. A useful high-level relationship is

$$R_b \approx \eta B_\mathrm{s},$$

where R_b is the tributary bitrate, B_s is the occupied bandwidth, and η is the spectral efficiency, measured in bit/s/Hz. For a given modulation and coding scheme, a higher bitrate therefore typically translates into a larger occupied bandwidth.

If the available physical channel provides a total bandwidth B_ch, while each tributary requires approximately B_s, a first-order estimate of the number of independent frequency channels is

$$N \approx \frac{B_\mathrm{ch}}{B_\mathrm{s}}.$$

The relationship is intuitive: a wider physical channel can accommodate more frequency channels, whereas tributaries requiring larger spectral width reduce the number of signals that can be transmitted simultaneously.

An important consequence of FDM concerns the bitrate that must be handled by the electronic equipment.

Assume that N tributaries each carry a digital stream at B bit/s. With FDM, these tributaries are not serialized into a single digital stream operating at NB bit/s. Each tributary continues to be generated, modulated, and processed by an electronic chain operating at approximately its original rate B bit/s. The resulting signals occupy different frequency bands and are transmitted in parallel over the same physical medium.

This is technologically important because increasing the bitrate of a single electronic processing chain becomes progressively more difficult. A higher bitrate requires electronic circuits to operate on increasingly short time scales: serializers, modulators, receivers, clock-recovery circuits, and signal-processing stages must all handle faster transitions and tighter timing constraints. Signal integrity also becomes more critical, while circuit complexity, power consumption, and cost generally increase.

FDM therefore offers a different way of increasing the aggregate capacity of a transmission system. Instead of building one electronic channel operating at the sum of all tributary bitrates, several lower-rate electronic channels operate in parallel and are mapped onto different portions of the available spectrum.

Time Division Multiplexing

Time Division Multiplexing (TDM) separates different tributary streams in the *time domain*. Unlike FDM, where several tributaries are transmitted simultaneously using different portions of the frequency spectrum, in TDM the tributaries share the same transmission channel by using it during different time intervals.

To organize this sharing mechanism, the time axis is divided into a periodic sequence of *frames*. Each frame is further divided into a number of *time slots*, and each tributary is assigned a specific slot within every frame. The same allocation is repeated periodically, so that the position assigned to each tributary is known in advance.

Consider N tributaries with the same bitrate R_b. In a simple synchronous TDM system, each frame contains N time slots, one for each tributary. Let T_F denote the frame period and assume that each tributary can place L bits into its assigned slot during every frame.

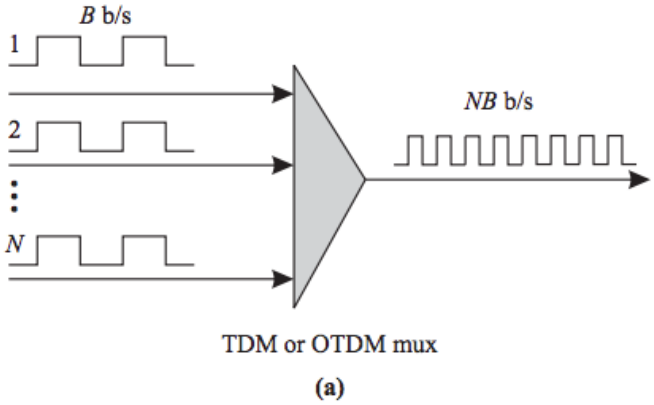


Figure 2.3: Principle of Time Division Multiplexing (TDM). Multiple tributary digital streams share the same transmission channel by using different time slots within a periodic frame.

As shown in Figure 2.3, the allocation is static: each tributary receives a predefined portion of every frame, independently of whether the corresponding source is continuously active.

The frame structure can be directly related to the bitrate supported by each tributary. Since a tributary can contribute L bits during every frame period T_F, its bitrate is

$$R_b = \frac{L}{T_F}.$$

This relation provides a simple way of interpreting the dimensioning of a TDM system. For a fixed number of bits L assigned to a tributary in each frame, supporting a larger bitrate requires frames to repeat more frequently, that is, a smaller value of T_F. Alternatively, more bits can be allocated to that tributary during each frame.

The multiplexed output must carry the contributions of all N tributaries. Since every frame contains N slots and each slot carries L bits, the total amount of tributary data transported in one frame is

$$NL.$$

Ignoring framing and synchronization overhead, the bitrate of the aggregate stream is therefore

$$R_{\text{out}} = \frac{NL}{T_F} = NR_b.$$

This is the fundamental scaling property of synchronous TDM: combining N equal-rate tributaries produces a serial output stream whose bitrate is approximately N times the bitrate of one tributary.

For example, suppose that four tributaries each operate at

$$R_b = 1 \text{ Gbps},$$

and that each tributary contributes $L = 1000$ bits during every frame. The frame period must then satisfy

$$T_F = \frac{L}{R_b} = \frac{1000}{10^9} = 1\ \mu\text{s}.$$

The complete frame contains

$$4 \times 1000 = 4000 \text{ bits},$$

which must be transmitted during the same 1 μs interval. Consequently,

$$R_{\text{out}} = \frac{4000}{1\ \mu\text{s}} = 4 \text{ Gbps}.$$

Each slot lasts

$$T_S = \frac{T_F}{4} = 250 \text{ ns},$$

and the output interface must transmit the 1000 bits associated with one tributary during that interval.

This example highlights the main technological difference with respect to FDM. In TDM, multiplexing creates a single serial stream whose electronic bitrate increases with the number of tributaries. The multiplexer, transmitter, receiver, clock-recovery circuitry, and subsequent electronic processing must therefore operate at the aggregate rate R_{out}.

As N or R_b increases, the available time for processing and transmitting the aggregate data becomes progressively shorter. Designing electronics that operate reliably at these higher rates becomes more challenging in terms of timing, signal integrity, power consumption, and cost.

2.2.2 Frequency and Wavelength

Before introducing WDM in more detail, it is useful to clarify what is meant by *wavelength* and how it relates to frequency.

Light is an electromagnetic wave and can be described either by its frequency f or by its wavelength λ. In vacuum, these two quantities are related by

$$c = \lambda f,$$

where $c \simeq 3 \times 10^8$ m/s is the speed of light. Therefore,

$$\lambda = \frac{c}{f}.$$

A higher optical frequency corresponds to a shorter wavelength, while a lower frequency corresponds to a longer wavelength.

Inside a fiber, the propagation velocity is lower than c because of the refractive index of the material. Strictly speaking, the wavelength inside the fiber is therefore shorter than the corresponding wavelength in vacuum, whereas the frequency remains unchanged. In optical communications, however, carriers are conventionally identified by their wavelength in vacuum. This allows values such as 1310 nm or 1550 nm to be used as standard references independently of the specific propagation conditions inside the fiber.

For example, an optical carrier with wavelength

$$\lambda = 1550 \text{ nm}$$

corresponds to a frequency of approximately

$$f = \frac{3 \times 10^8}{1550 \times 10^{-9}} \simeq 193.5 \text{ THz}.$$

A carrier at 1310 nm has a correspondingly higher frequency, since a shorter wavelength implies a larger value of f.

The wavelengths used in optical communication systems are not chosen arbitrarily. Fiber attenuation depends on wavelength, and some spectral regions provide significantly more favorable transmission conditions than others.

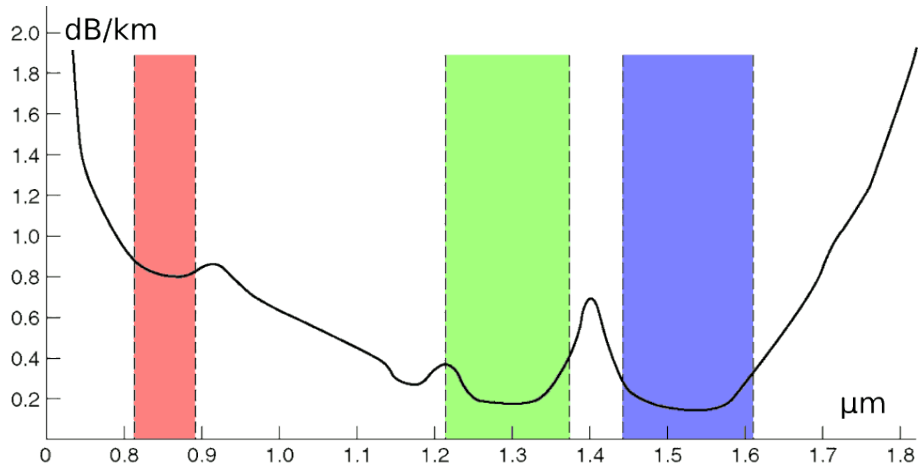


Figure 2.4: Qualitative attenuation characteristic of an optical fiber as a function of wavelength. The highlighted regions indicate important transmission windows exploited by optical communication systems. In particular, the region around 1550 nm is especially relevant because fiber attenuation reaches very low values.

Figure 2.4 shows the qualitative dependence of fiber attenuation on wavelength. The regions around 1310 nm and 1550 nm correspond to important transmission windows of optical fiber.

The 1310 nm region has historically been widely used because it offers favorable propagation characteristics. The region around 1550 nm is particularly important for long-distance transmission because attenuation becomes very low, allowing signals to propagate over longer distances before amplification or regeneration is required.

The same optical carrier can therefore be identified either by its wavelength or by its frequency. In optical networking, the wavelength representation is especially common, while channel plans and spectral allocations are often specified in terms of frequency spacing.

Figure 2.5 illustrates this point through an example of regularly spaced optical carriers.

In the example, adjacent optical carriers are separated by 100 GHz. Since frequency and wavelength are inversely related, a constant spacing in frequency does not correspond to a constant spacing in wavelength. This can be observed directly from the wavelength values associated with the different carriers in Figure 2.5.

The figure also highlights another important distinction. The *channel spacing* indicates the separation between the nominal center frequencies of two adjacent optical carriers, whereas the *signal bandwidth* represents the spectral region actually occupied by the modulated signal around each carrier.

These two quantities should therefore not be confused: a WDM system may use a regular grid of carrier frequencies while each individual optical signal occupies only a fraction of the spacing allocated between adjacent channels.

The important point for the following discussion is that frequency and wavelength are simply two equivalent ways of identifying an optical carrier. WDM exploits this property by assigning different tributary signals to different optical carriers. These carriers may be specified by their center frequencies, but in optical networking they are commonly referred to by their wavelengths, such as λ_1, λ_2, and λ_3.

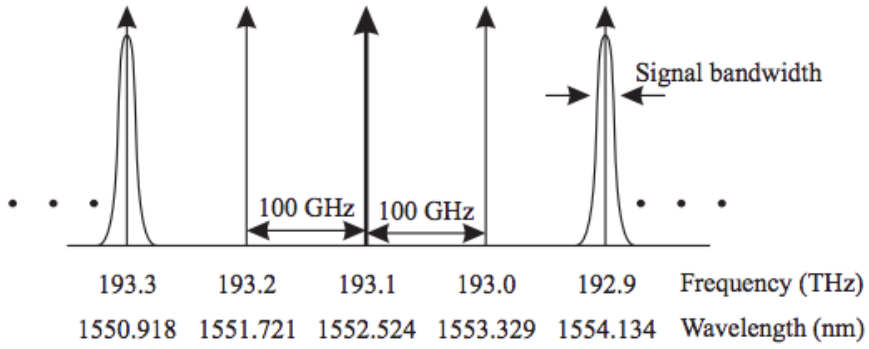


Figure 2.5: Example of regularly spaced optical carriers in a WDM system. Adjacent carriers are separated by 100 GHz in the frequency domain. Because frequency and wavelength are related by $c = \lambda f$, equally spaced frequencies do not correspond to equally spaced wavelengths. Each optical carrier also occupies a finite signal bandwidth around its nominal center frequency.

2.2.3 Wavelength Division Multiplexing

The principle introduced with FDM can be directly applied to optical communications. An optical fiber provides a very large transmission bandwidth, which can be divided into several optical channels. In Wavelength Division Multiplexing (WDM), each tributary is assigned to a different optical carrier.

Since an optical carrier can be identified either by its frequency f_i or by its corresponding wavelength λ_i, with

$$f_i = \frac{c}{\lambda_i},$$

optical channels are conventionally denoted by their wavelengths:

$$\lambda_1, \lambda_2, \ldots, \lambda_N.$$

Each wavelength carries an independent tributary signal. An optical multiplexer combines the different wavelengths onto the same fiber, while a demultiplexer at the receiving side separates them again.

In modern transport networks, a common implementation is Dense Wavelength Division Multiplexing (DWDM), where optical carriers are placed relatively close to each other in the frequency spectrum. Standard DWDM grids define the nominal central frequencies of the channels and their spacing. Typical fixed-grid channel spacings are 100 GHz, 50 GHz, 25 GHz, and 12.5 GHz.

It is important to distinguish *channel spacing* from the actual bandwidth occupied by a transmitted signal. For example, a 50 GHz DWDM grid means that the central frequencies of adjacent optical channels are separated by 50 GHz; the signal carried by each channel must fit within the spectral resources allocated around that central frequency. Its actual occupied bandwidth depends on the bitrate and modulation format used by the transponder.

Figure 2.6 shows that the tributaries are transmitted in parallel over the same fiber and remain distinguishable because they occupy different regions of the optical spectrum.

This is particularly attractive in optical transport because the usable bandwidth of the fiber is much larger than the bitrate that can conveniently be handled by a single electronic processing chain. Exploiting the whole fiber capacity through TDM alone would require an electronic stream operating at an increasingly high aggregate bitrate, which

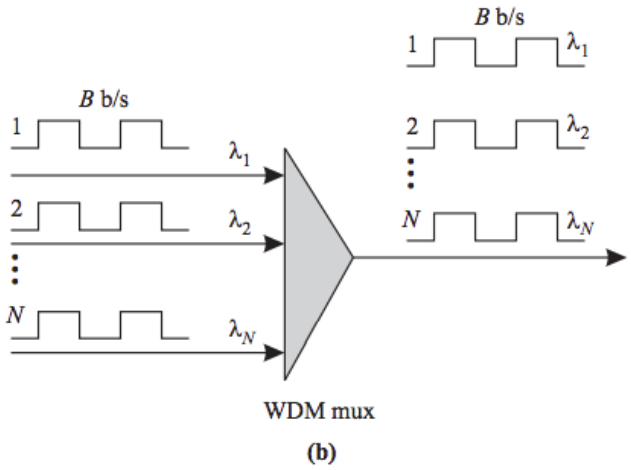


Figure 2.6: Principle of Wavelength Division Multiplexing (WDM). N tributary streams, each operating at bitrate R_b, are mapped by independent optical transmitters or transponders onto different optical carriers, labeled λ_1, λ_2, ..., λ_N. An optical multiplexer combines these wavelength channels onto a single fiber, while a demultiplexer at the receiving side separates them again.

makes the transmitter, receiver, clocking, and signal processing more difficult and expensive to implement.

WDM instead distributes the total capacity across several parallel optical channels. If N channels each carry a tributary at bitrate R_b, the aggregate transported capacity is approximately

$$C_{\text{fiber}} \simeq NR_b,$$

while each transponder processes only the bitrate associated with its own optical channel.

Different wavelengths are often represented using different colors because this makes the idea of parallel optical channels intuitive. In practical transport systems, however, the wavelengths used are mainly in the near-infrared region and are not visible to the human eye.

DWDM increases the number of channels that can be accommodated in a given optical spectrum by reducing the spacing between adjacent carriers. For example, a 50 GHz grid can accommodate approximately twice as many nominal channel positions over the same spectral interval as a 100 GHz grid. Denser spacing, however, also requires tighter control of the optical carrier frequencies and transmission characteristics.

Modern systems can also use a flexible DWDM grid, where the spectral width allocated to a channel is not necessarily the same for every connection. This allows wider frequency slots to be assigned to signals requiring more spectrum, for example because of a higher bitrate or a different modulation format.

2.3 Optical Networks as Connection-Oriented Networks

The multiplexing techniques discussed so far explain how several traffic flows can share the same physical transmission infrastructure. They do not yet explain, however, how an end-to-end transport service is provided to a client.

The optical transport service considered in this chapter is *connection-oriented*. Before traffic can be transmitted between two endpoints, the network must first establish a connection between them. Once the connection has been created, the resources and

switching state associated with that connection remain in place while the service is active. When the service is no longer required, the connection can be removed and the corresponding resources released.

The lifetime of an optical connection can therefore be described through four main phases: connection request, connection setup, connection maintenance, and connection teardown.

A connection starts with a request from a client of the optical network. At a minimum, the request identifies the two endpoints that must be interconnected and the amount of capacity required. Depending on the service, additional requirements may also be specified, such as performance or protection constraints.

The optical network must then determine how to realize the requested connection using its own physical resources. This is the *connection setup* phase. The first step is to identify a path between the two endpoints, that is, a sequence of network nodes and physical links that can be used to carry the connection. Finding a path, however, is not sufficient. Since the underlying infrastructure is shared among multiple connections, the network must also reserve a transmission resource on each link of the selected path.

The nature of this resource depends on the multiplexing technology being used. In a time-multiplexed system it may correspond to a *time slot*; in a frequency-multiplexed system it may correspond to a *frequency sub-band*; and in a WDM optical network it corresponds to an *optical channel*, typically identified by its wavelength λ.

A connection is therefore characterized not only by the sequence of links that it traverses, but also by the physical channel allocated to it on each of those links. The required resources must be available along the complete path before the connection can be successfully established.

Finally, the intermediate network devices must be configured so that the reserved channel entering a node is correctly switched toward the corresponding outgoing link. Once the path has been selected, the required resources have been reserved, and the switching state has been installed, the connection is ready to carry client traffic.

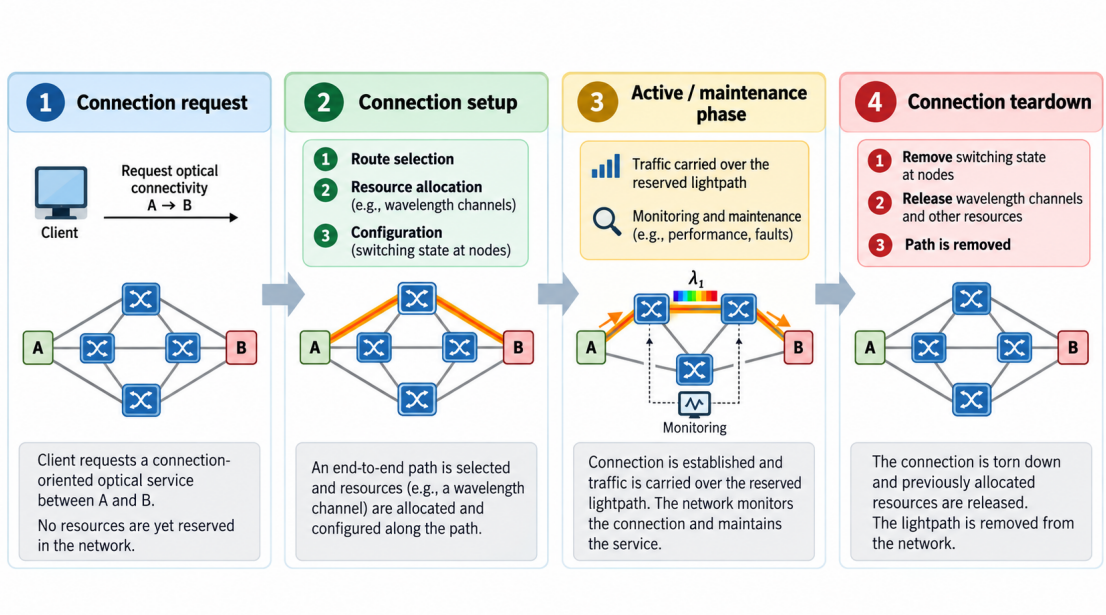


Figure 2.7: Lifecycle of a connection-oriented optical service.

Figure 2.7 summarizes this lifecycle. Once setup has been completed, the connection enters its active phase and client traffic can be transported. The network must retain the configuration associated with the connection and monitor the service while it remains active.

If failures or performance degradation occur, management and control functions may need to intervene to preserve or restore the service.

Finally, when the connection is no longer required, it is torn down. The switching state associated with it is removed and the transport resources can be reused for other connections.

2.4 From First-Generation to Second-Generation Optical Networks

The connection-oriented model describes the service offered by the transport network, but it does not specify how that service is implemented. The role of optical technology inside the network has changed substantially over time, and a useful way to understand this evolution is to distinguish between first-generation and second-generation optical networks.

2.4.1 Evolution toward Wavelength-Routed Optical Networks

In first-generation optical networks, optical fiber was used primarily as a high-capacity transmission medium. The intelligence of the network remained in the electrical domain: multiplexing, switching, routing, and other processing functions were performed electronically.

A typical approach was therefore to aggregate several lower-rate tributaries through TDM, producing a higher-rate electrical stream. This aggregate stream was then converted into an optical signal and transmitted over the fiber.

Whenever the signal reached a node where switching or other processing was required, it had to be converted back into the electrical domain. The node could then inspect or manipulate the digital signal, perform the required switching operation, and finally convert the resulting signal back into optical form for transmission over the next fiber section.

The fundamental role of optics in this architecture was therefore *transmission*. Optical fiber provided very high capacity and allowed signals to travel over long distances, but the decisions concerning how traffic was multiplexed and switched remained the responsibility of electronic equipment.

Second-generation optical networks introduced a fundamentally different use of the optical domain. With Wavelength Division Multiplexing, several independent optical channels can coexist on the same fiber, each associated with a different wavelength.

More importantly, these wavelength channels can be manipulated directly in the optical domain. Intermediate optical nodes can add, drop, or switch selected wavelengths without first converting all the transported information into electrical form.

The optical network therefore evolves from a system in which optics is used mainly for transmission into a network in which part of the switching functionality is also performed directly in the optical domain.

This makes it possible to establish an end-to-end optical connection through a sequence of optical nodes. Such a connection is referred to as a *lightpath*. From the client perspective, the lightpath provides a logical connection between two endpoints, while the optical network determines how that connection is realized across the underlying physical topology.

The introduction of lightpaths changes the problem faced by the transport network. Establishing a connection is no longer only a matter of selecting a sequence of links between two endpoints. The network must also determine which optical resources will be used along that route.

2.4.2 Lightpath Establishment and Wavelength Resources

When a new lightpath is requested, the optical network must perform two closely related tasks. It must first select a route between the source and the destination, and it must then allocate a suitable optical channel on every link of that route.

In the simplest wavelength-routed network, intermediate nodes do not perform wavelength conversion. A lightpath entering a node on a given wavelength must therefore leave that node on the same wavelength. Consequently, the same wavelength must be available on every link of the selected route.

This requirement is known as the *wavelength-continuity constraint*.

Suppose, for example, that a connection from node A to node D follows the route

$$A \to B \to C \to D.$$

If the connection is assigned wavelength λ_1, then λ_1 must be available on all three links:

$$(A, B), \qquad (B, C), \qquad (C, D).$$

Finding a physical route is therefore not sufficient. A route can be perfectly valid from a topological point of view and still be unusable because no single wavelength is simultaneously available on all of its links.

The joint problem of selecting a route and assigning a wavelength to an optical connection is commonly referred to as the *Routing and Wavelength Assignment* (RWA) problem.

If the wavelength-continuity constraint cannot be satisfied along a candidate route, the network has different possibilities. It can search for another route on which a common wavelength is available, or, if the network supports wavelength conversion, it can allow the connection to use different wavelengths on different portions of the path.

Wavelength conversion removes the requirement that the same optical carrier be used throughout the complete lightpath. For example, a connection may travel on λ_1 over the first part of its route and continue on λ_2 after an intermediate node. The information carried by the connection remains the same; only the optical carrier used to transport it changes.

A straightforward way to implement this function is through an optical-electrical-optical (O/E/O) conversion. The incoming signal on λ_1 is first converted into an electrical representation. The same digital information is then used to generate a new optical signal on λ_2.

Wavelength conversion therefore increases the flexibility of resource allocation and can allow a lightpath to be established even when no single wavelength is available along the complete route. The price is that an O/E/O conversion interrupts the optical transparency of the connection, since the transported signal must temporarily return to the electrical domain.

O/E/O conversion is not the only possible way to perform wavelength conversion. All-optical techniques can also change the wavelength of a signal without recovering the transported information in the electrical domain. For the purposes of this chapter, however, O/E/O conversion provides the most direct conceptual example of how the wavelength-continuity constraint can be relaxed.

The establishment procedure must therefore consider together the physical route and the optical resources available along it. A candidate route is usable if a continuous wavelength is available end to end or, where supported, if wavelength conversion can bridge portions of the route that require different optical carriers.

If neither a common wavelength nor a suitable wavelength-conversion capability is available, and no alternative feasible route can be found, the requested lightpath cannot be established and the connection request must be blocked.

2.5 Putting the Concepts Together: A Wavelength-Routed Network

The concepts introduced so far can now be combined in a single network example. Figure 2.8 shows a second-generation optical network in which several client devices are interconnected through a shared WDM infrastructure.

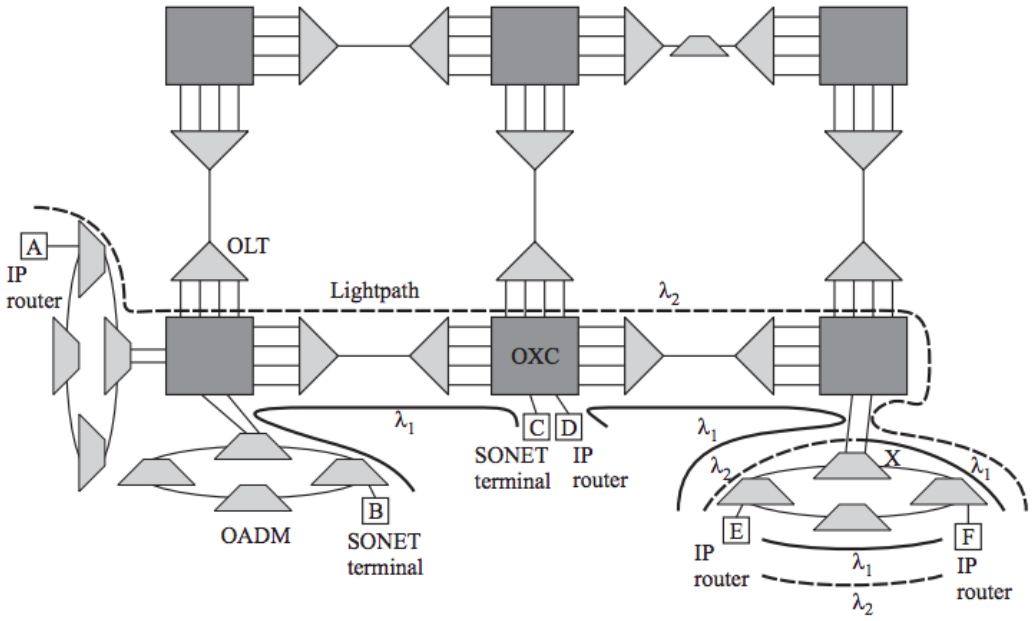


Figure 2.8: Example of a second-generation wavelength-routed optical network. Multiple client devices are interconnected through a shared WDM infrastructure. End-to-end lightpaths are established across a meshed topology, while intermediate optical nodes switch, add, or drop wavelength channels.

The first aspect to observe in Figure 2.8 is the network topology. Different portions of the optical transport infrastructure may adopt different topological organizations according to their role.

Toward the network periphery, optical rings are commonly used to collect and distribute traffic among multiple sites. In this context, Optical Add/Drop Multiplexers (OADMs) allow selected wavelengths to be locally added or dropped, while the remaining wavelength channels continue along the ring.

The backbone, instead, is typically more richly interconnected and can be organized as a mesh. Optical Cross-Connects (OXCs) are well suited to this role because they can switch wavelength channels among multiple input and output fibers. As a result, several physical routes may exist between the same pair of backbone nodes.

An end-to-end optical connection can therefore traverse different topological regions: traffic may be collected through an OADM-based ring, transported across an OXC-based meshed backbone, and eventually delivered through another peripheral ring or directly to the destination client.

Consider, for example, one of the highlighted end-to-end connections. A client attached to one side of the network requests connectivity toward a remote client. From the client's perspective, the result is simply an end-to-end transport service. Inside the optical network, however, the connection is implemented as a *lightpath*: a sequence of optical channels carried over multiple fiber links and switched through intermediate optical nodes.

A second important aspect concerns the distinction between the client domain and the

optical domain. An Optical Line Terminal (OLT) marks the boundary at which the client signal is adapted to the WDM transport system. Once the signal has entered the optical domain, the portion of the network between optical terminations can be regarded as an *optical link*: the client information is transported as an optical signal and can traverse intermediate optical components without being interpreted as a sequence of client-layer bits.

In the all-optical portions of the network, an intermediate OADM or OXC can therefore manipulate the optical channel while leaving the client signal in the optical domain. An OADM may add or drop a selected wavelength, while an OXC may switch a wavelength from an incoming fiber to a selected outgoing fiber. Unless an O/E/O regeneration or conversion stage is required, the client signal does not need to return to the electrical domain while traversing these intermediate nodes.

This property leads to the concept of *transparency*. The optical network does not need to understand the protocol carried by the client signal in order to transport it. The client may, for example, be an IP router, an Ethernet device, or another type of communication equipment. Once the signal has been properly adapted at the network edge, the optical layer primarily sees an optical channel rather than the internal format of the client traffic.

Transparency can also concern the client bitrate. If the signal remains in the optical domain, intermediate optical switching elements do not need to process the individual bits and are therefore largely independent of the bitrate and frame format carried by the wavelength. The same optical switching infrastructure can consequently transport different types of client signals without redesigning the intermediate nodes for each specific service.

The highlighted lightpath is only one of the connections carried by the infrastructure. Other clients can establish their own lightpaths simultaneously, using different wavelengths and possibly sharing part of the same physical route. WDM is what makes this possible: a single fiber can carry several independent optical channels, each potentially associated with a different end-to-end connection.

The physical topology and the set of established lightpaths should therefore not be confused. The physical topology describes the fibers and optical nodes available in the provider infrastructure. The lightpaths describe the connection-oriented services currently established on top of that infrastructure. Different lightpaths may share physical links, while the same pair of client nodes may potentially be connected through different physical routes.

Finally, the example also illustrates wavelength assignment. A lightpath must be associated with an optical channel on every link of its route. If the same wavelength is available throughout the complete path, the wavelength-continuity constraint can be satisfied directly. If wavelength conversion is available, the connection may instead use, for example, λ_1 on one portion of the route and λ_2 on another.

If neither of these conditions can be satisfied, the lightpath cannot be established. In other words, if no common wavelength is available along the selected route and wavelength converters are not available at the required intermediate nodes, the connection request must be blocked or rejected. The example shown in Figure 2.8 includes exactly this type of situation: one connection can be successfully set up, whereas another request cannot be accommodated because the required optical resources are not available under the wavelength-continuity constraint.

2.6 Optical Network Components

The preceding discussion treated optical nodes in an abstract way. We can now look more closely at the main components used to build the network.

2.6.1 Optical Line Terminals

An Optical Line Terminal (OLT) is located at the boundary between a client network and the WDM transport infrastructure. Its main role is to provide the interface between the client domain and the optical line system.

At a high level, an OLT is composed of three main functional elements: *transponders*, *wavelength multiplexers/demultiplexers*, and *optical amplifiers*. The transponders adapt the client signals to the optical transport system, the multiplexer combines several wavelength channels onto the same fiber, and the optical amplifier compensates for the losses introduced by the transmission system.

In the opposite direction, the same functions are performed in reverse. The incoming multi-wavelength signal is amplified when necessary, the demultiplexer separates the individual optical channels, and each transponder converts the corresponding line signal into a representation suitable for the client device.

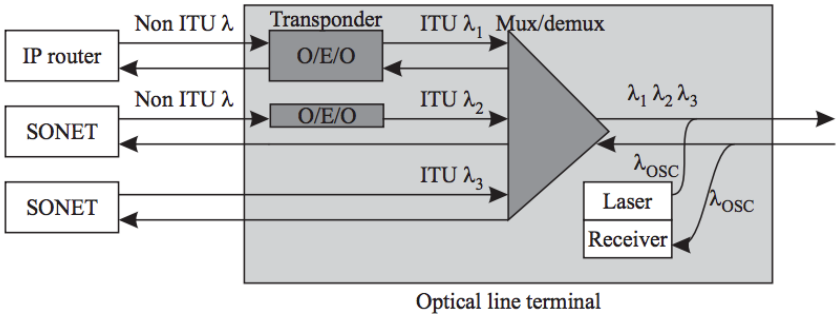


Figure 2.9: Architecture of an Optical Line Terminal (OLT).

Figure 2.9 summarizes the role of the OLT. The three main components operate at different levels: the transponders work on individual client signals, the multiplexer and demultiplexer operate on the set of wavelength channels, while the amplifier operates on the combined optical signal carried by the fiber.

Transponders

The transponder is the element that interfaces a single client signal with one optical channel of the WDM line system. Its role is broader than a simple electrical-to-optical conversion, because it performs several adaptation, monitoring, and transport functions.

Client adaptation. A client device is not necessarily able to generate directly the optical signal required by the WDM system. In the simplest case, the client interface may be electrical, and an electrical-to-optical conversion is therefore required. Even when the client already provides an optical signal, its wavelength or other physical characteristics may not be compatible with the optical line system.

For example, a client interface may operate around 1.3 μm (approximately 1310 nm), while the transport system may require a precisely controlled wavelength in the region around 1.55 μm. The transponder receives the client signal and generates a new optical signal suitable for transport inside the WDM network.

This client-side optical signal is sometimes described as a *gray* interface, in contrast with the precisely defined "colored" wavelengths used inside a DWDM system. From this perspective, the transponder can be viewed as the element that maps a generic client interface onto a specific optical line channel.

OTN adaptation and overhead. Since the client information becomes available in the electrical domain, the transponder can adapt it to the framing structure used by the

Optical Transport Network (OTN). Additional overhead can be introduced to identify and supervise the optical connection, signal failures, and support performance monitoring.

Forward Error Correction (FEC) information can also be added at this stage. FEC introduces controlled redundancy into the transmitted information so that the receiver can detect and correct a certain number of transmission errors.

FEC should not be confused with network protection. FEC protects the information carried over a given optical path against bit errors, whereas protection mechanisms react to network failures by moving traffic onto alternative resources or paths.

Performance monitoring. The transponder is also a natural monitoring point because the signal is available in the electrical domain. One of the most important performance parameters is the Bit Error Rate (BER), which provides an indication of the quality of the received digital signal. OTN overhead and FEC information can be used to evaluate error performance and determine whether the connection is operating within the required limits.

Depending on the technology, the equipment may also expose measurements such as transmitted and received optical power, loss-of-signal indications, FEC error counters, and additional signal-quality metrics.

Optical transmission and reception. At the physical level, the transmitter uses a laser to generate the optical carrier, while the receiver uses a photodetector, typically implemented with a photodiode, to convert the received optical power back into an electrical signal.

A simple introductory model is On–Off Keying (OOK), where different optical power levels represent the binary information. In the most intuitive interpretation, the presence of light represents one binary value and the absence, or a much lower optical power, represents the other. Modern high-capacity coherent systems can use substantially more sophisticated modulation formats, but OOK provides a useful first model for understanding how digital information can modulate an optical carrier.

Traffic grooming. Up to this point, we have implicitly assumed a simple correspondence between clients and wavelengths:

$$\text{one client signal} \longrightarrow \text{one optical channel.}$$

This assumption is useful for explaining the basic operation of an optical network, but it is not a fundamental requirement. The capacity available on a wavelength can be significantly larger than the bit rate generated by an individual client. Assigning an entire wavelength to every low-rate client would therefore result in poor utilization of the available optical capacity.

Since the client information is available in the electrical domain at the transponder, several lower-rate client signals can be combined into a single higher-rate digital stream before the optical carrier is generated. The resulting aggregate signal is then transmitted over a single wavelength. This operation is referred to as *traffic grooming*.

For example, instead of transporting four low-rate clients over four distinct wavelengths,

$$C_1 \to \lambda_1, \qquad C_2 \to \lambda_2, \qquad C_3 \to \lambda_3, \qquad C_4 \to \lambda_4,$$

their traffic can be multiplexed in the electrical domain and transported over a common optical channel:

$$C_1, C_2, C_3, C_4 \longrightarrow \text{electrical aggregation} \longrightarrow \lambda_1.$$

The important point is therefore that a wavelength is not necessarily associated with a single client. It represents an optical transport channel, whose capacity can be shared by multiple client signals.

A device that combines the transponder function with the aggregation of multiple lower-rate client signals is commonly referred to as a *muxponder*. From the line-side perspective, the result is still one optical channel on one wavelength, but the information carried by that channel belongs to several client tributaries.

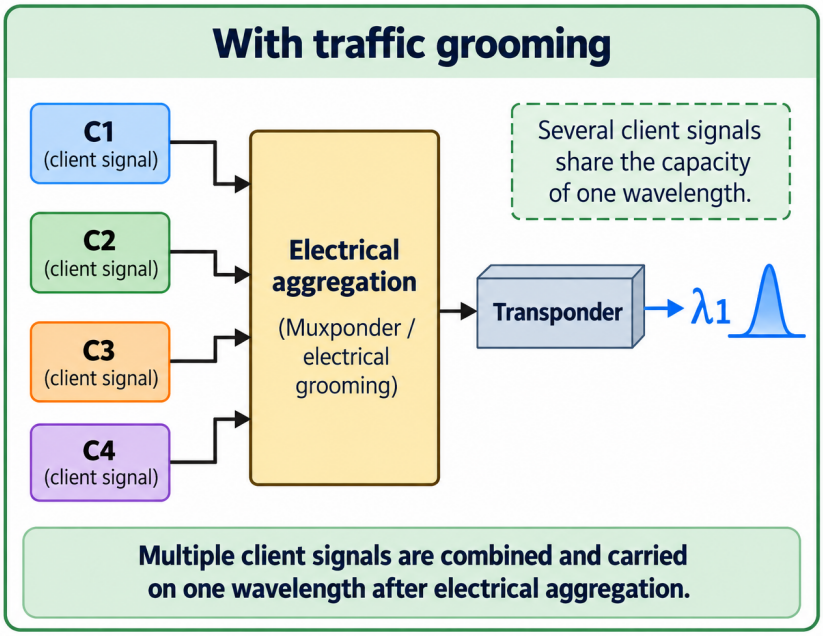


Figure 2.10: Traffic grooming at the transponder side. Several lower-rate client signals are first aggregated in the electrical domain by a grooming function or muxponder. The resulting higher-rate electrical signal is then converted by the transponder into a single optical channel and transmitted over one wavelength, here represented by λ_1.

Figure 2.10 also highlights an important distinction between traffic grooming and wavelength multiplexing. Grooming combines client traffic in the *electrical domain* before the line-side optical signal is generated. Wavelength multiplexing, discussed in the next section, instead combines already generated optical channels so that several wavelengths can propagate simultaneously over the same physical fiber.

Cost, footprint, and scalability. A conventional transponder provides the line-side transmission and reception functions associated with one optical channel. In the basic WDM architecture, each active wavelength therefore requires a corresponding transponder instance.

If an OLT has to transmit

$$\lambda_1, \lambda_2, \ldots, \lambda_N,$$

the basic architecture requires N line-side transponder functions before their outputs can be combined by the wavelength multiplexer.

This is an important practical consideration because transponders typically account for a substantial fraction of the cost, physical footprint, and power consumption of an OLT. Increasing the number of wavelengths therefore increases not only the capacity transported by the fiber but also the amount of terminal equipment required.

The relation "one transponder = one wavelength" is best understood as a functional model. Modern hardware can integrate several transponder functions on the same physical card, and muxponders can aggregate several lower-rate client signals before transmitting them over a higher-rate optical channel. Nevertheless, each independently generated line-side

optical carrier still requires the corresponding transmission and reception functionality.

Wavelength Multiplexers and Demultiplexers

Once the individual transponders have generated their respective optical channels, the wavelengths must be combined so that they can share the same physical fiber. This is the role of the wavelength multiplexer.

If the transponders generate the channels

$$\lambda_1, \lambda_2, \ldots, \lambda_N,$$

the multiplexer receives these optical signals on separate input ports and combines them into a single WDM signal. All wavelengths then propagate simultaneously over the same fiber.

At the opposite end of the optical link, the demultiplexer performs the inverse operation. The composite WDM signal is separated according to wavelength, and each channel is directed toward the corresponding line-side receiver.

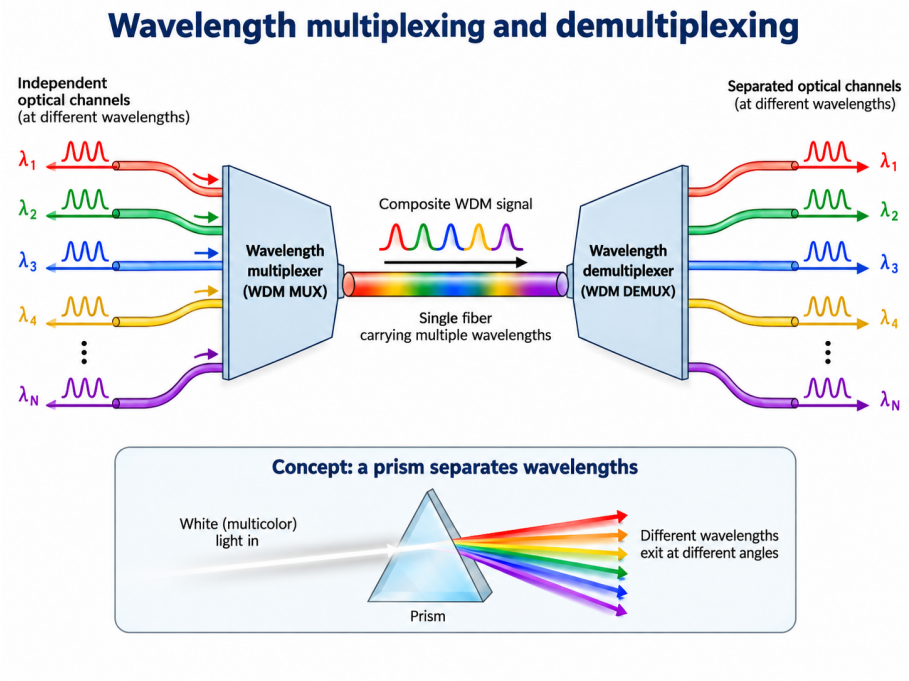


Figure 2.11: Wavelength multiplexing and demultiplexing.

Figure 2.11 illustrates the basic operation. Multiplexing does not merge the client information into a single faster electronic bit stream. The optical channels remain distinct and coexist on the same physical medium.

A useful physical analogy is a prism. When light containing several wavelengths passes through a prism, different wavelengths are refracted by different amounts and therefore emerge in different directions. Conversely, the same principle can be imagined in reverse to combine different wavelength components.

A practical WDM multiplexer is not necessarily implemented using a simple prism—real systems can employ devices such as diffraction gratings, optical filters, or arrayed-waveguide gratings—but the prism provides an intuitive picture of the essential operation: optical signals can be separated or combined according to their wavelength.

The multiplexer and demultiplexer therefore implement the physical counterpart of the WDM principle introduced earlier. The transponders create the individual optical channels;

the multiplexer allows those channels to share the same fiber; and the demultiplexer separates them again before they are delivered to the corresponding receivers.

Optical Amplifiers

Even though optical fiber provides very favorable transmission characteristics, the power of an optical signal decreases as it propagates along the fiber. Additional losses are also introduced by passive optical components such as connectors, multiplexers, demultiplexers, and switching elements.

Optical amplifiers are used to compensate for these losses. Their key characteristic is that amplification can be performed directly in the optical domain: the incoming light signal is amplified without first being converted into an electrical representation.

This is particularly useful in a WDM system because a suitable optical amplifier can amplify several wavelength channels simultaneously. The device does not have to recover the individual client bit streams before increasing their optical power.

The amplifier therefore differs conceptually from a transponder or electronic regenerator. A transponder terminates the optical signal and makes the client information available in the electrical domain. An optical amplifier instead acts directly on the optical signal and preserves the transparency of the transported channels.

In long-distance systems, optical amplifiers can be placed at appropriate points along the transmission path to compensate for accumulated attenuation. A common example in the 1550 nm region is the Erbium-Doped Fiber Amplifier (EDFA), which can amplify a group of WDM channels directly in the optical domain.

Optical Supervisory Channel

In addition to the client-carrying wavelengths, the OLT also terminates an *Optical Supervisory Channel* (OSC). The OSC is a separate optical channel used for the operation and supervision of the optical line system rather than for ordinary client traffic.

At the transmitting terminal, the OSC can be added to the optical signal sent over the line. At the receiving terminal, it is extracted and processed by the network equipment. The supervisory channel can therefore provide communication among the devices responsible for managing and monitoring the optical infrastructure.

Conceptually, the signal leaving an OLT can thus be viewed as the combination of several client-carrying wavelength channels and an additional supervisory channel dedicated to the operation of the optical network.

2.6.2 Optical Add/Drop Multiplexers

Optical Line Terminals are appropriate when all the wavelength channels carried by a fiber have to be terminated at a node. In many transport-network scenarios, however, this is not the case. A large fraction of the traffic crossing an intermediate node may simply be travelling toward more distant destinations.

Terminating these *passthrough* wavelength channels would be inefficient. Each channel would have to be converted from the optical domain to the electrical domain and then retransmitted optically on the next link, even though the intermediate node does not need to access the transported client information.

An *Optical Add/Drop Multiplexer* (OADM) avoids this unnecessary termination. It allows selected wavelength channels to be *dropped* from the optical line toward a local client and other wavelength channels to be *added* by the local client, while the remaining channels pass through the node directly in the optical domain.

From a logical perspective, an OADM therefore has four interfaces. The *input* and *output* interfaces connect the device to the optical transport infrastructure, whereas the *drop* and

add interfaces serve the local traffic. Incoming wavelengths that are not locally terminated travel from the input toward the output without undergoing an O/E/O conversion.

Why OADMs Are Useful: Passthrough Traffic

The advantage of an OADM becomes clearer by considering a simple example. Assume that three sites, denoted by A, B, and C, have to be interconnected through an optical transport network. The sites may represent, for example, three geographically separated branches of the same organization.

Suppose that the required logical connectivity is the following: one optical channel is required between A and B, one optical channel between B and C, while the traffic between A and C is larger and requires three independent wavelength channels.

Hence,

$$N_{AB} = 1, \qquad N_{BC} = 1, \qquad N_{AC} = 3.$$

Assume also that the physical transport infrastructure follows the sequence

$$A \longrightarrow B \longrightarrow C.$$

The three A–C lightpaths must therefore physically cross node B, even though their traffic is not destined to that node.

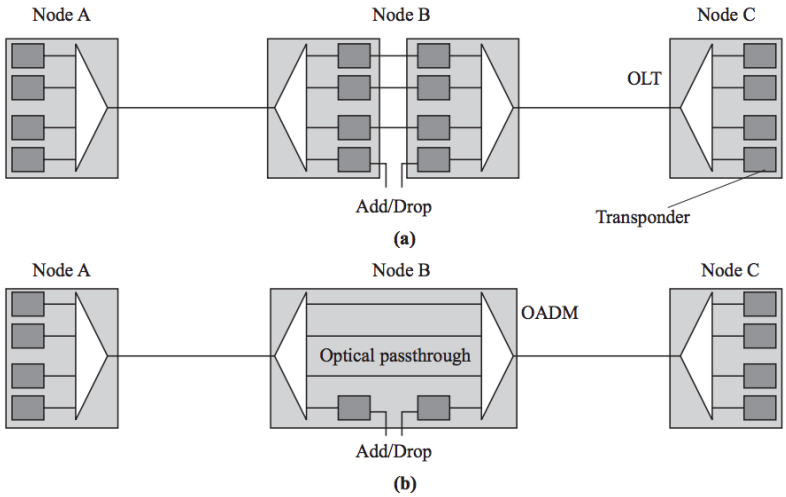


Figure 2.12: Handling passthrough traffic with and without an OADM.

Consider first the solution in which every wavelength is terminated at each intermediate node. Four wavelengths travel on the A–B link: one belongs to the A–B connection and three belong to the A–C connection. Similarly, four wavelengths travel on the B–C link.

Using the simplified model in which each wavelength termination requires one transponder function, node A requires four transponders and node C also requires four. At node B, however, all four incoming wavelengths are terminated and four wavelengths must subsequently be transmitted on the outgoing link. The overall number of transponder functions is therefore

$$N_{\mathrm{TP}}^{\mathrm{termination}} = 4 + 8 + 4 = 16.$$

The inefficient part of this architecture is immediately visible. Of the four wavelengths arriving at B from A, only one actually carries traffic destined to B. The other three

channels contain A–C traffic and are terminated only because the intermediate equipment operates in the electrical domain.

If node B is equipped with an OADM, the three A–C wavelength channels do not need to be terminated. They can pass directly from the incoming fiber to the outgoing fiber while remaining in the optical domain. Only the channel associated with local traffic has to be dropped, and the locally generated B–C channel has to be added.

With the same simplified counting, the terminal equipment at A and C still requires four transponder functions each, while only two transponder functions are needed for the local add/drop traffic at B. Hence,

$$N_{\mathrm{TP}}^{\mathrm{OADM}} = 4 + 2 + 4 = 10.$$

The OADM therefore eliminates six transponder functions in this example, corresponding to a reduction of

$$\frac{16 - 10}{16} = 37.5\%.$$

More importantly, all three passthrough channels remain entirely in the optical domain at the intermediate node.

The saving becomes even more significant as the fraction of passthrough traffic increases. If N wavelength channels traverse an intermediate node without being locally terminated, a conventional O/E/O architecture requires two additional transponder functions at that node for every passthrough channel: one to receive the incoming optical signal and one to retransmit it. An OADM removes these intermediate terminations. Consequently, when passthrough traffic dominates, the reduction in the number of transponder functions can approach one half with respect to a terminate-and-retransmit architecture.

OADM Architecture

The internal operation of an OADM can be understood by recalling the wavelength multiplexing principle introduced earlier. The incoming fiber carries a composite WDM signal containing several optical channels,

$$\lambda_1, \lambda_2, \ldots, \lambda_N.$$

A demultiplexing stage first separates the optical channels according to their wavelength. Selected wavelengths are directed toward the *drop* ports and delivered to the local equipment. The remaining wavelengths are not terminated: they bypass the local electronic domain and continue toward the output side of the device.

At the same time, locally generated optical channels can enter through the *add* ports. Finally, the passthrough wavelengths and the newly added channels are multiplexed again onto the outgoing fiber.

The basic OADM considered here is a *fixed* device. The set of wavelengths that are added or dropped is determined by its physical configuration. Once the device has been configured, the same wavelength channels continue to be added and dropped until the configuration is manually modified.

This behavior is appropriate when the optical topology and the established lightpaths change infrequently, but it provides little flexibility for dynamic reconfiguration. For example, an OADM configured to drop λ_1 and λ_2 cannot instantaneously decide to drop a different set of wavelengths according to changes in traffic demand.

This limitation motivates more flexible optical switching devices. In the following sections, Optical Cross-Connects will provide a much larger switching capability in meshed backbone networks, while reconfigurable add/drop architectures can remove the static wavelength assignment imposed by a fixed OADM.

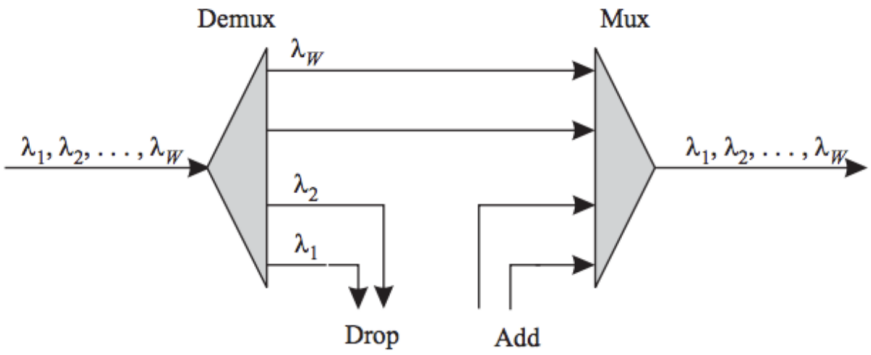


Figure 2.13: Internal architecture of a fixed Optical Add-Drop Multiplexer (OADM).

2.6.3 Optical Cross-Connects

Optical Cross-Connects (OXCs) provide the switching functionality required in the core of a wavelength-routed optical network. While OADMs are particularly suited to peripheral portions of the infrastructure, where a limited number of wavelength channels are locally added or dropped, OXCs are designed to handle a larger number of fibers, wavelength channels, and possible switching configurations. For this reason, they are particularly well suited to the meshed backbone portion of the optical network.

From a functional point of view, an OXC can be regarded as an *optical switch*. A wavelength channel arriving on one input fiber can be redirected toward one of several possible output fibers. This is precisely the flexibility required in a meshed topology, where an intermediate node may be connected to several neighboring nodes and a lightpath can therefore continue along different possible directions.

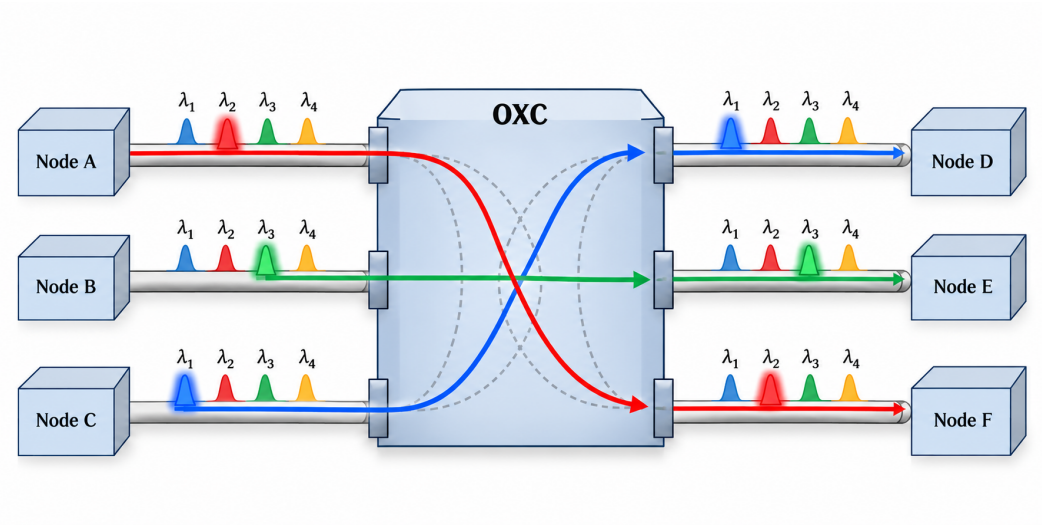


Figure 2.14: Role of an Optical Cross-Connect in a meshed wavelength-routed network. The OXC receives multiple wavelength channels on several input fibers and switches each channel toward the output fiber required by the corresponding lightpath.

Figure 2.14 highlights the basic operation of the device. Each fiber can carry several wavelength channels, and the OXC establishes a mapping between an incoming optical channel and a selected outgoing direction. For example, a channel carried on λ_2 and arriving from one neighboring node may be switched toward a different fiber in order to continue along the path assigned to its lightpath.

A useful way to understand this function is to compare an OXC with a router, since

routers are the switching devices that students are usually most familiar with. A router receives packets in digital form and can inspect the binary information contained in their headers. In particular, it can read the destination IP address and use that information to determine the appropriate output interface.

An OXC operates in a different domain. What the device receives is an optical signal carried on a given wavelength. The OXC does not directly access the binary information transported by that signal and therefore cannot base its switching operation on fields such as an IP address.

The switching behavior is instead established when the optical connection is configured. The node is instructed to associate a given incoming wavelength channel with the output fiber that allows the lightpath to continue toward its destination. Switching therefore takes place at the *wavelength level*: the optical channel itself, rather than the client information transported inside it, is the entity being switched through the node.

OXCs are modular devices whose exact internal architecture depends on the functions and interfaces installed in the equipment. At an abstract level, however, their fundamental component is a *switching matrix* that creates the required connections between input and output optical channels.

A first possibility is to perform switching in the electrical domain. In this case, incoming optical signals are converted into electrical signals, switched electronically, and then converted back into the optical domain. This approach provides considerable processing flexibility, but it requires O/E/O conversion and therefore reduces the transparency of the optical path.

A second possibility, which is particularly attractive in optical transport networks, is to perform the switching directly in the optical domain. In this case, the incoming WDM signal is first separated into its wavelength components, and the desired optical channel is then redirected toward the selected output fiber without interpreting the client information carried by that wavelength.

One important implementation of this idea is based on *Micro-Electro-Mechanical Systems* (MEMS). In a MEMS-based OXC, the switching matrix is realized using arrays of microscopic movable mirrors. Each mirror can be electronically controlled so as to change its orientation and, consequently, the direction in which the light beam is reflected.

The key idea is that the control is electronic, but the transported signal remains optical. After wavelength separation, the optical channel to be routed is physically steered through the switching matrix by the MEMS mirrors until it reaches the desired output port.

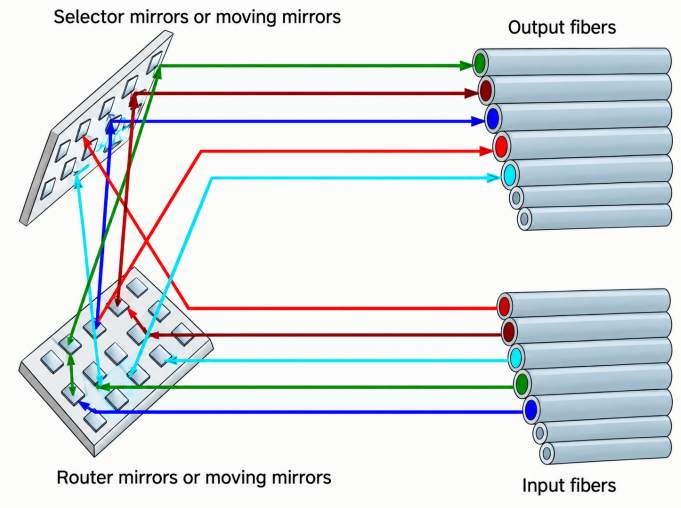


Figure 2.15: Conceptual MEMS-based optical switching matrix.

Figure 2.15 illustrates this principle. The incoming multi-wavelength optical signal is first separated into individual wavelength channels. A selected wavelength can then be directed through the MEMS switching matrix toward the required output fiber. In this way, the OXC provides a reconfigurable optical switching function while preserving the signal in the optical domain.

MEMS-based switching is a particularly intuitive example of how an OXC can be implemented, but it is not the only possible solution. The essential point is that the switching matrix establishes the mapping between input and output optical channels, and that this can be done without requiring the client information carried by the wavelength to be interpreted.

Configuration and Lightpath Provisioning

The switching state of an OXC is not determined autonomously by inspecting the traffic that crosses the device. Instead, it is configured by a higher-level management system.

Optical transport networks are therefore typically supervised by a *Network Management System* (NMS), which provides a centralized view of the network infrastructure and can interact with the individual optical devices. Through the NMS, the operator can configure the network resources and modify the switching state of the OXCs.

When a client requests the establishment of a new optical connection, the NMS receives the request and determines how the corresponding lightpath should be created. In particular, it selects a route through the optical topology and identifies the wavelength resources that must be used along that route.

Once the lightpath has been determined, the NMS configures the OXCs located along the selected path. At each intermediate node, the switching matrix is programmed so that the incoming wavelength channel is connected to the appropriate outgoing fiber. The sequence of these local configurations creates the end-to-end optical path requested by the client.

In the case of a MEMS-based OXC, this configuration ultimately determines the position of the micro-mirrors inside the optical switching matrix. Once the matrix has been configured, the optical signal is physically directed toward the selected output port according to the established lightpath.

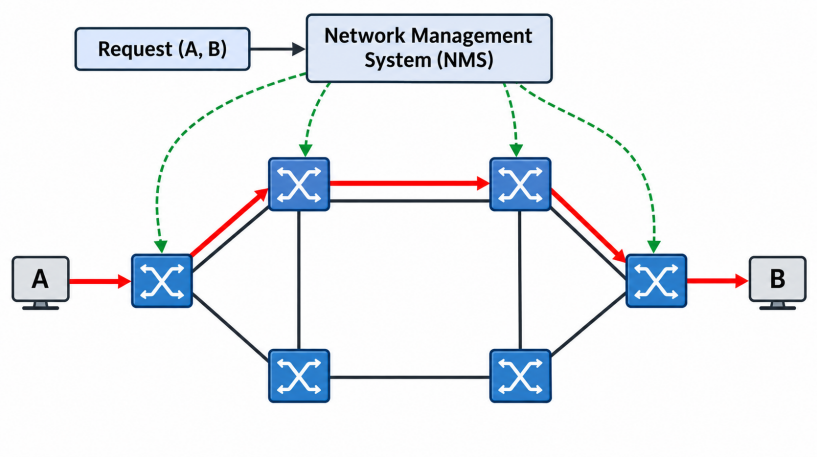


Figure 2.16: Lightpath provisioning through a Network Management System. A client requests an optical connection; the NMS determines the route and configures the switching matrices of the OXCs along the selected path.

Figure 2.16 summarizes this process. The client does not directly configure the optical devices. The connection request is handled by the NMS, which translates the desired

end-to-end connectivity into a set of configurations applied to the OXCs along the path.

If the lightpath has to be modified, for example because a different route is selected, the NMS can compute a new configuration and update the switching state of the affected OXCs. In this way, the optical connectivity can be reconfigured without manually changing the physical fiber connections inside the network.

2.7 Signal Degradation and Regeneration

Keeping a signal in the optical domain avoids unnecessary O/E/O conversions, but it does not mean that the signal can propagate indefinitely without degradation. As an optical signal travels along a lightpath, several physical effects progressively modify the waveform received at the destination.

To understand the problem, consider a simple digital signal in which a binary one is represented by an optical pulse of duration T. The signal generated by the transmitter can be represented as $s(t)$. After propagation over a sufficiently long optical path, the received signal $r(t)$ may differ substantially from the original waveform.

Three different effects are particularly important. First, attenuation reduces the optical power of the signal. Second, noise, interference, and propagation impairments can distort the shape of the pulse. Finally, dispersion can broaden the pulse in time, so that its duration may become larger than the original duration T.

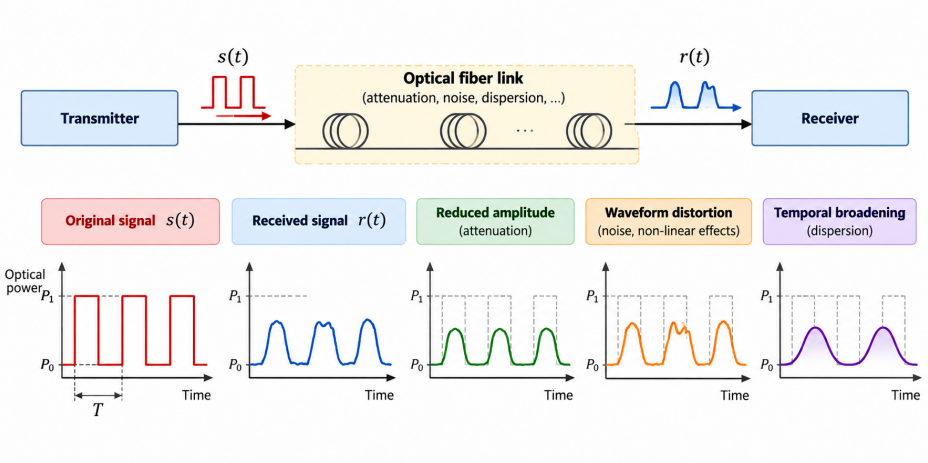


Figure 2.17: Conceptual degradation of a digital optical pulse during propagation. Compared with the original signal $s(t)$, the received signal $r(t)$ may exhibit reduced amplitude, waveform distortion, and temporal broadening.

Figure 2.17 illustrates these effects. The received pulse may therefore be weaker, less regular, and longer than the pulse originally generated by the source.

This degradation eventually affects the ability of the receiver to correctly recover the transmitted binary information. In a simple threshold-based receiver, for example, the received optical level is compared with a decision threshold in order to distinguish between a binary zero and a binary one. If the signal becomes too weak or too distorted, the probability of making an incorrect decision increases.

Regeneration is therefore required whenever the accumulated degradation becomes too large. Depending on how much of the original signal has to be reconstructed, three different levels of regeneration are commonly distinguished: 1R, 2R, and 3R.

2.7.1 1R Regeneration: Reamplification

The simplest form of regeneration is usually referred to as *1R*. Its purpose is essentially to restore the optical power of the signal.

A typical example is an optical amplifier. The incoming signal is amplified directly in the optical domain, increasing its power without recovering the digital information carried by the optical waveform.

If the degraded input signal is denoted by $i(t)$, the output $o(t)$ has a larger amplitude, but its shape and temporal characteristics remain essentially those of the received signal.

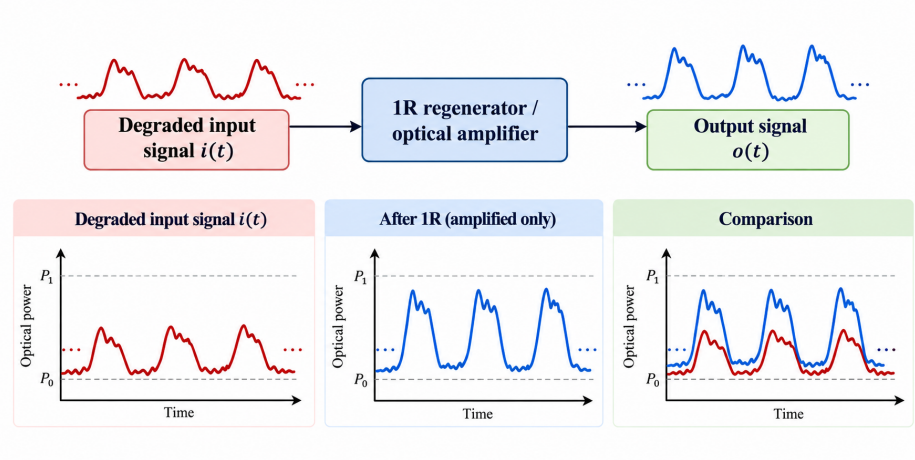


Figure 2.18: Conceptual 1R regeneration. Optical amplification restores the signal power, but does not reconstruct the original pulse shape or timing.

The main advantage of 1R regeneration is that it can be performed entirely in the optical domain. The amplifier does not need to understand the information carried by the signal and can therefore operate on a broad range of signals, including analog signals.

Its limitation is that amplification corrects only the loss of optical power. Any distortion already accumulated by the waveform is amplified together with the signal. Similarly, temporal broadening is not corrected. For sufficiently long lightpaths, restoring power alone is therefore no longer sufficient.

2.7.2 2R Regeneration: Reamplification and Reshaping

A more complete form of regeneration is obtained with *2R*, which performs both *reamplification* and *reshaping*.

The basic idea is that the regenerator knows the expected shape of the digital signal. If the transmitted information is represented by square optical pulses, for example, a distorted received pulse can be interpreted and a new pulse having the expected shape and amplitude can be generated.

Compared with 1R, the regenerated signal is therefore a much better approximation of the waveform originally generated by the source. However, this additional functionality has an important consequence: the regenerator must access and interpret the signal in the electrical domain. An O/E/O conversion is therefore required.

This reduces transparency. A purely optical amplifier can operate without knowing the format of the transported signal, whereas a 2R regenerator must make assumptions about the waveform that has to be reconstructed. A device designed to reshape one particular signal format cannot necessarily regenerate a completely different one.

At the same time, 2R does not explicitly recover the timing reference of the original signal. It can therefore remain transparent to the bit rate: the regenerator reconstructs

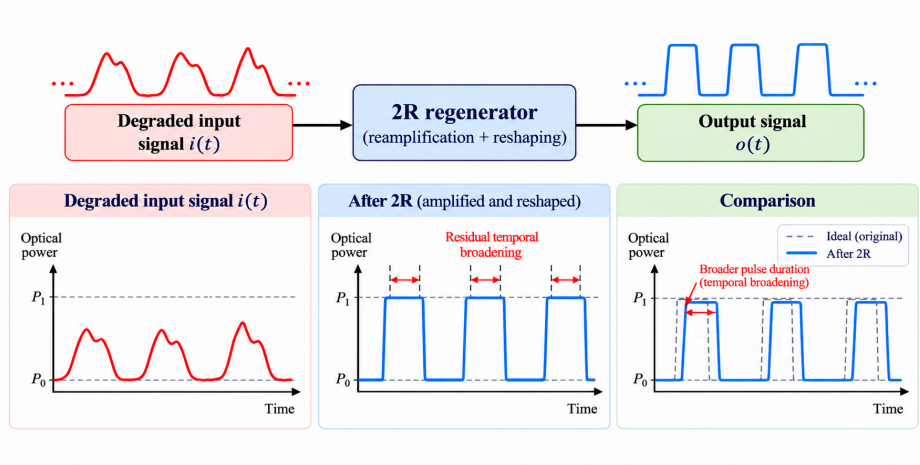


Figure 2.19: Conceptual 2R regeneration. The degraded signal is amplified and reshaped toward the expected digital waveform, while timing errors are not explicitly corrected.

the waveform shape without necessarily requiring knowledge of the exact duration of each transmitted bit.

The drawback is that timing errors are not removed. If several 2R regeneration stages are crossed along a path, temporal deviations (known as jitter) can progressively accumulate.

2.7.3 3R Regeneration: Reamplification, Reshaping, and Retiming

The most complete form is *3R regeneration*, which combines *reamplification*, *reshaping*, and *retiming*.

In addition to restoring the amplitude and shape of the pulse, the regenerator recovers the timing reference of the transmitted signal. It therefore knows when each bit interval should begin and end and can generate a new pulse aligned with the correct timing position.

Retiming is particularly important because propagation can introduce temporal misalignment between the expected and received pulses. This timing variation is commonly referred to as *jitter*. If the pulse moves sufficiently far from its expected temporal position, the receiver may sample the signal at the wrong instant and introduce bit errors.

A 3R regenerator removes this accumulated timing uncertainty by resynchronizing the regenerated signal. The output can therefore be regarded as a "fresh" copy of the original digital signal: power, waveform shape, and timing have all been reconstructed.

The price paid for this higher regeneration quality is a further reduction in transparency. Retiming requires knowledge of the bit rate in order to recover the clock and determine the duration of the transmitted symbols. Consequently, a 3R regenerator is no longer transparent to arbitrary bit rates and framing formats.

2.7.4 Regeneration and Transparency

The three regeneration levels therefore represent progressively deeper access to the transported signal. There is therefore a fundamental trade-off between *regeneration quality* and *transparency*. A 1R stage performs only a physical operation on the optical signal and can therefore support a very broad range of client signals. A 2R stage reconstructs the expected waveform and consequently imposes some constraints on the signal format. A 3R stage additionally recovers the timing information and therefore depends also on the bit rate and framing characteristics of the client signal.

This explains why an optical transport network cannot always remain completely optical from source to destination. Whenever the accumulated signal degradation requires deeper

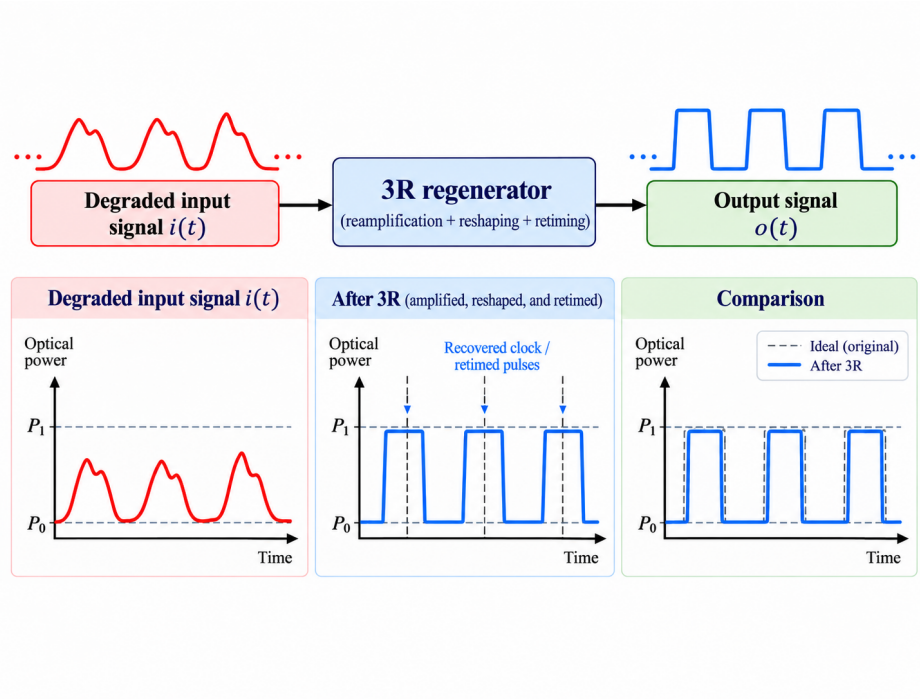


Figure 2.20: Conceptual 3R regeneration. In addition to restoring power and shape, the signal is retimed according to a recovered clock reference, producing a fresh digital copy of the transmitted signal.

regeneration, the network may need to temporarily enter the electrical domain. The resulting infrastructure can therefore be viewed as a sequence of transparent optical regions separated, when necessary, by electronic regeneration points.

2.8 Evolution of Optical Transmission Systems

The concepts introduced so far can also be used to understand how optical transmission systems evolved over time. Improvements in optical fibers, light sources, transmission windows, amplification, and regeneration progressively increased the distance over which optical signals could be transported and eventually made wavelength-division multiplexing practical.

Figure 2.21 summarizes this evolution through four representative technological stages. The first three configurations belong to the evolution of first-generation optical networks, where optics was mainly used for transmission while multiplexing and most processing functions remained in the electrical domain. The final configuration represents the transition to second-generation optical networks, where multiple wavelength channels can be transported simultaneously and remain in the optical domain over large portions of the network.

2.8.1 Early Systems: Multimode Fiber and LED Sources

The earliest configuration in Figure 2.21 consists of a transmitter and a receiver connected through an optical fiber, with several regeneration stages placed along the path. The large number of regenerators immediately reveals an important limitation of the technology available at that time: the optical signal degraded significantly over relatively short distances.

One of the main causes was the use of *multimode fiber*. An optical fiber consists of a central *core*, surrounded by a *cladding*. In a multimode fiber, several spatial propagation modes can coexist. From an intuitive geometrical perspective, the different contributions

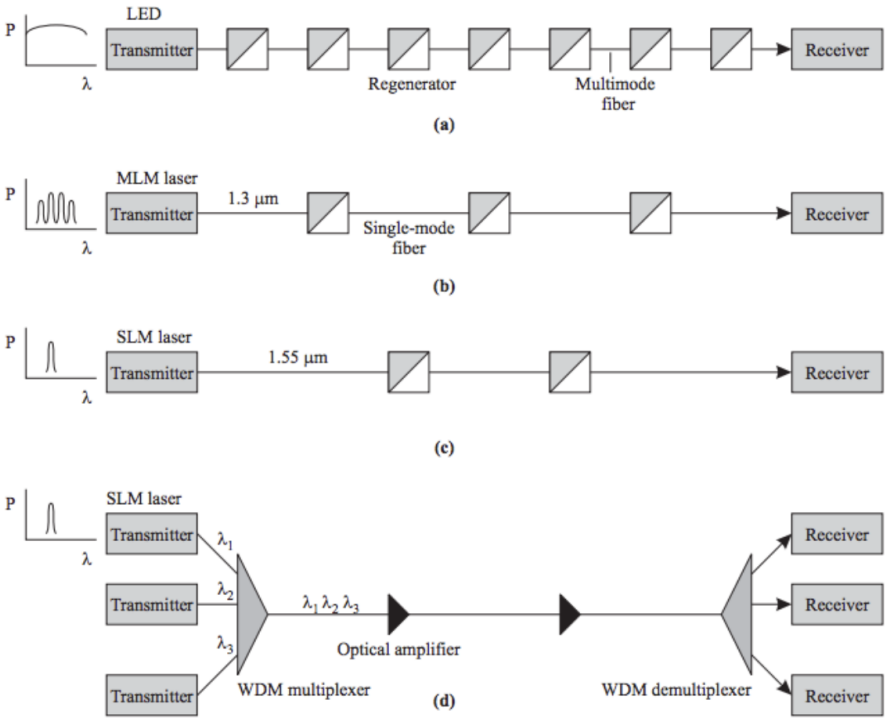


Figure 2.21: Evolution of optical transmission systems.

associated with the same transmitted pulse may therefore follow different paths inside the fiber.

Some paths are shorter, whereas others are longer. As a consequence, different contributions of the same transmitted pulse can reach the receiver at different times. The received pulse becomes wider than the pulse generated by the transmitter. This phenomenon is referred to as *modal dispersion*.

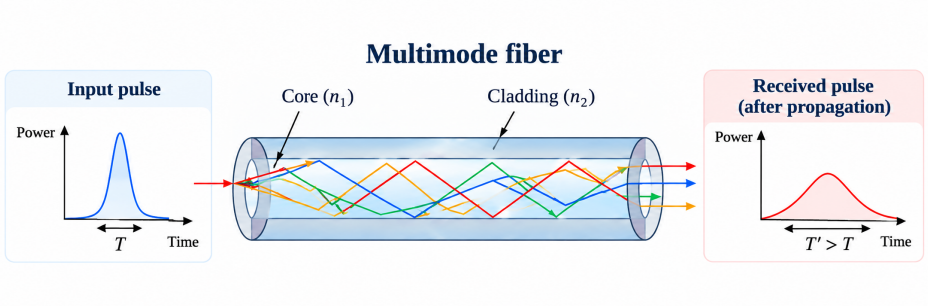


Figure 2.22: Conceptual illustration of modal dispersion in a multimode fiber. Different spatial propagation modes associated with the same transmitted pulse travel along different paths and reach the receiver at different times, producing temporal broadening.

Figure 2.22 provides a physical interpretation of the pulse broadening discussed in the previous section. Even if the transmitter generates a relatively short pulse, the energy associated with that pulse is spread over a longer time interval at the receiver.

The optical source represented another important limitation. Early systems used light-emitting diodes (LEDs), whose transmitted optical power was distributed over a relatively broad range of wavelengths. The source was therefore not sufficiently selective to generate a large number of narrow and well-separated optical carriers.

For this reason, wavelength-division multiplexing was not yet an effective solution for

sharing the transmission system among many users. First-generation optical systems relied mainly on *Time Division Multiplexing* (TDM), where different client signals were assigned different time intervals.

2.8.2 Single-Mode Fiber and Laser Sources

A major technological improvement was the introduction of *single-mode fiber*. Compared with multimode fiber, the core is much smaller and the fiber is designed so that only the fundamental spatial mode propagates.

The consequence is important: the different spatial propagation paths responsible for modal dispersion are removed. The pulse no longer suffers from the temporal spreading caused by different spatial modes, and substantially longer distances can be covered before regeneration becomes necessary.

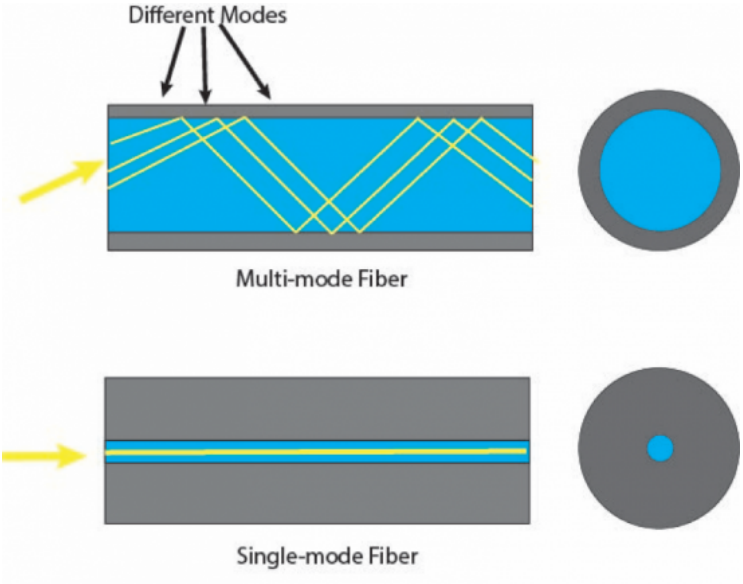


Figure 2.23: Conceptual comparison between multimode and single-mode fiber. A multimode fiber supports several spatial propagation modes, while a single-mode fiber confines propagation to the fundamental mode, thereby eliminating modal dispersion.

This improvement can also be recognized in Figure 2.21: the number of regeneration stages is considerably smaller than in the previous configuration. The signal can travel a longer distance before its quality becomes insufficient.

At the same time, the optical source evolves from an LED to a laser. The emitted spectrum becomes narrower, but the laser used at this intermediate stage can still contain several longitudinal spectral components. The transmitted optical signal therefore still occupies a finite range of wavelengths.

This introduces another form of pulse spreading. Even though all the optical energy now propagates in the same spatial mode, the different wavelength components do not necessarily propagate through the fiber at exactly the same velocity. They can consequently arrive at the receiver at slightly different times. This effect is called *chromatic dispersion*.

Figures 2.22 and 2.24 therefore illustrate two different physical mechanisms that can produce a similar observable effect: temporal broadening of the received pulse. Modal dispersion is caused by different spatial propagation modes, whereas chromatic dispersion is caused by different wavelength components of the optical signal.

The transition to single-mode fiber removes the first problem, but does not automatically remove the second one.

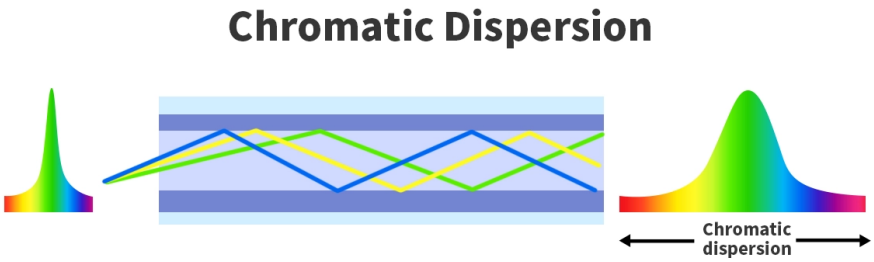


Figure 2.24: Conceptual illustration of chromatic dispersion. Different wavelength components of the same optical pulse propagate in the same spatial mode but experience different propagation delays, causing the received pulse to broaden in time.

2.8.3 Narrower Optical Spectra and the 1550 nm Window

A further technological step concerns both the optical source and the transmission window. Earlier systems operated around 1.3 μm. Later systems moved toward the region around 1.55 μm, where optical fiber provides lower attenuation and therefore allows signals to travel over longer distances.

At the same time, laser technology continued to improve. The emitted optical power could be concentrated over a much narrower spectral region. A narrower spectrum reduces the range of wavelength components that experience different propagation delays and therefore limits the temporal broadening caused by chromatic dispersion.

The combined effect of lower attenuation, single-mode propagation, and a narrower optical spectrum further increases the distance between consecutive regeneration points. This progression is visible in the third configuration of Figure 2.21.

2.8.4 From TDM to WDM: Second-Generation Optical Networks

The final configuration in Figure 2.21 marks the transition to second-generation optical networks. At this point, the technological ingredients required for effective wavelength-division multiplexing are available.

Different transmitters can generate narrow optical carriers centered at different wavelengths,

$$\lambda_1, \lambda_2, \ldots, \lambda_N.$$

A WDM multiplexer combines these channels onto the same optical fiber. The channels propagate simultaneously through the optical infrastructure and are separated again by a WDM demultiplexer at the receiving side.

The transition from TDM to WDM is therefore the result of several technological advances rather than an isolated change in the multiplexing technique. It requires suitable single-mode fibers, sufficiently selective laser sources, favorable transmission windows, and optical amplification.

Optical amplifiers are particularly important because they can restore the power of the complete WDM signal without requiring the individual wavelength channels to be converted into the electrical domain. Multiple optical channels can consequently remain in the optical domain over long portions of the transmission path.

This represents the fundamental architectural transition between the two generations of optical networks. In first-generation systems, optical fiber primarily provides transmission capacity, while multiplexing and intelligent processing functions are largely electronic. In

second-generation systems, wavelength channels themselves become network resources that can be multiplexed, amplified, switched, and routed in the optical domain.

The resulting network can contain long transparent optical regions separated, when necessary, by points at which electronic processing is performed for functions such as regeneration, wavelength conversion, or client adaptation.

This technological evolution also creates a new operational problem. Once a large number of lightpaths and optical resources can be established across the network, mechanisms are needed to configure them, monitor their performance, detect failures, and coordinate the optical devices. This provides the transition toward the control and management functions discussed next.

2.9 Management and Control of Optical Networks

The previous sections have mainly described how optical signals are generated, multiplexed, switched, amplified, regenerated, and transported. Operating a real optical network, however, requires additional functions able to configure the infrastructure, establish and remove lightpaths, monitor the quality of the provided services, and react to failures.

A useful conceptual distinction is that among the *management plane*, the *control plane*, and the *data plane*.

The *data plane* is the physical infrastructure that actually transports the traffic. In an optical network, it includes devices such as OLTs, OADMs, ROADMs, optical amplifiers, OXCs, and the fibers interconnecting them.

The *control plane* contains the logic used to determine how network resources should be used. It includes functions such as maintaining information about the network topology and available resources, computing suitable paths, and coordinating the establishment of connections.

The *management plane* is responsible for operating and configuring the network. Although an individual device could be configured manually, this approach is clearly unsuitable for a carrier network composed of many optical elements. Management operations are therefore typically coordinated through a *Network Management System* (NMS).

The NMS can be regarded as the main point through which the operator interacts with the network. It maintains a network-wide view of the infrastructure and coordinates the software functions required to operate it. From a high-level perspective, several important tasks can be associated with the NMS.

Client and service interface. The NMS can expose an interface between the optical transport network and its clients. A client layer can request the establishment of an optical connection between two endpoints and may also specify service requirements such as bandwidth, target Bit Error Rate (BER), protection requirements, maximum delay, or jitter constraints.

In the centralized model considered here, the request is therefore directed to the NMS. This should not necessarily be interpreted as a human user directly interacting with the management platform: the client can be another network layer or a customer-facing software system that requests a connectivity service from the optical infrastructure.

Topology and resource information. In order to establish new connections, the management and control system must have information about the optical topology and about the resources currently available in the network. This includes, for example, the optical nodes and links that can be traversed and the wavelength resources that can be assigned to new lightpaths.

Lightpath provisioning. When a new connection is requested, the network must determine how the corresponding lightpath can be established. A suitable route has to be selected and the required optical resources must be identified along that route.

In a centralized architecture, the algorithms supporting these decisions can be executed under the coordination of the NMS. Once a suitable lightpath has been selected, the corresponding configuration has to be installed in the optical devices along the path.

For example, an OXC may have to be configured so that a wavelength arriving on a particular input port is redirected toward a selected output port. The switching matrix is therefore configured before the client traffic is transported. Once all the required nodes have been configured, their local switching states collectively create the end-to-end lightpath.

Device configuration and Element Management Systems. The NMS does not necessarily communicate directly with every optical device. Optical equipment is often vendor-specific, and different vendors may provide their own software platforms and interfaces for configuring and monitoring their equipment.

An intermediate management layer can therefore be provided by an *Element Management System* (EMS). An EMS manages a set of network elements, typically belonging to a particular vendor or equipment family, and hides many of the device-specific mechanisms required to interact with them.

The NMS can consequently provide a network-wide abstraction above several EMSs. From the operator perspective, the NMS coordinates the overall infrastructure, while the EMSs translate higher-level management operations into the vendor-specific actions required by the underlying network elements.

Performance monitoring. Establishing a lightpath is only the first part of the management process. Once the connection is active, the operator must verify that the requested service quality is actually being provided. Relevant parameters may include BER, received optical power, delay, jitter, and other performance indicators available at suitable monitoring points.

The management system can collect this information from the network elements and use it to detect degradation before the quality of the provided service becomes unacceptable.

Fault management. The NMS also plays an important role when failures occur. Network devices can generate alarms reporting problems such as link failures, loss of signal, or equipment malfunction. The management system must collect and correlate this information, identify the affected resources, and support the actions required to restore the service.

Figure 2.25 summarizes this organization. The optical devices constitute the data plane at the bottom of the architecture. Above them, EMSs can provide device- or vendor-specific management functions, while the NMS provides the operator with a global view of the infrastructure and coordinates network-wide operations.

The same architecture also highlights the interaction between management and control. A client request arrives at the management system, the network state and available resources are considered to determine a suitable lightpath, and the resulting decision is translated into configuration operations on the relevant optical devices.

The exact boundary between management and control functions depends on the network architecture. In a simple centralized model, several control functions, including path computation and connection setup, can be coordinated through the NMS. More advanced architectures may delegate some of these functions to dedicated or distributed control-plane components, particularly when fast connection establishment or restoration is required.

These high-level functions provide a useful roadmap for the following sections. Connection management concerns the establishment, maintenance, and removal of lightpaths. Performance management concerns the observation of the quality of the active connections, while fault and alarm management deal with the detection and handling of failures. Supporting these functions also requires a structured representation of the optical and electronic transport layers, which will be introduced later.

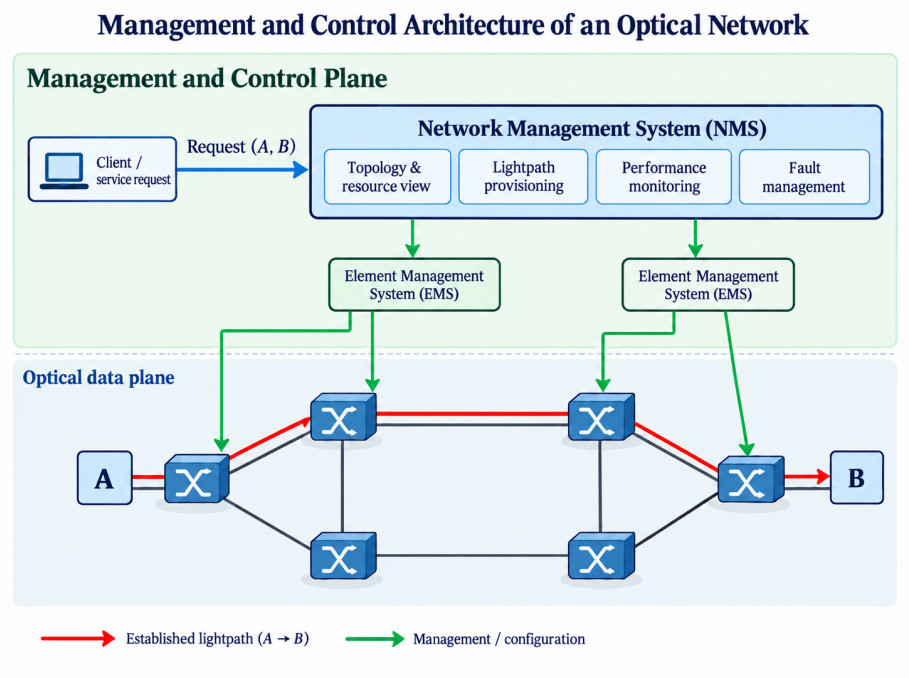


Figure 2.25: Conceptual management and control architecture of an optical network. Client-layer requests are handled at the network-management level. The NMS maintains a network-wide view, coordinates lightpath provisioning and monitoring, and interacts with groups of optical devices either directly or through vendor-specific Element Management Systems.

2.10 Layers of the Optical Transport Network

An optical transport network can be regarded, from the perspective of the traditional protocol stack, as part of the physical infrastructure. However, this does not mean that its internal organization consists of a single undifferentiated layer.

Several different functions have to be performed inside the transport network: client signals must be adapted, encapsulated, converted into optical signals, associated with wavelength channels, multiplexed with other channels, and finally transmitted over physical fibers. It is therefore useful to represent the optical transport infrastructure itself through a layered architecture.

The basic idea is similar to the layered models already familiar from computer networks. At different layers, the same communication is observed at different levels of abstraction. As we move downward through the architecture, the representation becomes progressively closer to the physical infrastructure.

Consider, for example, two routers that have to communicate through the optical transport network. From the client perspective, the information to be transported may simply be an IP packet. The optical network must accept this client information and progressively transform it into a signal that can be carried over the physical fiber.

Figure 2.26 illustrates this principle. At the highest level considered here, the client information is represented in digital form. As we move downward, additional details of the transport infrastructure become visible. Eventually, the representation changes from digital frames in the electrical domain to optical signals propagating through the physical network.

Client adaptation and the OPU. The starting point is the client signal. Different client technologies may provide information in different formats and at different bit rates. Before this information can be transported through OTN, it has to be adapted to the format expected by the transport system.

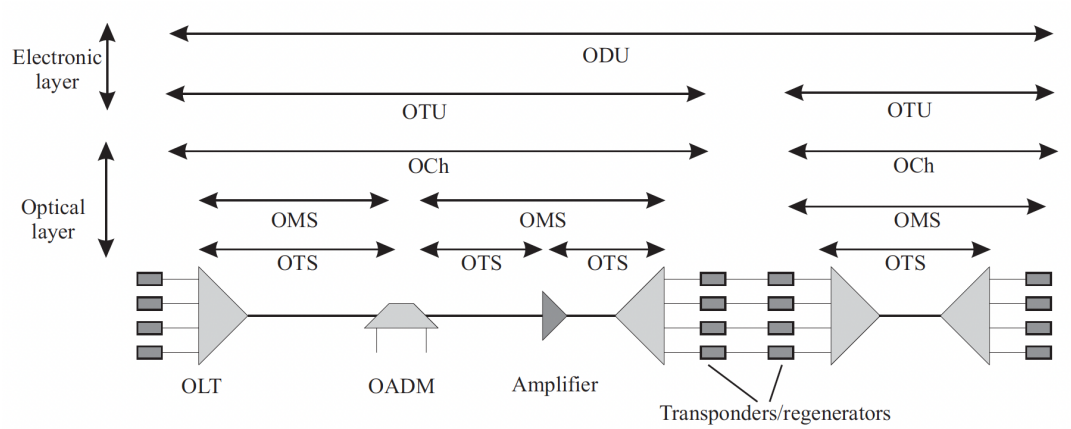


Figure 2.26: Progressive view of a client connection through the OTN layered architecture. At the upper levels the connection is represented as a logical end-to-end service, while lower levels progressively expose optical channels, multiplexed wavelength signals, amplifiers, and physical fiber sections.

The *Optical channel Payload Unit* (OPU) performs this adaptation function. Conceptually, the client information becomes the payload of an OPU, and additional overhead can be introduced to describe how the client signal has been mapped into the OTN structure.

At this level, the network is still operating entirely on a digital representation of the information. The OPU does not correspond to an optical wavelength: it is part of the electronic framing used before the client signal is converted into an optical carrier.

The Optical channel Data Unit. The OPU is then carried inside an *Optical channel Data Unit* (ODU). Additional overhead is introduced around the adapted client information.

The ODU provides a logical representation of the transported client connection. From this perspective, the internal physical organization of the optical infrastructure can still be largely hidden. Two endpoints can be viewed as connected through an end-to-end transport service even though the actual path may cross several optical subnetworks and intermediate devices.

This is analogous to what happens in other layered architectures: a high-level logical connection can hide a much more complex lower-level infrastructure.

The Optical channel Transport Unit. The ODU is further encapsulated into an *Optical channel Transport Unit* (OTU). The OTU introduces additional overhead and Forward Error Correction (FEC) information.

This layer is particularly important from the perspective of transmission monitoring. Since the signal is still represented digitally, bit errors are well-defined and can be detected through the information carried in the OTU frame.

An important difference between ODU and OTU concerns the portion of the network that they represent. The ODU can describe a higher-level end-to-end client connection, whereas an OTU is associated with a transmission section bounded by points where the signal becomes available again in the electrical domain.

For example, a long ODU connection may traverse multiple all-optical subnetworks. At the boundary between two such regions, the signal can be converted back into the electrical domain, regenerated, checked, and then transmitted again through a new optical region. A new OTU section can therefore begin at that point.

This distinction is useful for performance monitoring. Measuring the BER only at the end of a very long connection would reveal that the connection has degraded, but would provide little information about which portion of the network caused the problem. Monitoring at OTU boundaries provides a finer localization of transmission impairments.

From the electrical domain to the optical domain. Up to this point, the information is represented through digital frames and binary fields. Moving below the OTU layer introduces a fundamental transition: the client information is converted from the electrical domain into an optical signal.

The transponder performs this conversion and generates the optical carrier used inside the transport network. From this point onward, it is no longer useful to think of encapsulation as simply adding another binary header. The transported entity is now a physical optical signal.

This distinction is fundamental. In the electrical domain, information can be organized into frames and headers whose individual bits can be inspected. In the optical domain, instead, the network operates directly on properties of the light signal, such as its wavelength and physical propagation path.

The Optical Channel layer. The first layer in the optical domain is the *Optical Channel* (OCh) layer.

At this level, each client optical connection can still be distinguished individually because it is associated with a particular wavelength channel. For example, two client signals transported on λ_1 and λ_2 appear as two distinct optical channels.

The OCh layer therefore provides the natural abstraction for considering the end-to-end routing of individual lightpaths through the optical network.

It is important to distinguish the client information from the optical channel that transports it. At the OCh layer, the network does not inspect the client packets carried by the wavelength. It distinguishes the optical channels themselves.

The Optical Multiplex Section. Moving one level further down exposes the fact that several optical channels are normally carried together over the same fiber.

The *Optical Multiplex Section* (OMS) represents a section in which multiple wavelength channels have been combined by a WDM multiplexer. At this level, the physical signal propagating through the fiber is a multiplexed optical signal containing several wavelength contributions.

This introduces an important change of perspective. At the OCh layer, λ_1, λ_2, and λ_3 can be considered as three distinct optical channels associated with different client connections. At the OMS layer, the same wavelengths are observed as components of one multiplexed WDM signal.

$$\underbrace{\lambda_1, \lambda_2, \lambda_3}_{\text{individual OCh channels}} \quad \longrightarrow \quad \underbrace{\{\lambda_1, \lambda_2, \lambda_3\}}_{\text{one OMS}}$$

This is conceptually similar to the effect of moving downward in other layered architectures: information that can be distinguished at a higher layer can appear aggregated when observed at a lower layer.

The Optical Transmission Section. The lowest optical layer considered here is the *Optical Transmission Section* (OTS).

At this level, the representation focuses on the physical transmission infrastructure itself. Individual fiber sections and optical amplification stages become visible. An OTS therefore represents the portion of transmission infrastructure between consecutive optical termination or amplification points.

The progression from OCh to OMS and finally to OTS can therefore be interpreted as an increasing visibility of the physical infrastructure. OCh focuses on individual wavelength channels, OMS exposes the multiplexed WDM signal, and OTS exposes the physical transmission sections over which that signal propagates.

The key point of the presented architecture is that each layer provides a different abstraction of the same communication. At the upper layers, the network retains detailed information about individual client connections but hides much of the physical infrastructure.

Moving downward progressively exposes multiplexers, optical amplifiers, and individual fiber sections. This is particularly useful for network management: a high-level layer is convenient for managing client services, while a lower layer provides finer visibility when a specific physical component has to be monitored or a failure has to be localized.

The complete transformation can therefore be summarized as

$$\text{Client} \rightarrow \text{OPU} \rightarrow \text{ODU} \rightarrow \text{OTU} \rightarrow \text{OCh} \rightarrow \text{OMS} \rightarrow \text{OTS}.$$

However, this sequence should not be interpreted simply as a collection of protocol headers. OPU, ODU, and OTU belong to the electronic representation of the transported information and can therefore introduce digital overhead. OCh, OMS, and OTS instead describe progressively lower abstractions of the optical signal and of the physical transport infrastructure.

This distinction also explains why functions such as BER monitoring naturally belong at points where the signal is available in the electrical domain. Within a transparent optical region, the network manipulates light directly and does not have access to the individual binary values carried by the client signal.

The layered architecture thus serves two complementary purposes. It describes how client information is progressively transformed into an optical signal, and it provides different levels of visibility for operating and managing the network. These abstractions will become particularly important when considering performance monitoring, fault detection, and service restoration.

2.10.1 Performance and Fault Management

Establishing a lightpath is only the first step in providing an optical transport service. Once a connection is active, the network operator must ensure that the service continues to satisfy the requirements agreed with the client.

A customer may, for example, require guarantees in terms of delay, jitter, throughput, or Bit Error Rate (BER). Providing such guarantees therefore requires the operator not only to configure the connection, but also to continuously observe its behavior and take corrective actions whenever its quality becomes unacceptable.

This is the role of *performance management*. Its objective is to monitor the relevant performance parameters of active connections and support the actions required to maintain the expected quality of service.

Fault management addresses a related but different problem. Its purpose is to detect failures in the network, report them to the management system, identify the component responsible for the problem, and support service restoration.

The distinction is useful. A connection may still be operational but exhibit progressively degrading performance, for example because the optical signal is becoming too weak or distorted. In other situations, a physical failure such as a fiber cut may completely interrupt one or more lightpaths.

Bit Error Rate and performance monitoring. One of the most important performance parameters associated with a digital lightpath is the *Bit Error Rate* (BER). An increase in BER indicates that the received digital signal is being progressively corrupted and can therefore be used as an indication that the quality of the transmission path is degrading.

However, the layered architecture introduced in the previous section reveals an important limitation. BER is meaningful only where the transported information is available as a digital electrical signal. Inside a transparent optical region, the network handles an analog optical waveform rather than a sequence of binary values. It is therefore not meaningful

to associate a specific number of bit errors with an optical amplifier, an OADM, or a transparent optical switch.

BER monitoring is consequently performed at points where the optical signal is converted back into the electrical domain, typically at transponders or regenerators.

The OTU layer is particularly suitable for this purpose. The OTU frame includes Forward Error Correction (FEC) information that can be used to detect transmission errors and assess the quality of the received digital signal.

This also explains why monitoring at the boundaries of all-optical subnetworks is useful. Consider an end-to-end connection traversing several transparent optical regions. Measuring the BER only at the final destination would reveal that the connection has degraded, but would provide little information about which portion of the network introduced the impairment.

By checking the signal whenever it becomes available again in the electrical domain, the operator can instead associate the degradation with a smaller portion of the infrastructure.

Optical trace. Performance and fault management also require the management system to know which connection or physical resource is being observed.

A lightpath may cross several nodes and several internal components before reaching its destination. It is therefore useful to associate a unique identifier with the connection. This identifier is referred to as an *optical path trace*.

The trace allows the management system to identify a particular lightpath, verify its connectivity, and associate monitoring information with the correct connection.

The concept of a trace is not necessarily limited to the complete lightpath. Identifiers can also be associated with lower-level resources, such as individual optical transmission sections. This becomes particularly important when the management system must determine exactly which portion of the infrastructure is affected by a problem.

Fault Management: Detection and Restoration. Fault management is concerned with maintaining service continuity when a failure affects the optical infrastructure. At a high level, the process can be divided into two main phases: *failure detection* and *service restoration*.

The first phase consists of detecting that an abnormal condition has occurred and determining which network component is responsible for it. This task is particularly important because the same physical failure may be observed by several elements of the network and may consequently generate a large number of alarms.

Once the failed component has been identified, the second phase is *restoration*. At this point, the problem becomes primarily a control-plane task. The control plane operates on the updated network state, computes an alternative path that avoids the failed resource, and coordinates the reconfiguration required to move the affected connection onto the new path.

Restoration can therefore be viewed as a detour around the failed component. The specific mechanisms used to compute and install the alternative path depend on the control architecture adopted by the network, but the general objective is always the same: re-establish connectivity while excluding the unavailable resource.

The two phases are strictly related. Restoration can only be performed correctly once the network has detected the failure and has obtained sufficiently accurate information about its location.

Failure Detection and Alarm Generation. A network element that detects an abnormal condition reports the problem by generating an *alarm* toward the network management system. Depending on its position and function, an element may observe the

failure in different ways. A device adjacent to a broken fiber may directly detect loss of the optical signal, while a transponder farther away may observe that an active lightpath is no longer available or that its signal quality has severely degraded.

The fundamental problem is that these observations may all originate from the same physical failure. Consequently, a single fault can cause many different network elements to report alarms to the NMS.

Consider the example in Figure 2.27. A fiber carries 32 lightpaths, each of which traverses additional intermediate network elements. If that fiber is cut, the device closest to the failed link detects the physical failure, but the interruption is also observed at several other points along the affected lightpaths. In the reference example, a single physical failure can result in a total of 129 alarms.

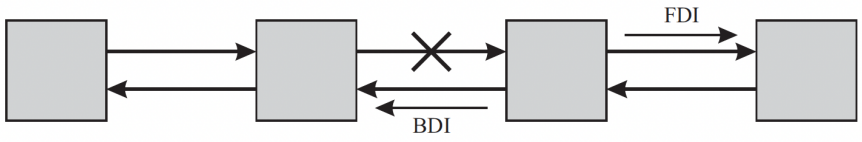


Figure 2.27: Example of alarm proliferation caused by a single fiber failure.

The reception of many alarms therefore does not necessarily indicate the presence of many independent failures. The NMS must determine which alarm corresponds to the *root cause* and which alarms are only consequences of that failure.

This distinction is essential for failure localization. Knowing that many lightpaths are down is useful to understand the impact of the event, but it is not sufficient to determine which physical resource must be repaired or avoided during restoration.

Alarm Suppression and Defect Indication. To reduce the number of redundant alarms and help the NMS identify the actual origin of a failure, optical networks provide an *alarm suppression* mechanism.

The basic idea is that the network element closest to the failed component detects the abnormal condition and reports the corresponding alarm toward the NMS. At the same time, information about the already detected failure is propagated through the network so that other elements do not interpret the consequences of the same event as independent faults.

Two defect indications are used for this purpose. The *Forward Defect Indicator* (FDI) propagates in the downstream direction with respect to the failure, whereas the *Backward Defect Indicator* (BDI) propagates in the opposite direction.

An important aspect of this mechanism is that defect information does not necessarily propagate end to end at a single OTN layer. Different network elements observe the connection at different levels of abstraction, and a defect indication must therefore be propagated at the layer that is relevant to the elements that have to be notified.

Consider the example in Figure 2.28. The failure occurs inside one all-optical subnetwork. The element immediately adjacent to the failed physical section can directly observe that the optical transmission has been interrupted. At this point, the failure is naturally associated with the lowest optical level, the *Optical Transmission Section* (OTS).

The failure is physically located in a specific *Optical Transmission Section* (OTS). The other OTS sections of the same optical subnetwork are not the faulty resources and should therefore not be interpreted as independent failures.

At the *Optical Multiplex Section* (OMS) level, however, the situation is different. An OMS represents the transport of the complete multiplexed optical signal between its termination points. If one of the OTS sections composing that OMS fails, the multiplexed signal can no longer be correctly delivered across the whole section.

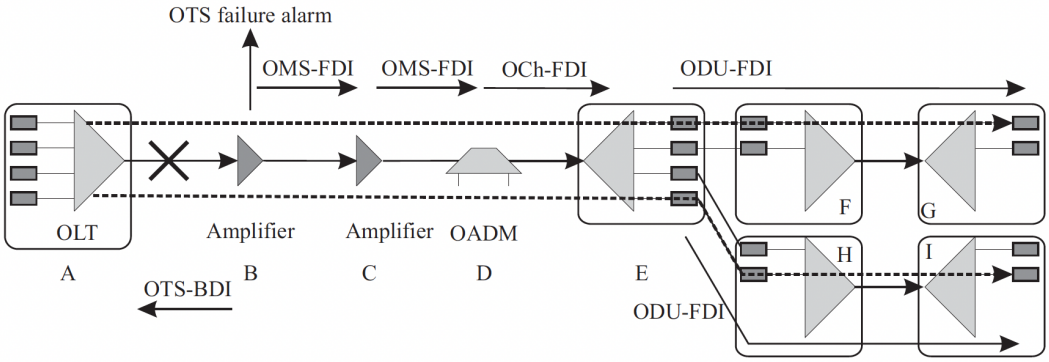


Figure 2.28: Propagation of defect indications across different OTN layers. Close to the physical failure, defect information is associated with the optical transmission and multiplex sections. As the indication reaches higher-level termination points, it can be propagated at the OCh level and, after returning to the electrical domain, through ODU-level information. Only the elements that may otherwise generate secondary alarms need to be notified.

As a consequence, the devices terminating the OMS observe that the multiplexed signal is no longer available and may independently raise alarms. These alarms do not identify new failures: they are secondary effects of the original OTS failure.

An OMS-level defect indication is therefore propagated toward these devices to inform them that the problem has already been detected at a lower layer. Upon receiving the defect indication, they can suppress their redundant alarms instead of reporting the same physical failure again to the NMS.

At the wavelength level, the consequences of the failure may also be observed by the termination points of individual lightpaths. A transponder, for example, may detect that the optical channel that was previously active is no longer available. The defect information can therefore also appear at the *Optical Channel* (OCh) level so that such channel-level alarms can be suppressed.

This progression is important because the same physical failure is being observed through different abstractions of the transport network:

$$\text{OTS} \longrightarrow \text{OMS} \longrightarrow \text{OCh}.$$

At each level, the purpose is the same: inform the elements that would otherwise raise an additional alarm that the anomaly they observe is a consequence of an already identified failure.

Eventually, the defect indication reaches a boundary of the all-optical subnetwork, where the signal is converted back into the electrical domain. From this point onward, there is no reason to propagate the failure through every optical element of another all-optical subnetwork if those elements are not affected by the original fault.

Instead, the information can be carried at the higher electronic layer toward those termination points that may still observe the failure and generate an alarm. In the OTN hierarchy, this means that defect information can continue at the *ODU* level.

The overall propagation can therefore be interpreted conceptually as

$$\text{physical failure} \rightarrow \text{OTS} \rightarrow \text{OMS} \rightarrow \text{OCh} \rightarrow \text{ODU},$$

although not every failure necessarily requires all these stages. The relevant level depends on which network elements can observe the consequences of the failure and therefore need to suppress their alarms.

The same reasoning applies in the opposite direction through the BDI. The network does not blindly distribute the failure indication to every component. Instead, defect information is propagated backward and forward only as far as necessary to reach the elements that would otherwise interpret the same root cause as an independent fault.

Once the defect has been localized and the redundant alarms have been suppressed, the NMS has a much clearer indication of the root cause. The detection phase can then be considered complete, and the control plane can start the restoration procedure by computing and establishing a path that avoids the failed resource.

2.11 Optical Layer Overhead

The discussion of fault management raises an important question. We have seen that defect indications may have to propagate through an all-optical subnetwork in order to suppress redundant alarms. At the same time, however, we have repeatedly emphasized that devices operating in the optical domain do not directly process binary frames and cannot simply add a conventional header containing management information.

Some form of additional information is nevertheless required. The network must be able to identify optical resources, propagate defect information, and monitor the quality of the transported signals. These functions are supported through different forms of *overhead*.

The form taken by this overhead depends on the domain in which the signal is being processed. When the signal is available in the electrical domain, management information can be inserted directly into digital OTN frames. Inside an all-optical subnetwork, instead, conventional binary headers cannot be used and the overhead must itself be represented in optical form.

Two mechanisms are particularly important in the optical domain: the *Optical Supervisory Channel* (OSC) and the *pilot tone*.

Optical Supervisory Channel. From a physical point of view, the OSC is an additional optical wavelength dedicated to supervision and management. It does not carry client traffic. A dedicated transponder generates this monitoring wavelength at the beginning of a fiber section, and the OSC is multiplexed together with the wavelengths carrying the client signals.

At the opposite end of the fiber, the OSC wavelength is extracted from the multiplexed optical signal and processed by the corresponding supervisory equipment. If the optical path continues through another fiber section, a new OSC is generated for that section.

The OSC therefore operates on a section-by-section basis:

$$\text{OSC insertion} \rightarrow \text{fiber section} \rightarrow \text{OSC extraction}.$$

This organization is particularly important because it provides a monitoring and signaling channel even while the client signals remain entirely in the optical domain. No conversion of the client wavelengths into electrical form is required simply to exchange management information.

At the lowest optical layer, the OSC can be associated with an *Optical Transmission Section* (OTS). Since an OTS corresponds to a physical transmission section between consecutive optical termination or amplification points, the OSC can be used to supervise that portion of the infrastructure and to carry information identifying the section.

For example, a unique *trace* can be associated with the OTS and carried through the OSC. The receiving equipment can then verify that the expected optical section is actually connected and can associate alarms or management information with the correct physical resource.

The same supervisory mechanism can also support management functions at the *Optical Multiplex Section* (OMS) level. An OMS contains the complete WDM signal composed of

several wavelength channels and may span multiple lower-level OTS sections. Although the OSC is physically inserted and terminated section by section, the information transported through successive OSC channels can be used to supervise the OMS and to propagate defect information associated with the multiplexed optical signal.

Hence, the OSC provides an optical-domain management channel that supports supervision at both the OTS and OMS levels. The important distinction is that the OSC is associated with the shared optical infrastructure and with the multiplexed signal, rather than with one particular client wavelength.

This last point also explains its limitation. Since the OSC is not associated with an individual client channel, it cannot provide fine-grained monitoring of the optical power experienced by a specific wavelength. For that purpose, a different mechanism is required: the *pilot tone*.

Pilot tone. Different monitoring requirements arise when management information must remain associated with one specific optical channel.

Consider the signal generated by a transponder. The client information is converted into an optical signal centered around a particular wavelength λ_s. The transmitted signal occupies a finite spectral region around that wavelength.

A small, low-power optical signal can be introduced together with the client channel. This additional signal is referred to as a *pilot tone*.

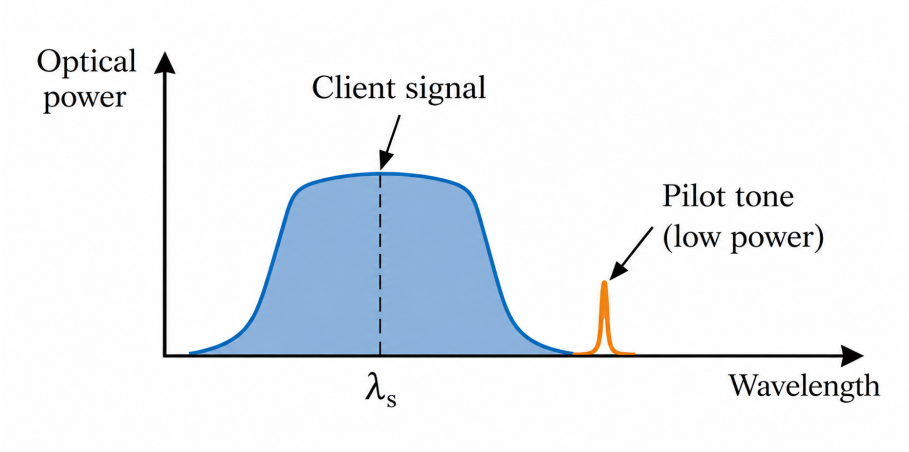


Figure 2.29: Conceptual representation of a pilot tone associated with a client optical channel. The transponder generates the client signal around the assigned wavelength, while a much lower-power optical component is introduced for monitoring purposes.

The important difference with respect to the OSC is that the pilot tone is associated with a particular client wavelength. Each optical channel can therefore have its own pilot tone, which travels through the all-optical subnetwork together with the corresponding client signal.

This property makes the pilot tone suitable for client-specific monitoring. For example, its received power can be compared with the transmitted value in order to estimate the attenuation experienced by the associated optical channel. Similarly, the pilot tone can provide a trace associated with the individual optical channel or lightpath.

The same property, however, also introduces an important limitation. Suppose that a fiber carrying the client signal is completely cut. Since the pilot tone travels together with that signal, it is also lost at the failure point. It cannot therefore be used to propagate a defect indication beyond the failed fiber.

This explains why defect indication and client-channel monitoring require different mechanisms. The OSC can support the propagation of defect information associated with

the optical infrastructure, whereas the pilot tone is more appropriate for monitoring an individual wavelength channel.

Electronic OTN overhead. Optical overhead is required while the signal remains inside a transparent optical region. At the boundaries of such a region, however, a transponder or regenerator converts the signal back into the electrical domain. Conventional digital overhead can then be used again.

The ODU and OTU structures introduced previously provide such overhead. Because their information is represented digitally, they can carry connection identifiers and defect indications associated with higher-level transport services.

The OTU is also the natural location for BER monitoring. Its Forward Error Correction information allows the receiver to evaluate errors in the digital signal and therefore estimate the quality of the corresponding transmission section.

The different mechanisms therefore provide management visibility at different levels of the OTN architecture. The same management requirement cannot necessarily be supported by the same mechanism everywhere in the network: the appropriate form of overhead depends on whether the information concerns a physical optical section, an individual wavelength channel, or a connection represented in the electrical domain.

Table 2.1: Relationship between management functions and the main forms of overhead used in an optical network.

Function	**OSC**	**Pilot tone**	**Electronic OTN overhead**
Trace	OTS / physical section	OCh / optical channel	OTU / ODU connection
Defect indication	OTS, OMS, OCh	Not used	OTU, ODU
Performance monitoring	None	Optical power	BER

Table 2.1 summarizes the resulting organization. The OSC provides section-related overhead within the optical infrastructure. The pilot tone provides information associated with a particular optical channel. Finally, OTN frame overhead becomes available whenever the signal returns to the electrical domain and supports functions such as connection identification, defect indication, and BER monitoring.

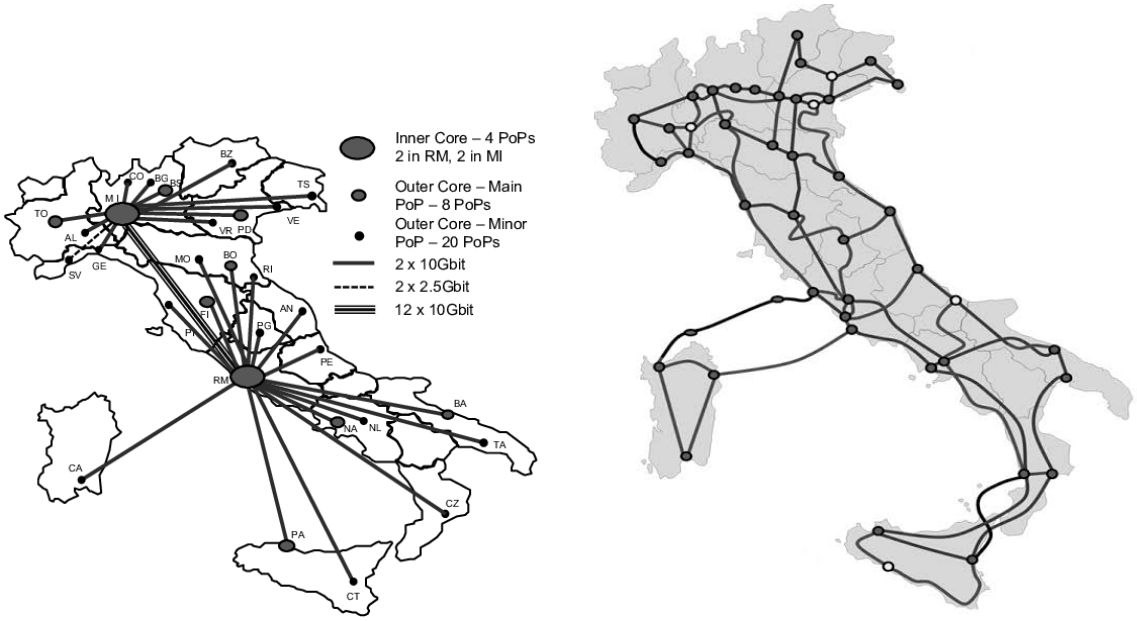
2.12 Optical Networks as an Underlay for Packet Networks

We can now return to one of the main ideas introduced at the beginning of the chapter: the topology observed by a packet network does not necessarily correspond to the physical topology used to transport its traffic.

Figure 2.30 provides a concrete example by showing the IP backbone and the underlying optical transport network of Telecom Italia. The two infrastructures are clearly different. The IP layer is organized around routers and logical links connecting IP Points of Presence, whereas the optical layer contains its own nodes, fiber links, and transport topology.

The comparison highlights an important point. A link between two routers in the IP topology should not be interpreted as a physical fiber directly joining those two routers. From the IP-layer perspective, the two devices are simply adjacent. The transport network may instead realize that logical adjacency through a lightpath crossing several optical nodes and several physical fiber links.

The optical layer therefore hides much of its internal complexity from the packet layer. When an IP connection has to be supported, the transport network selects a suitable optical route, allocates the required wavelength resources, configures the optical switching elements



(a) IP backbone **(b) Optical transport network**

Figure 2.30: Example of the IP and optical transport infrastructures of Telecom Italia. The IP backbone in (a) provides the logical packet-layer connectivity among IP Points of Presence, while the optical network in (b) provides the underlying physical transport infrastructure. Logical IP links can therefore be implemented through lightpaths established across the optical network.

along the path, and establishes the corresponding lightpath. From the viewpoint of the IP routers, all these operations result in what appears to be a direct logical link.

The concepts developed throughout this chapter describe what happens below that apparently simple adjacency. The client signal is adapted and represented through the OTN electronic layers, converted into an optical channel by a transponder, multiplexed with other wavelength channels, transported across the optical infrastructure, and switched through intermediate optical nodes. Management and monitoring functions are then used to supervise the connection while it remains active.

Figure 2.30 also shows why the two layers should be considered as distinct network infrastructures. The optical topology can offer several possible physical routes between different geographical areas, while the packet-network operator can construct a different logical topology on top of the connectivity exported by the transport network. A single optical infrastructure can consequently support many logical IP links whose topology does not reproduce the topology of the underlying fibers.

This separation also provides flexibility in network design. Changes in the optical route used to implement a lightpath do not necessarily require a corresponding change in the logical IP topology. Conversely, the packet layer can be redesigned by establishing, removing, or modifying lightpaths without having to reproduce the physical topology of the optical infrastructure.

There is also an important aggregation role associated with the packet layer. A wavelength represents a valuable transport resource, and only a finite number of wavelength channels can be carried by a fiber. Assigning a dedicated optical channel to every individual user or small traffic flow would therefore be inefficient. Packet networks aggregate many smaller flows into larger traffic demands, which can then be transported efficiently over the optical infrastructure.

The overall architecture can therefore be summarized as a hierarchy. The optical

network provides high-capacity transport connectivity, while the IP network builds a logical packet-forwarding topology on top of that connectivity. The physical route followed by a lightpath and the logical route observed by an IP packet are related, but they belong to different layers and need not have the same topology.

This observation provides the transition to the next chapter. Once the optical underlay has provided logical links between routers, the next question concerns how traffic is forwarded across the resulting packet backbone and how that forwarding can be engineered. These problems lead naturally to the study of Multiprotocol Label Switching (MPLS).

2.13 Chapter Summary

An optical transport network provides connection-oriented connectivity between remote client devices. A client requests a connection, while the provider selects a route, assigns optical-channel resources, configures intermediate switching elements, and creates an end-to-end lightpath.

WDM allows multiple optical channels to share the same fiber by assigning them different wavelengths. Second-generation optical networks exploit WDM and perform important functions directly in the optical domain, reducing repeated O/E/O conversion and improving transparency.

The main components considered in this chapter are transponders and Optical Line Terminals, Optical Add/Drop Multiplexers, and Optical Cross-Connects. Transponders adapt client signals to the optical transport system; OADMs add and drop selected wavelengths; OXCs provide more general optical switching. Traffic grooming allows several low-rate client flows to share one optical channel, while regeneration restores a signal when propagation impairments become excessive.

OTN and the optical-layer hierarchy provide different levels of visibility over the transport service. ODU and OTU operate in the electrical domain, while OCh, OMS, and OTS describe progressively lower optical scopes. These layers support monitoring, identification, and fault localization.

Finally, operating the optical infrastructure requires management and control functions. The provider must provision connections, monitor performance, detect and localize failures, suppress redundant alarms, and carry management information in both electrical and optical domains. From the packet layer, however, this complexity can remain hidden: a complete optical lightpath may appear simply as a direct IP link between two routers.

Further Reading

The material presented in this chapter draws primarily from Chapters 1, 7, and 8 of:

- R. Ramaswami, K. N. Sivarajan, and G. H. Sasaki, *Optical Networks: A Practical Perspective*, 3rd ed., Morgan Kaufmann, 2009.

Chapter 3

Multiprotocol Label Switching

3.1 From IP Forwarding to MPLS

The previous chapter described how an optical transport network can provide high-capacity connectivity between geographically distant routers. From the perspective of the IP layer, two routers may appear to be directly connected through a logical link, even though that link is actually implemented by a lightpath traversing a much more complex optical infrastructure.

We now move one layer upward and focus on the packet network that uses those transport services. In particular, we consider the backbone of an Internet Service Provider (ISP), whose infrastructure is primarily composed of routers interconnected through logical IP links.

The ISP backbone typically spans a wide geographical area and provides connectivity to several access networks. These access networks collect traffic from residential users, companies, mobile networks, and other customers, while the backbone transports this traffic over long distances and provides connectivity toward other networks and, eventually, the rest of the Internet.

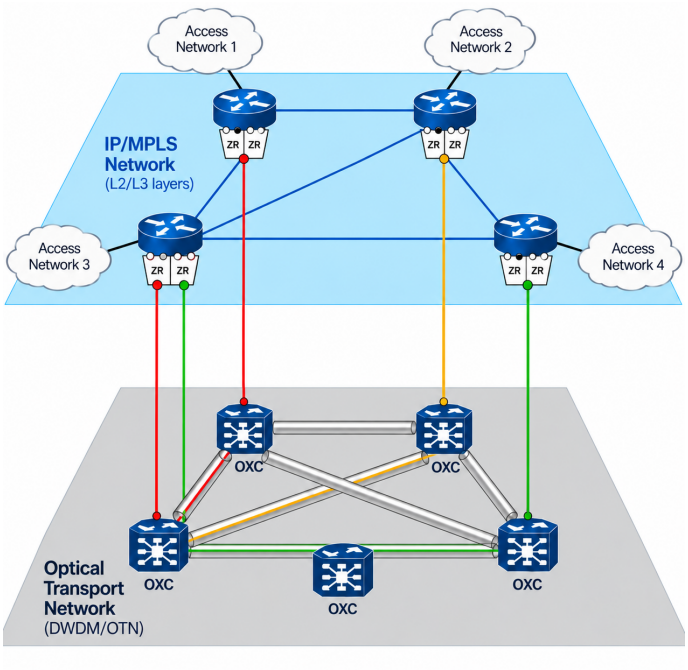


Figure 3.1: An IP/MPLS backbone over an optical transport infrastructure.

3.1.1 The Role and Requirements of an Internet Service Provider

At this point, a reasonable question is why smaller access providers or individual customers do not simply obtain connectivity directly from the optical transport provider.

One reason is economic. An optical lightpath represents a high-capacity transport resource and its provisioning involves optical equipment and network resources that make it relatively expensive. Allocating an entire wavelength to a small traffic demand would generally result in poor utilization of that resource.

There is also a scalability issue. The number of wavelength channels that can be simultaneously transported over a fiber is finite and much smaller than the number of users and traffic flows that may need connectivity. The optical network therefore cannot reasonably dedicate an independent wavelength to every individual traffic source.

The ISP plays an important aggregation role between these two worlds. Many relatively small traffic flows entering from different access networks are aggregated into much larger traffic demands inside the packet backbone. These aggregated demands can then make efficient use of the high-capacity transport services provided by the optical network.

This concept is closely related to the traffic-grooming principle introduced in the previous chapter. There, several lower-rate client signals could be aggregated before being transported over a high-capacity optical channel. Here, aggregation takes place at the packet-network level: a large number of user flows contribute to the traffic exchanged between different portions of the ISP backbone.

Consider, for example, all the traffic entering the provider network at one edge router and leaving it through another edge router. Rather than reasoning about every individual user packet, we can initially regard this traffic as an *ingress-to-egress traffic demand*. Such demands can be very large and therefore justify the use of high-capacity transport resources between the corresponding areas of the network.

However, efficiently transporting traffic is only one of the objectives of a telecommunications provider. The provider operates a network in order to offer services to its customers, and the quality of those services may be regulated through *Service Level Agreements* (SLAs). The network must therefore be operated so that the promised level of service can be provided while the available infrastructure is used efficiently.

This introduces several requirements. The provider would like to control how traffic is distributed across the backbone, differentiate the treatment provided to traffic with different Quality-of-Service requirements, offer services such as Virtual Private Networks (VPNs), and rapidly react when failures affect the infrastructure.

The question is therefore whether a conventional IP network provides enough control to implement these functions efficiently.

3.1.2 Limitations of Conventional IP Forwarding

To answer this question, consider a traffic demand entering an IP backbone through an ingress router and leaving it through an egress router. Two different functions determine how the corresponding packets cross the network: *routing* determines the path toward the destination, while *forwarding* determines how each router sends an incoming packet toward the next hop.

In the conventional IP model considered here, the routing system searches for a least-cost path toward each destination. If all links are assigned the same cost, this reduces to selecting a shortest path, namely a path traversing the minimum number of links.

Once the routing system has selected the route, each router installs the corresponding information in its forwarding table. Packet forwarding is then based primarily on the destination IP address. When a packet arrives, the router searches its forwarding table using the destination prefix and selects the corresponding next hop and output interface.

The behavior considered in this chapter can therefore be summarized through three important properties:

- routing follows a *least-cost*, or in the simplest case *shortest-path*, criterion;
- a selected path is used to carry the traffic toward a given destination;
- forwarding decisions are *destination based*.

These properties make conventional IP routing simple and highly scalable, but they also limit the degree of control available to the network operator.

Consider the example in Figure 3.2. Several traffic flows enter the backbone from different locations but eventually travel toward the same destination region. Because routing attempts to minimize the path cost, the selected paths may progressively converge toward the same set of links.

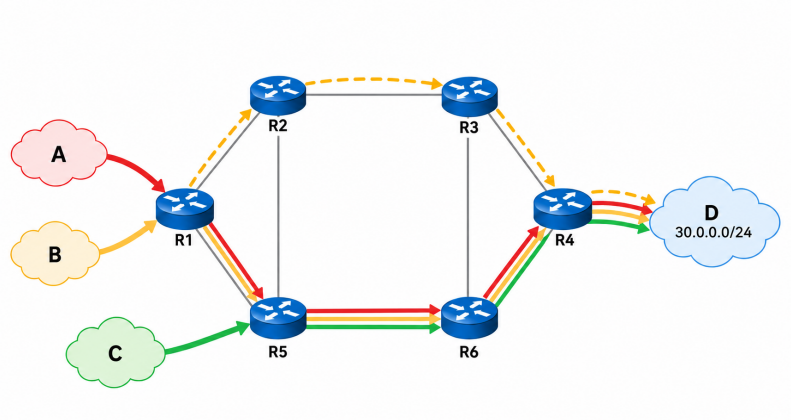


Figure 3.2: Traffic aggregation caused by shortest-path IP routing. Traffic demands originating from networks A, B, and C are directed toward destination network D and progressively converge onto the lower path through routers R_1, R_5, R_6, and R_4. The dashed yellow path shows an alternative route available to traffic $B \rightarrow D$ that could reduce the load on the shared lower links, but is not selected by conventional shortest-path forwarding.

This behavior can produce a significant imbalance in the utilization of the network. Some links may carry several large traffic demands while alternative paths remain almost unused. As link utilization increases, congestion becomes more likely, which in turn can increase packet delay and packet loss.

For best-effort applications this degradation may sometimes be acceptable. Other applications, however, can have much more stringent performance requirements. Delay-sensitive or real-time services may require the provider to offer a more predictable level of performance.

Suppose, for example, that two traffic flows reach the same destination. One is ordinary best-effort traffic, while the other belongs to an application with strict delay requirements. From the provider perspective, it could be useful to keep the first flow on the conventional shortest path and steer the second one onto a different, less congested route.

A purely destination-based forwarding rule does not naturally provide this degree of differentiation. If the packets have the same destination prefix, the forwarding table tends to associate them with the same next hop, independently of the application that generated them or of the quality of service they require.

One possible approach is to modify the link costs used by the routing protocol. Changing the cost of a link can indeed influence the paths computed by the routing system and can therefore be used to improve the overall distribution of traffic.

The limitation is granularity. Changing a routing metric potentially affects all the destinations and traffic flows whose routes depend on that link. The operator cannot easily say: "move only this particular traffic demand onto a different route, while leaving all the other traffic unchanged."

Modern IP networks can also exploit mechanisms such as Equal-Cost Multi-Path (ECMP) forwarding to distribute traffic across multiple equal-cost routes. The fundamental issue considered here, however, remains the same: conventional destination-based routing provides considerably less explicit, per-traffic-class control over the complete end-to-end path than a connection-oriented mechanism.

What the provider would like to perform is *Traffic Engineering*: instead of allowing traffic placement to be only a consequence of shortest-path routing, the operator should be able to control how traffic demands are mapped onto the available network resources.

The same need for greater control appears when Quality-of-Service differentiation is considered. Different classes of traffic may require different treatment, and the provider may want their packets to follow different paths even when their IP destinations are the same.

Finally, providers must support additional services that cannot be obtained from basic destination-based IP forwarding alone. One important example is the creation of Virtual Private Networks, in which different customers share the same provider infrastructure while retaining logically separate private networks.

Traffic Engineering, Quality-of-Service differentiation, VPNs, and fast recovery from failures therefore point toward the same general requirement: the provider needs more control over the way traffic crosses its backbone.

3.1.3 Connection-Oriented Packet Networks

A possible way to obtain this additional control is to introduce a *connection-oriented* forwarding model into the packet network.

The idea is conceptually similar to the connection-oriented behavior already encountered in the optical network. Before traffic is transported, a logical connection is established between an ingress point and an egress point. The path followed by the connection is therefore determined in advance.

In a packet network, this connection can be viewed as a *tunnel* crossing the backbone. Once traffic has been associated with the tunnel, the packets follow the path established for that connection until they reach its egress point.

This does not transform the packet network into a circuit-switched network. The transported information is still organized into individual packets, and many packet flows can share the same physical links. The important difference is that forwarding can now be associated with a previously established logical connection rather than being determined independently from the destination address at every stage of the path.

This provides an important form of *path differentiation*. Two traffic classes entering at the same point and directed toward the same destination can be associated with different connections and can therefore follow different routes through the backbone.

The provider can consequently separate the problem of classifying traffic from the problem of transporting it. At the network edge, packets can be classified according to the service that they require. Inside the backbone, the corresponding connection determines the path along which they are forwarded.

This is the basic idea that motivates *Multiprotocol Label Switching* (MPLS). MPLS introduces a connection-oriented forwarding mechanism into an IP packet network while preserving the packet-based nature of the service.

Rather than replacing IP, MPLS adds an additional forwarding mechanism between the IP layer and the underlying link-layer technology. For this reason, it is often informally

described as a "Layer 2.5" technology.

Its purpose is not simply to replace one packet header with another. The important point is that MPLS gives the provider a mechanism for associating traffic with predetermined paths through the backbone. This additional degree of control will provide the foundation for the services considered later in this chapter, including Traffic Engineering, Virtual Private Networks, and fast protection mechanisms.

The next section introduces the architectural elements that make this connection-oriented forwarding model possible.

3.2 MPLS Architecture

The previous section identified a missing element in a conventional IP backbone. The provider would like to retain IP connectivity while gaining greater control over the path followed by selected traffic classes and over the services offered to its customers.

Multiprotocol Label Switching (MPLS) was introduced to provide this additional functionality without replacing IP. From a protocol-stack perspective, MPLS can be regarded as an intermediate forwarding layer between the network layer and the underlying link-layer technology.

In a typical modern backbone, IP packets may therefore be transported according to a stack conceptually represented as

$$\text{IP} \longrightarrow \text{MPLS} \longrightarrow \text{Ethernet} \longrightarrow \text{optical transport.}$$

For this reason, MPLS is sometimes informally described as a *Layer 2.5* technology. It does not replace the routing intelligence of IP, nor does it replace the Layer-2 technology used to carry packets over an individual link. Instead, it introduces an additional forwarding mechanism between them.

The term "Layer 2.5" is only an informal description and does not correspond to an official OSI layer. It is useful because MPLS forwarding takes place below the IP forwarding function while remaining independent of the particular Layer-2 technology used on a link.

The term *Multiprotocol* reflects this architectural independence. MPLS was designed so that the forwarding mechanism is not intrinsically tied to a particular client network protocol or to a particular underlying link technology. In the remainder of this chapter, however, we will consider the most common case in which MPLS transports IP packets over Ethernet links.

3.2.1 MPLS Forwarding and Label Switching

An MPLS-capable router can be conceptually divided into two components: a *control component* and a *forwarding component*. The former retains the Layer-3 intelligence of a conventional IP router and is responsible for routing information, label allocation and binding, and the exchange of label information with other MPLS nodes. The latter operates directly on packets and implements the label-switching forwarding paradigm.

Figure 3.3 summarizes this organization. MPLS should therefore not be regarded as a separate device placed next to an IP router. Rather, MPLS forwarding extends the capabilities of a conventional IP router, resulting in an integrated IP/MPLS node.

From the perspective of the forwarding component, an MPLS node implements three fundamental data-plane actions on packets: *PUSH*, *SWAP*, and *POP*. These three operations correspond to the three main situations encountered by a packet while traversing an MPLS domain.

The first situation occurs when an ordinary IP packet reaches the ingress of the MPLS domain. As discussed previously, the ingress LER classifies the packet into a Forwarding

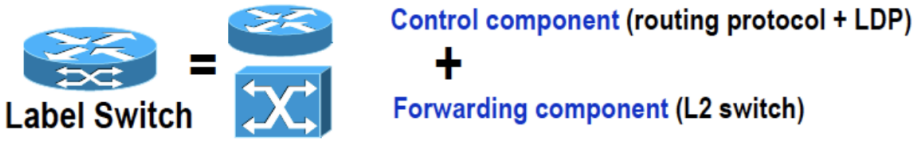


Figure 3.3: Conceptual organization of an IP/MPLS router. The control component combines conventional IP routing functions with MPLS label allocation and distribution, while the forwarding component implements label-based packet switching.

Equivalence Class (FEC) and determines the corresponding Label Switched Path (LSP). To steer the packet into that LSP, the router performs a *PUSH* operation: an MPLS header containing the appropriate label is inserted in front of the original IP packet. Conceptually,

$$\boxed{\text{IP packet}} \xrightarrow{\text{PUSH}} \boxed{\text{MPLS label} \mid \text{IP packet}}.$$

After this operation, the packet no longer needs to be forwarded through the MPLS domain according to its destination IP address. The label identifies the forwarding treatment that has already been selected for the packet.

Once inside the MPLS domain, the packet reaches intermediate Label Switching Routers. At each of these nodes, the forwarding decision is based on the MPLS label. The router identifies the appropriate forwarding entry and replaces the incoming label with the label expected on the next segment of the LSP. This is the *SWAP* operation. For example, a packet may arrive at an LSR carrying label 45 and leave the same router carrying label 78:

$$\boxed{45 \mid \text{IP packet}} \xrightarrow{\text{SWAP}} \boxed{78 \mid \text{IP packet}}.$$

The fact that the label can change from one hop to the next is important: MPLS labels have local significance. The same end-to-end LSP is therefore not identified by a single label value that remains unchanged throughout the network.

The third operation occurs when MPLS forwarding must terminate. In that case, the router performs a *POP* operation and removes the MPLS header:

$$\boxed{\text{MPLS label} \mid \text{IP packet}} \xrightarrow{\text{POP}} \boxed{\text{IP packet}}.$$

At this point, the resulting packet can be handed back to the conventional IP forwarding function. In other words, POP does not merely remove a header: it can mark the transition from MPLS forwarding back to destination-based IP forwarding. Once the packet has been redirected to the IP forwarding component, the router again uses the destination IP address to determine how the packet should continue.

To understand how an MPLS router selects one of these actions, it is useful to return to the general *match-and-action* abstraction used by packet switching devices. A conventional IP router performs a lookup using the destination IP address and obtains an action such as forwarding the packet through a specific next hop or output interface. An MPLS router applies the same general principle, but uses label-based forwarding state.

In a simplified view, the forwarding information used by MPLS can be expressed as a label lookup table. An incoming MPLS packet is matched against its label, and the corresponding entry determines the required forwarding action. In the most common case, the router sends the packet to another MPLS next hop and, if necessary, replaces the incoming label with a new outgoing label. This is the standard label-switching behavior associated with SWAP.

However, an MPLS forwarding entry may also indicate that MPLS processing must terminate. In that case, the label lookup does not return another LSR as next hop, but

instead redirects the packet to the *IP layer*. Operationally, this means that the packet undergoes a POP operation, the MPLS header is removed, and the resulting IP packet is then processed by the traditional IP routing table.

This behavior is illustrated in Figure 3.4. The upper part of the figure shows conventional IP forwarding: the destination address contained in the IP header is matched against the IP routing table in order to select the next hop. The lower part shows MPLS forwarding through a label lookup table. Some entries map an input label to an outgoing label and an MPLS next hop, thereby implementing ordinary label switching. Other entries, instead, map the incoming MPLS packet back to the IP layer (e.g., for best effort traffic).

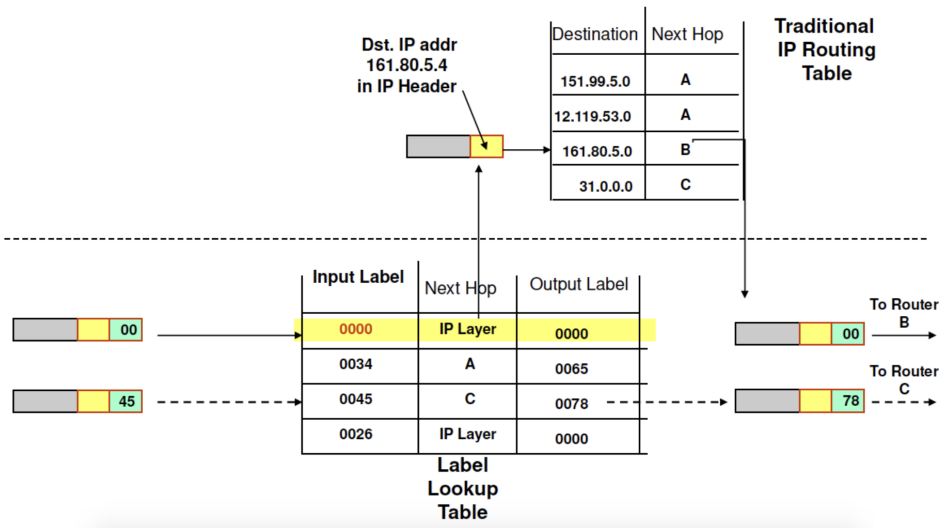


Figure 3.4: Relationship between traditional IP forwarding and MPLS label forwarding. In the upper part, forwarding is based on the destination IP address and a conventional routing table. In the lower part, forwarding is based on the incoming MPLS label and a label lookup table.

This makes it clear that MPLS forwarding and IP forwarding coexist in the same router. An incoming packet may enter the node as a plain IP packet and be pushed into an LSP, or it may arrive as an MPLS packet and be either switched to another MPLS hop or returned to the IP layer. The forwarding component must therefore be capable of handling both forms of processing.

We can now follow the complete forwarding process from one side of the MPLS domain to the other.

Consider an IP packet arriving from an access network at the ingress LER. The router first classifies the packet and identifies its FEC. From the FEC, it determines the LSP to be used. If the corresponding forwarding state indicates that the packet has to enter the LSP with label 10, the ingress LER performs a PUSH operation and forwards the resulting MPLS packet into the domain.

The next LSR receives an MPLS packet rather than an ordinary IP packet. Assume that the packet arrives carrying label 10. Its label lookup table may instruct the router to replace label 10 with label 20 and forward the packet to the next hop. A subsequent LSR may then replace label 20 with label 100 before forwarding the packet further along the LSP.

At the end of the LSP, the forwarding entry associated with label 100 indicates that MPLS processing must terminate. The router therefore performs POP, obtaining the original IP packet. The packet is then passed to the IP forwarding function, which examines its destination address and forwards it toward the destination network.

Figure 3.5 provides the complete view of this process. It is useful to observe that only the ingress LER needs to perform the initial classification into FECs. Once the packet has entered the MPLS domain, the intermediate routers forward it according to the label-based state already installed in their forwarding tables. When required, the last MPLS-aware node can finally redirect the packet back to the IP layer.

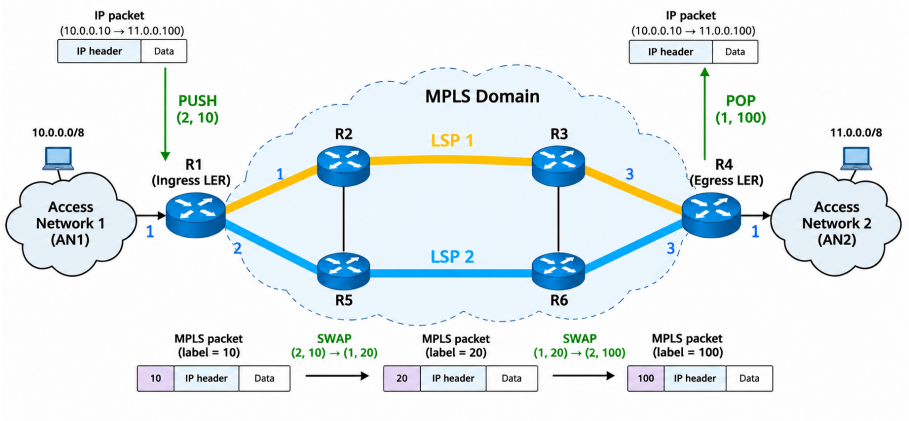


Figure 3.5: End-to-end forwarding along an MPLS Label Switched Path.

The overall process can therefore be summarized as

$$\begin{aligned} \text{Ingress:} &\quad \boxed{\text{IP forwarding}} \rightarrow \boxed{\text{classification / FEC}} \rightarrow \boxed{\text{PUSH}} \\ \text{MPLS domain:} &\quad \boxed{\text{SWAP}} \rightarrow \cdots \rightarrow \boxed{\text{SWAP}} \\ \text{Egress:} &\quad \boxed{\text{POP}} \rightarrow \boxed{\text{IP forwarding}}. \end{aligned}$$

This summary highlights an important property of an IP/MPLS router: IP forwarding and MPLS forwarding are not separate worlds, but two forwarding modes that coexist in the same node. A packet can enter the MPLS forwarding process, traverse an LSP through label switching, and later be returned to the IP forwarding process when MPLS processing is no longer required.

So far, however, we have assumed that all the necessary label bindings and forwarding entries are already available in the routers. The next step is therefore to understand how this state is created and how an LSP is established before data packets begin to traverse it.

3.2.2 Building a Label Switched Path

In the previous section, we followed packets along an MPLS Label Switched Path and assumed that every router already contained the forwarding entries required to perform the appropriate PUSH, SWAP, and POP operations.

Before an LSP can actually be used, however, this forwarding state must be created. This is a control-plane operation usually referred to as *LSP setup*.

Consider an LSP associated with a particular Forwarding Equivalence Class. At the ingress, the router must know which label has to be pushed onto packets belonging to that FEC. At every intermediate router, an incoming label must be associated with an outgoing interface and, normally, with a new outgoing label. Finally, at the end of the LSP, the forwarding state must indicate that the MPLS encapsulation has to be removed.

The control plane must therefore create a coherent sequence of forwarding entries across all the routers belonging to the LSP.

The association between a Forwarding Equivalence Class and a label is referred to as a *label binding*. A label has only local significance: a router is free to select the label

that it uses to identify a particular FEC on one of its incoming links. The same FEC can consequently be represented by different labels at different points of the LSP.

Once a router has selected a label for a FEC, this information must be communicated to the neighboring router that will send packets toward it. Otherwise, the upstream router would not know which label the downstream router expects to receive.

A *Label Distribution Protocol* (LDP) provides the control-plane mechanisms required to exchange this information and establish the label bindings needed along an LSP. We are not interested here in the detailed format of the protocol messages. The important point is that LSP setup is a signaling procedure through which the routers progressively create the state required by the MPLS data plane.

The basic setup procedure can be understood through two signaling messages, which we will refer to as *PATH* and *RESV*.

The procedure starts at the ingress router. A PATH message is sent toward the egress along the path that is intended to become the LSP. The PATH message is therefore propagated in the same direction in which data packets will later travel.

As the PATH message crosses an intermediate router, the node creates a temporary local state associated with the connection being established. This state identifies the LSP under construction and stores the information needed to recognize the signaling exchange and to continue it toward the next node.

This temporary information is usually referred to as *flow state*. It allows the router to remember that a PATH message for a particular LSP has crossed the node and, in particular, from which direction the setup request arrived and toward which next hop it was propagated.

The state created during this phase is not intended to remain indefinitely in the router. It is maintained as *soft state*: unless it is periodically refreshed, it eventually expires.

The reason for using soft state is closely related to the distributed nature of the control procedure. A router may fail, a link may disappear, or the signaling process itself may be interrupted. If the state created during an incomplete LSP setup were permanent, obsolete information could remain inside the routers even though the corresponding LSP no longer existed. By associating the state with a lifetime, stale information is automatically removed when it is no longer refreshed.

The PATH message therefore progresses from the ingress toward the egress, creating the temporary state required to remember the path followed by the setup request.

Once the PATH message reaches the egress router, the reverse phase of the setup procedure begins. A *RESV* message is sent back toward the ingress.

The RESV message follows the path previously established by the PATH message, but in the opposite direction. This is the phase in which the label bindings needed by the data plane are progressively established.

Consider two adjacent routers along the selected path. The downstream router selects the label that it wants to receive for the FEC associated with the LSP and communicates this information to its upstream neighbor through the RESV message.

The upstream router can then configure the outgoing part of its label-switching entry. It now knows that packets belonging to that LSP must be sent toward that downstream router using the label that has just been advertised.

The same router also selects the label that it wants to receive from its own upstream neighbor. When the RESV message continues toward the ingress, this new label is communicated to the preceding router.

The process is repeated hop by hop in the upstream direction. As a result, each intermediate router obtains the two pieces of information required for the SWAP operation:

$$(\text{incoming interface}, \text{incoming label}) \longrightarrow (\text{outgoing interface}, \text{outgoing label}).$$

For example, if a router has decided to receive packets of the LSP with label 20 and its downstream neighbor has indicated that it expects label 100, the resulting forwarding entry can be represented as

$$(1, 20) \longrightarrow (3, 100).$$

Once the RESV message finally reaches the ingress router, all the routers along the path contain the forwarding state required to realize the LSP. The ingress LER knows which label must be pushed, the intermediate LSRs know how to perform their SWAP operations, and the final router knows how the MPLS forwarding process terminates.

The overall setup procedure can therefore be summarized as

PATH: ingress → intermediate routers → egress,

creation of temporary flow state,

RESV: egress → intermediate routers → ingress,

label binding and installation of forwarding state.

At the end of this exchange, the control plane has transformed the selected sequence of routers into an operational Label Switched Path that can be used by the data plane.

An additional question remains: how is the path followed by the PATH message selected?

Two different approaches can be distinguished: *hop-by-hop routing* and *explicit routing*.

In the *hop-by-hop* case, the setup procedure relies on the conventional IP routing information already available at each router. When a PATH message reaches a node, that node independently determines the next hop toward the destination according to its routing table. The signaling message therefore progresses through the network following the same routing decisions that would be used for ordinary IP traffic.

For example, if IP routing determines the path

$$R_1 \to R_2 \to R_3 \to R_4,$$

the PATH message follows that sequence of routers and the subsequent RESV message returns along the same path in the opposite direction, establishing the required label bindings.

The resulting LSP is therefore connection oriented from the data-plane perspective, but its route is still determined hop by hop by the underlying IP routing system.

This is also its main limitation. If IP routing selects a shortest or least-cost path, a hop-by-hop LSP naturally inherits that choice. MPLS label switching alone does not provide any additional control over the route.

With *explicit routing*, instead, the path to be followed is specified before the signaling procedure begins. The PATH message carries the information needed to force the setup through a predetermined sequence of routers rather than allowing every router to select the next hop independently.

For example, even if conventional IP routing would normally use

$$R_1 \to R_2 \to R_3 \to R_4,$$

the explicitly selected LSP may be required to follow

$$R_1 \to R_5 \to R_6 \to R_4.$$

The PATH message is then forwarded along this explicitly selected sequence, and the RESV message travels backward along the same path, establishing the corresponding label bindings.

Two forms of explicit route specification can be considered. A *strict explicit route* specifies the complete sequence of routers that must be traversed. A *loose explicit route*, instead, identifies only selected intermediate nodes that the LSP must cross, leaving conventional routing to determine how to travel between those specified points.

The distinction between hop-by-hop and explicit routing therefore concerns the selection of the path used during LSP setup:

Hop-by-hop:	each router uses its IP routing information to select the next hop of the PATH message,
Explicit routing:	the path, or a set of mandatory intermediate nodes, is specified before the setup procedure.

In both cases, once PATH and RESV signaling has completed, the result from the data-plane perspective is the same: the routers contain the label-switching state required to forward packets along the LSP.

Explicit routing, however, introduces the possibility of selecting a path different from the one naturally chosen by IP routing. This capability will be particularly important when discussing MPLS Traffic Engineering.

3.2.3 MPLS Label Stacking

So far, we have assumed that an MPLS packet carries a single label. MPLS, however, allows more than one label to be associated with the same packet. The resulting sequence of labels is called a *label stack*.

The basic idea can be understood by considering a packet that is already being forwarded through an LSP. Suppose that an IP packet has entered LSP_1 and therefore already carries the MPLS label used to identify that path.

At some point in the network, it may be useful to transport the entire LSP_1 through another Label Switched Path, denoted as LSP_2. Instead of removing the existing MPLS header, the ingress router of LSP_2 performs another PUSH operation and adds a second label on top of the existing one.

The packet therefore evolves from

$$\boxed{\text{Label}_1 \mid \text{IP packet}}$$

to

$$\boxed{\text{Label}_2 \mid \text{Label}_1 \mid \text{IP packet}}.$$

The newly inserted label becomes the *top label* of the stack. Routers belonging to LSP_2 operate only on this outer label and do not need to examine the label associated with LSP_1.

From their perspective, the complete content carried below the top label is simply the payload transported by LSP_2.

When the packet reaches the end of LSP_2, the outer label is removed through a POP operation:

$$\boxed{\text{Label}_2 \mid \text{Label}_1 \mid \text{IP packet}} \xrightarrow{\text{POP}} \boxed{\text{Label}_1 \mid \text{IP packet}}.$$

The original MPLS label is therefore exposed again, and forwarding can continue according to LSP_1.

This mechanism makes it possible to create a hierarchy of MPLS tunnels. An LSP can effectively transport packets that are already associated with another LSP, without

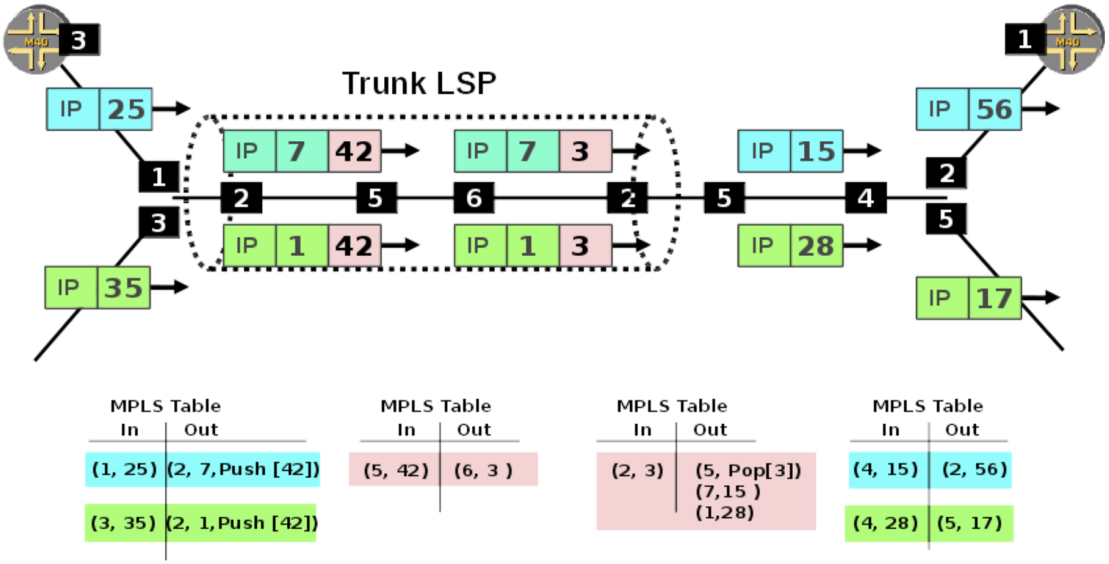


Figure 3.6: MPLS label stacking and nested Label Switched Paths.

requiring the intermediate routers of the outer tunnel to know the forwarding state of the inner one.

Figure 3.6 illustrates this nesting mechanism. The important point is that MPLS forwarding always operates on the label at the top of the stack. A PUSH operation adds a new label on top, while a POP operation removes the current top label and exposes the label immediately below it.

The MPLS header also contains an S (*Bottom of Stack*) bit that allows the router to determine whether the current label is the last label in the stack. The value

$$S = 1$$

identifies the bottom label, whereas labels placed above it have

$$S = 0.$$

For a packet carrying two labels, the structure can therefore be represented as

$$\underbrace{\text{Label}_2,\ S = 0}_{\text{top}} \mid \underbrace{\text{Label}_1,\ S = 1}_{\text{bottom}} \mid \text{IP packet}.$$

Label stacking therefore extends the tunnel abstraction introduced earlier: rather than having only packets inside an MPLS tunnel, MPLS also allows one logical forwarding context to be transported inside another MPLS tunnel.

3.3 Why MPLS? Services and Network Control

The previous sections introduced the basic mechanisms provided by MPLS: packets can be classified into Forwarding Equivalence Classes, associated with Label Switched Paths, forwarded through label switching, and transported through nested LSPs using label stacking.

These mechanisms are not an objective by themselves. Their importance becomes clear when we return to the requirements of a telecommunications provider introduced at the beginning of the chapter.

A provider must offer differentiated services to its customers while operating a shared backbone efficiently. At the same time, it must retain sufficient control over how traffic is transported, react rapidly to failures, and support different service requirements.

Conventional destination-based IP forwarding provides reachability, but leaves the provider with limited control over the path followed by individual traffic classes. MPLS introduces an additional forwarding abstraction that can be used to address this limitation.

In particular, MPLS has been widely used in provider backbones to support four main functions:

- **Virtual Private Networks (VPNs).** Multiple customers can share the same provider infrastructure while remaining logically separated. MPLS labels allow customer traffic to be transported across the backbone without requiring the core routers to participate directly in the customers' routing domains.
- **Traffic Engineering.** Selected traffic can be associated with explicitly routed LSPs rather than being constrained to the path selected by conventional IP routing. The provider can therefore control how traffic is distributed over the available network resources.
- **Protection and fast recovery.** Backup LSPs can be prepared in advance so that traffic can be rapidly moved away from failed links or nodes. This provides a recovery mechanism that does not necessarily require waiting for the complete convergence of the IP routing system.
- **Quality of Service support.** The classification of packets into different forwarding classes provides a basis for applying differentiated treatment to different types of traffic within the provider network.

These applications rely on the same architectural principle: the provider can separate the forwarding treatment applied to a packet from the conventional destination-based IP path and associate selected traffic with explicitly managed forwarding state.

The remainder of the chapter examines three of these applications in greater detail. We first consider how MPLS can be used to build Virtual Private Networks, then study its role in Traffic Engineering, and finally discuss protection and fast-recovery mechanisms.

3.4 MPLS Virtual Private Networks

One of the most important services that a provider can build over an MPLS backbone is a *Virtual Private Network* (VPN).

Consider a company with several geographically separated sites. From the customer perspective, these sites should behave as parts of the same private network, even though the infrastructure used to interconnect them is shared with many other customers.

A VPN therefore provides a logically separated network over a common provider infrastructure. Two requirements are particularly important. First, traffic and routing information belonging to different customers must remain isolated. Second, customers should be free to use their own IP addressing plans, including private addressing, without coordinating them with other customers.

Figure 3.7 illustrates the problem in a provider network serving two different customers. The provider operates a common backbone, represented by AS 100, while the customer networks are connected at different Points of Presence (PoPs).

An important detail in Figure 3.7 is that the same IP prefix, 10.0.0.0/24, appears in networks belonging to different customers. From the provider perspective, the destination IP prefix alone is therefore not always sufficient to identify the correct customer network. Additional information is required to preserve the separation between the different VPNs.

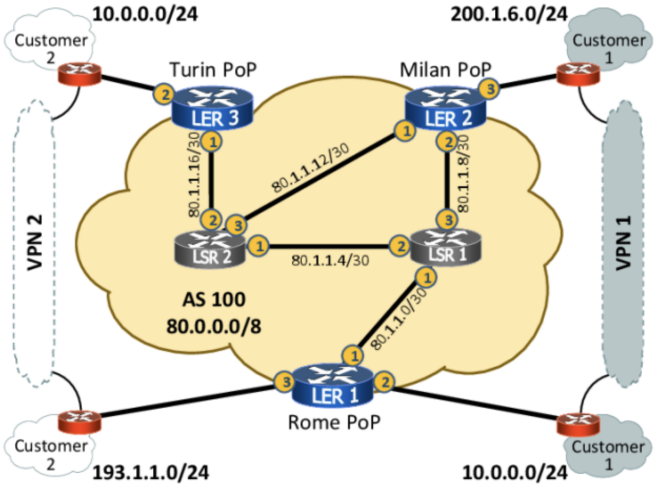


Figure 3.7: Problem statement for an MPLS Layer-3 VPN. Different customers connect geographically separated networks through the same provider backbone. Customer addressing is independent of the provider addressing space and may overlap across different VPNs.

The basic architecture used to provide this service is shown in Figure 3.8. Routers located at the customer sites are called *Customer Edge* (CE) routers. They connect to *Provider Edge* (PE) routers located at the boundary of the provider network. Routers operating exclusively inside the provider backbone are referred to as *P* routers.

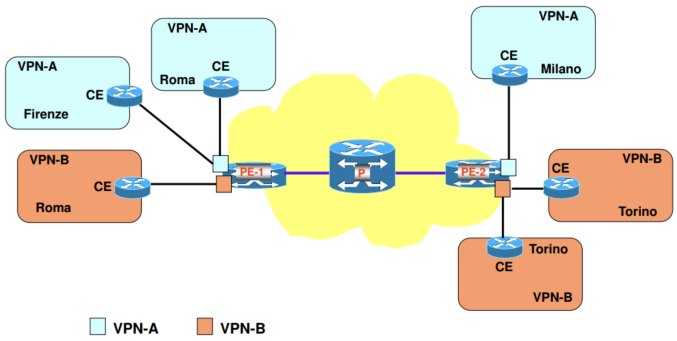


Figure 3.8: Example of two VPNs sharing the same provider infrastructure. Customer Edge routers belonging to VPN-A and VPN-B connect to the common backbone through Provider Edge routers.

The PE routers play a central role in this architecture. They interface with the customer networks, maintain customer-specific information, and inject customer traffic into the provider backbone. The P routers, instead, can remain independent of the individual customer routing domains and are mainly responsible for transporting packets across the core.

The construction of an MPLS Layer-3 VPN can be understood through three main steps:

1. establish any-to-any connectivity among the PE routers;
2. distribute customer reachability information among the PEs;
3. provide an MPLS transport mechanism between the PEs.

3.4.1 Building an MPLS L3VPN

Move 1: establish IP connectivity among the PE routers.

Before considering the customer networks, the provider must first guarantee connectivity inside its own backbone. In particular, each PE must be able to reach every other PE.

A common solution is to assign a stable loopback address to each PE router. Figure 3.9 shows an example in which the PE routers located in Rome, Milan, and Turin are identified respectively by

$$80.80.80.1/32, \qquad 80.80.80.4/32, \qquad 80.80.80.5/32.$$

These addresses belong to the provider addressing space and are completely independent of the prefixes used by the customers.

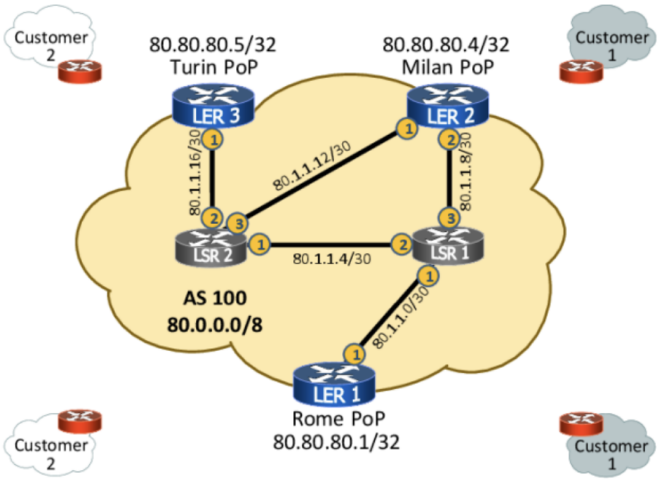


Figure 3.9: PE identification through loopback interfaces. Each Provider Edge router is assigned a stable /32 loopback address that uniquely identifies it inside the provider backbone.

Assigning the addresses is not sufficient: the provider must also distribute their reachability throughout the backbone. An Interior Gateway Protocol (IGP), such as OSPF or IS-IS, is therefore used to advertise the PE loopback addresses and the internal provider topology.

Figure 3.10 shows the result from the perspective of the Rome PE. Its routing table contains routes toward the loopback addresses of the remote PEs. The Milan and Turin PEs can therefore be reached through the next hop selected by the provider IGP.

At the end of Move 1, the provider has therefore constructed an IP network in which the PE routers can communicate with one another. However, a PE still does not know which customer prefixes can be reached through each remote PE.

Move 2: distribute customer reachability information.

The second step concerns the customer networks. Each CE must communicate to its attached PE the prefixes that can be reached through that customer site.

When BGP is used on the CE–PE connection, the CE and the PE establish an external BGP session. The CE can therefore advertise its customer prefixes to the directly connected PE.

The PE must then distribute this information to the other PEs that participate in the same VPN. This second exchange is performed using *Multiprotocol BGP* (MP-BGP). Within a single provider Autonomous System, the PE routers exchange this information through internal BGP sessions.

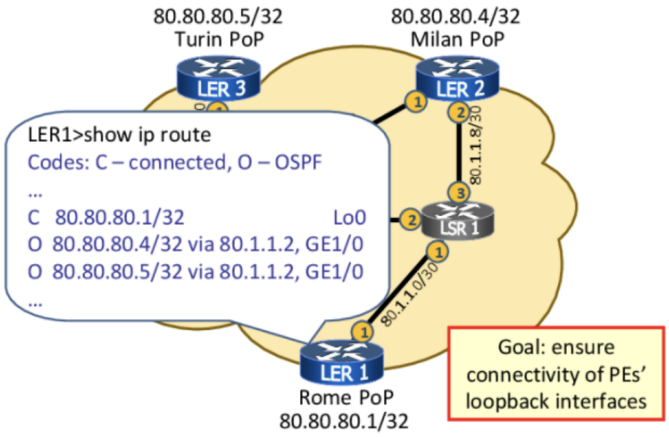


Figure 3.10: Any-to-any IP connectivity among the Provider Edge routers. The provider IGP distributes the PE loopback prefixes, allowing each PE to reach every other PE through the internal provider network. The example shows routes toward the remote PE loopbacks in the routing table of the Rome PE.

The two exchanges therefore serve different purposes:

$$\boxed{\text{CE}} \xrightarrow{\text{eBGP}} \boxed{\text{PE}} \xrightarrow{\text{MP-BGP / iBGP}} \boxed{\text{remote PE}}.$$

Figure 3.11 illustrates this mechanism. The Rome PE learns customer prefixes from its connected CE routers and advertises their reachability toward the other Provider Edge routers. In the example, the prefixes 10.0.0.0/24 and 193.1.1.0/24 are associated with different customers and must therefore remain distinguishable when their reachability information is propagated.

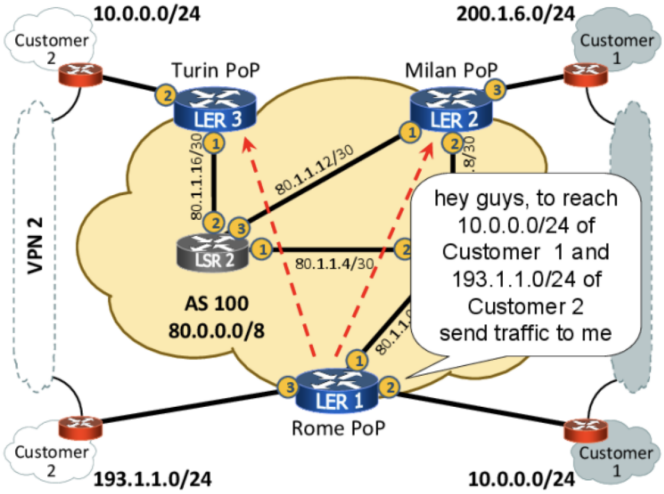


Figure 3.11: Distribution of customer reachability information. A PE learns customer prefixes from the attached CE routers and advertises the corresponding reachability information toward the other PEs. MP-BGP is used between the Provider Edge routers to distribute the VPN reachability information while preserving the association with the corresponding customer.

At this point it is useful to distinguish the two routing systems operating inside the provider:

Provider IGP: how can a PE reach another PE?

MP-BGP: which customer networks are reachable through that PE?

The provider IGP is therefore concerned with the topology and addresses of the provider infrastructure, whereas MP-BGP distributes customer reachability information among the edges of the network.

This separation also prevents the internal P routers from having to maintain the complete set of customer prefixes.

Move 3: provide MPLS transport between the PE routers.

Moves 1 and 2 give the ingress PE the information required to determine which remote PE provides access to a particular customer destination. A third mechanism is now required to actually transport the customer packet across the provider backbone.

MPLS provides this transport by establishing an LSP between the ingress and egress PE routers. This PE-to-PE LSP can be viewed as an MPLS trunk across the provider network.

However, reaching the correct egress PE is not sufficient. A PE can serve many different customers and VPNs. When the packet reaches the destination PE, that router must also determine the customer-specific forwarding action to apply.

This is where the label-stacking mechanism introduced earlier becomes useful.

Figure 3.12 illustrates the two different roles played by the MPLS labels.

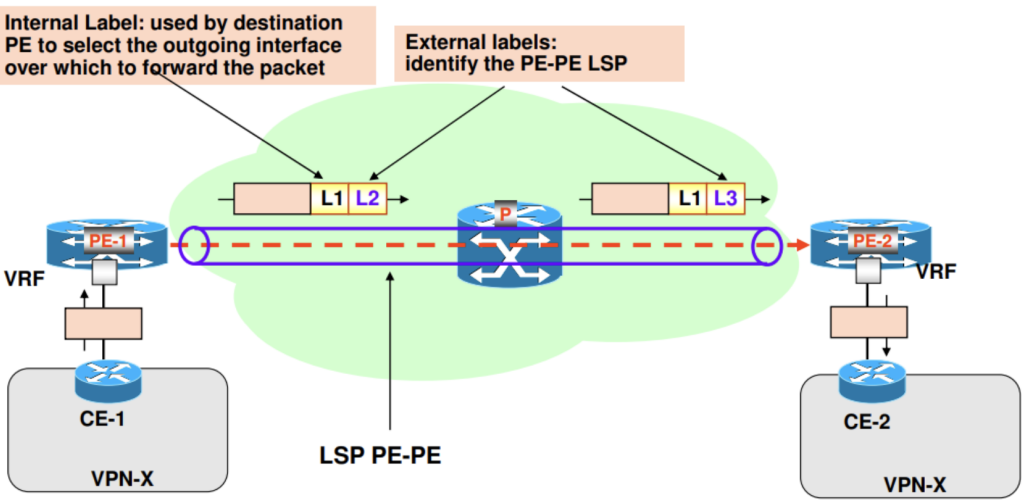


Figure 3.12: Use of MPLS label stacking for VPN transport. The external label identifies the PE-to-PE LSP and is processed by the routers in the provider core. The internal label is carried together with the customer packet and is used by the destination PE to select the appropriate customer-facing forwarding action.

The packet transported across the provider backbone can therefore be represented as

| external label | internal label | customer IP packet |.

The *external label* identifies the PE-to-PE LSP. This is the label used by the P routers to transport the packet through the provider backbone. As in any MPLS LSP, the value of this label may change from one hop to another through the usual SWAP operations.

The *internal label* serves a different purpose. It is not used to select the path across the provider core. Instead, it carries information that will be required by the destination PE once the packet reaches the end of the PE-to-PE LSP.

The two labels therefore separate two forwarding decisions:

External label:	which PE must the packet reach?
Internal label:	how must the destination PE process the packet?

The P routers only need to process the external label. They do not need to know the customer destination address or maintain the routing information of the individual VPNs.

At this point, the three basic mechanisms required to build the VPN have been introduced: PE-to-PE reachability, customer route distribution, and MPLS transport. We can now look more closely at how a PE keeps the information of different customers separated.

Routing tables and VRFs.

A single PE can simultaneously connect several customers to the provider network. Moreover, different customers are free to use the same private addressing space. The example introduced in Figure 3.7, for instance, contains the same 10.0.0.0/24 prefix in two different VPNs.

The PE therefore cannot store all customer routes in a single routing table. The same destination prefix may correspond to completely different networks depending on the customer to which the packet belongs.

MPLS L3VPNs solve this problem by maintaining separate routing and forwarding contexts called *VPN Routing and Forwarding* instances, or *VRFs*.

Figure 3.13 illustrates the organization of the routing information inside a PE. In addition to a *global routing table*, used for the provider network itself, the router maintains one or more customer-specific VRFs. In the example, VRF-A contains the routing information associated with VPN-A, while VRF-B contains the routes associated with VPN-B.

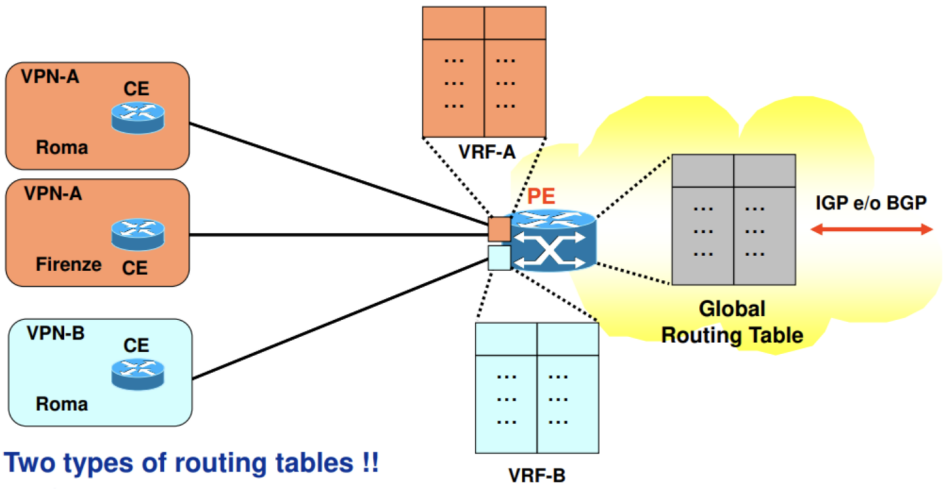


Figure 3.13: Routing information maintained by a Provider Edge router. The global routing table contains the routes required to operate within the provider backbone, while separate VRFs contain the customer-specific routing information associated with the different VPNs.

Customer-facing interfaces are associated with the appropriate VRF. As a consequence, the interface on which a packet arrives provides the context needed to select the correct forwarding table.

The forwarding process can therefore be represented conceptually as

incoming interface → VRF → IP destination lookup → VPN-specific forwarding action.

For example, the prefix 10.0.1.0/24 may appear in both VRF-A and VRF-B. Because the two entries belong to different forwarding contexts, they can identify different remote sites and can be associated with different MPLS labels.

This also clarifies the role of the internal label introduced in Figure 3.12. The external label is required to transport the packet to the correct destination PE. The internal label preserves the customer-specific forwarding information required when the packet reaches that PE.

The complete operation of an MPLS L3VPN can therefore be summarized as

Move 1: establish reachability among the PEs,

Move 2: distribute customer reachability information,

Move 3: transport customer packets through PE-to-PE MPLS LSPs.

VRFs complement these three mechanisms by maintaining separate customer forwarding contexts at the provider edge.

The resulting architecture achieves a clear separation of responsibilities. The provider IGP maintains connectivity inside the backbone, MP-BGP distributes customer reachability information among the PE routers, MPLS transports the packets across the core, and VRFs preserve the logical separation between the different customer networks.

3.5 MPLS Traffic Engineering

One of the motivations for introducing MPLS in a provider backbone is the possibility of controlling how traffic is distributed over the available network resources.

In a conventional IP network, packets are forwarded according to the paths computed by the routing protocol. In a link-state protocol such as OSPF, for example, the path toward a destination is determined by the configured link costs. All traffic associated with the same destination therefore tends to follow the same shortest path, independently of the actual amount of traffic that has to be transported.

This can lead to an inefficient use of the network capacity.

Consider the example in Figure 3.14. Three traffic demands have to be carried by the network: 1 Mbps from $R2$ to $R4$, 2 Mbps from $R1$ to $R3$, and 2 Mbps from $R1$ to $R4$. With conventional shortest-path routing, the flow from $R1$ to $R4$ follows the direct path through LSR1 and LSR2.

The link between LSR1 and LSR2, however, provides only 1.5 Mbps of capacity. The routing decision is therefore perfectly valid from the point of view of the routing metric, but it is incompatible with the traffic demand. At the same time, other links in the network provide substantially more capacity.

Figure 3.14 illustrates an important limitation of destination-based shortest-path routing: the routing protocol optimizes the routing metric, not the actual distribution of traffic over the available capacity.

A possible reaction would be to modify the IGP metrics. For example, the cost of the direct LSR1–LSR2 link could be increased so that the $R1$–$R4$ traffic is moved toward the alternative path through LSR3.

This solution is illustrated in Figure 3.15. Although the original bottleneck is avoided, changing an IGP metric affects all traffic whose shortest-path computation depends on that metric. The alternative path is therefore shared by several flows, while another physical link may remain unused.

The example shows why Traffic Engineering cannot always be reduced to the selection of suitable IGP metrics. A metric is associated with a link and influences the shortest paths of potentially many source-destination pairs. It does not directly express the requirement that a specific traffic demand should follow a particular path.

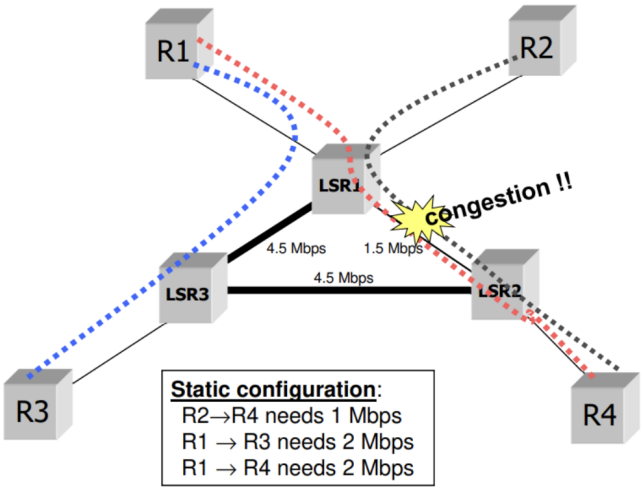


Figure 3.14: Traffic distribution with conventional shortest-path routing.

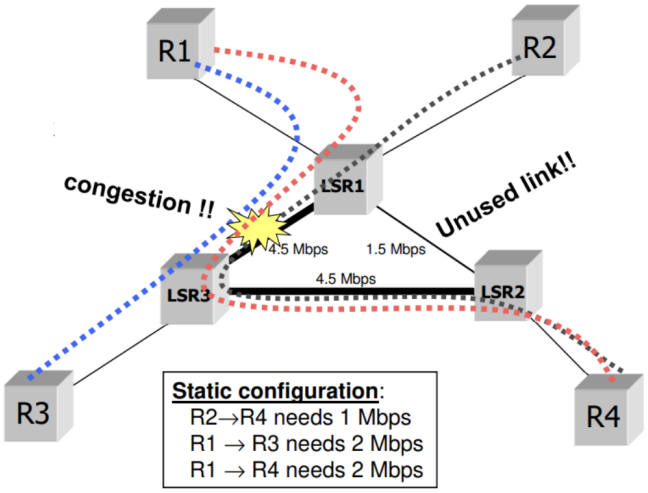


Figure 3.15: Traffic redistribution obtained through IGP metric tuning.

Even when multiple equal-cost paths are available, the amount of control remains limited. Figure 3.16 considers the use of Equal-Cost Multi-Path (ECMP) together with an appropriate configuration of the IGP link costs. Traffic can now be distributed over multiple shortest paths, but the forwarding decisions are still derived from the destination-based routing system rather than from an explicit assignment of individual traffic demands to network paths.

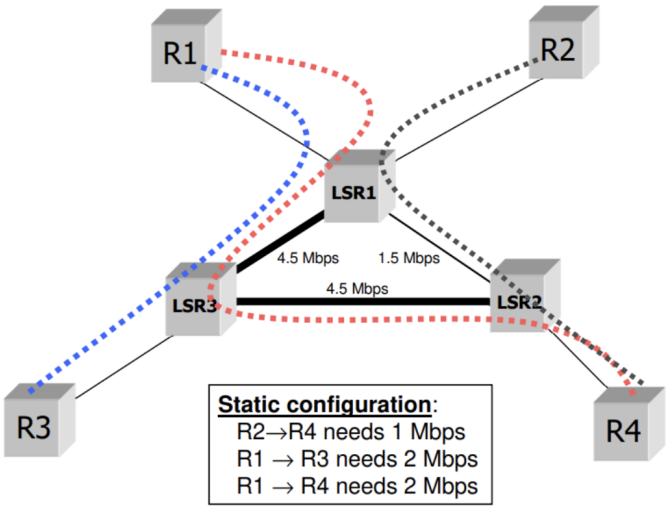


Figure 3.16: Traffic distribution with Equal-Cost Multi-Path (ECMP).

MPLS provides a different mechanism. Instead of attempting to influence all routing decisions indirectly through the IGP metrics, selected traffic can be associated with an explicitly chosen Label Switched Path.

In the example of Figure 3.17, the traffic from $R1$ to $R4$ is assigned to the LSP

$$R1 \rightarrow \text{LSR1} \rightarrow \text{LSR3} \rightarrow \text{LSR2} \rightarrow R4.$$

The path followed by this traffic is therefore no longer determined simply by the default shortest path toward $R4$. The provider explicitly selects a different sequence of network resources for this traffic demand.

The remaining traffic does not necessarily have to follow the same LSP. It can be associated with other LSPs or it can continue to use the paths determined by conventional IP routing.

This is the key principle behind MPLS Traffic Engineering: different traffic demands can be associated with different LSPs, allowing the provider to control their paths independently.

The distinction can be summarized as

$$\begin{aligned} \text{Conventional IP routing:} &\quad \text{destination} \rightarrow \text{shortest path}, \\ \text{MPLS Traffic Engineering:} &\quad \text{selected traffic} \rightarrow \text{selected LSP}. \end{aligned}$$

The use of an explicitly routed LSP does not imply that every packet in the network must be traffic engineered. MPLS and conventional IP forwarding can coexist: only the traffic for which specific path control is required needs to be associated with a traffic-engineered LSP.

Once Traffic Engineering is formulated in terms of LSPs, two additional questions arise: *who selects the path?* and *how is that path selected?*

Concerning the first question, the path may be selected by an external entity, which computes the desired route and then configures the routers accordingly. Alternatively, the routers themselves may participate in the path-selection process and autonomously

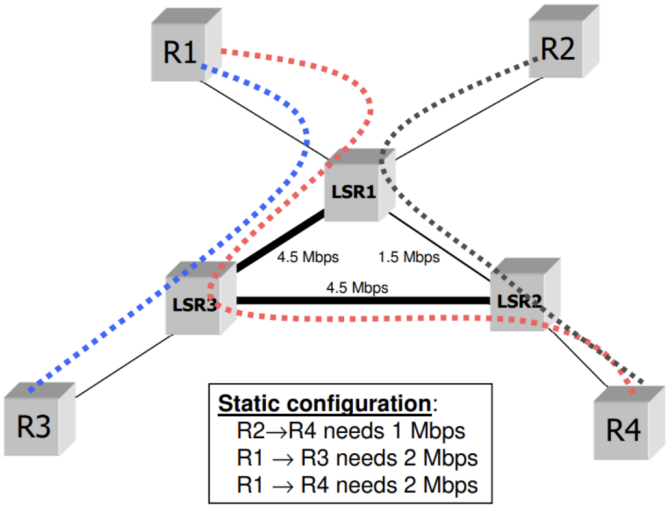


Figure 3.17: Traffic Engineering through an explicitly routed MPLS Label Switched Path.

determine a route satisfying the required constraints, for example constraints associated with the requested Quality of Service.

Concerning the second question, path computation can follow two general approaches. In an *online* approach, a path is computed whenever a new LSP has to be established. The decision is therefore taken for one request at a time, considering the network state available when that request arrives.

An alternative is an *offline* approach, in which path selection is formulated as a global optimization problem involving the set of paths in the network. Instead of selecting each new LSP independently, the objective is to find a traffic distribution that makes more effective use of the overall network resources.

The alternatives can therefore be summarized as

$$\text{Path control:} \begin{cases} \text{external path computation and router configuration,} \\ \text{autonomous path computation by the routers,} \end{cases}$$

$$\text{Optimization:} \begin{cases} \text{online: one new LSP is considered at a time,} \\ \text{offline: paths are considered through global optimization.} \end{cases}$$

Traffic Engineering therefore exploits one of the fundamental properties introduced by MPLS: the forwarding path of selected traffic can be controlled independently of the path that conventional destination-based routing would otherwise select. This provides the provider with a mechanism for distributing traffic more effectively over its infrastructure and for avoiding situations in which some resources become congested while alternative capacity remains available.

3.6 MPLS Fast Recovery and Protection

A further requirement of a provider network is the ability to maintain service continuity when links or network nodes fail.

Traditional IP routing already provides a recovery mechanism. When a failure changes the network topology, dynamic routing protocols such as OSPF detect the change, distribute updated topology information, recompute the affected routes, and install the resulting forwarding entries.

This mechanism is effective, but the complete convergence process may require from several seconds to tens of seconds. Such recovery times may be acceptable for Best Effort traffic, but they can be too long for services requiring strong continuity guarantees.

At lower layers, transport technologies have traditionally provided much faster protection mechanisms. Optical Transport Networks, for example, can support protection switching at time scales on the order of tens of milliseconds, commonly targeting recovery within approximately 50 ms.

However, protection provided by a lower, or *server*, layer cannot always be assumed to be available. The IP/MPLS layer is a client of the underlying transport infrastructure, but the service offered by that infrastructure may or may not include fast protection.

It is therefore important for the client layer to have its own failure recovery mechanisms. In this way, the IP/MPLS network can protect its traffic independently of the protection capabilities provided by the underlying transport technology.

One of the objectives of MPLS protection is consequently to provide fast recovery directly at the packet layer, reducing the interruption experienced by traffic when a failure occurs.

Traditional IP recovery: failure detection → routing convergence → new forwarding state,

Fast protection: failure detection → predefined alternative forwarding.

The fundamental difference is that fast protection does not require the network to complete a new routing convergence before forwarding can resume. Alternative forwarding state is prepared in advance and can be activated as soon as the relevant failure is detected.

Fast failure detection.

Fast recovery first requires fast failure detection. Waiting for the normal failure-detection mechanisms of an IGP such as OSPF would generally be too slow for protection mechanisms operating on a time scale of a few tens of milliseconds.

A protocol commonly used for this purpose is *Bidirectional Forwarding Detection* (BFD). BFD establishes a lightweight session between neighboring network elements and periodically exchanges control packets to verify that the forwarding path between them is still operational. If the expected packets are not received within the configured detection interval, the peer or forwarding path can rapidly be declared unavailable.

BFD and OSPF Hello messages therefore operate with different objectives and typically at very different time scales. BFD is designed specifically for rapid connectivity detection and can operate at sub-second, potentially millisecond, time scales. OSPF Hello messages, instead, are part of the routing protocol itself and are associated with neighbor maintenance and topology changes.

Making the routing protocol react to every transient event at extremely short time scales would be undesirable. Each topology change may trigger the distribution of new link-state information, new shortest-path computations, and changes in the forwarding tables. Excessively frequent changes could therefore generate routing instability.

The two mechanisms can consequently be viewed as complementary:

BFD: rapid detection of forwarding failures,

OSPF: topology dissemination and routing convergence.

Fast protection can react immediately to the information provided by BFD, while the routing protocol subsequently converges toward a new stable network state.

Protection at different network layers.

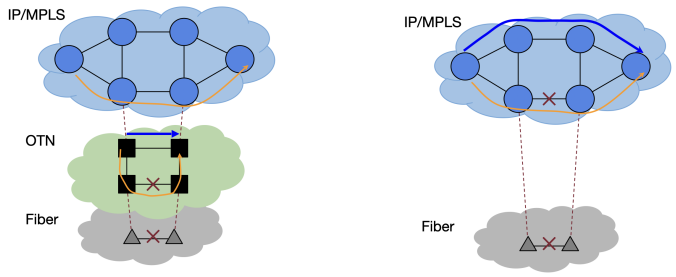


Figure 3.18: Protection at different layers of the transport infrastructure.

Failure recovery can be implemented at different layers of the transport infrastructure. Figure 3.18 illustrates this interaction by comparing protection performed by the optical transport layer with recovery performed directly by the IP/MPLS layer.

In the left-hand case of Figure 3.18, the IP/MPLS connection is supported by an OTN connection. When the physical resource used by that connection fails, the server layer can switch the connection onto alternative resources. From the perspective of the IP/MPLS client layer, logical connectivity can therefore remain available even though the underlying physical route has changed.

In the right-hand case, no such protection service is assumed from the server layer. The failure therefore becomes directly visible to IP/MPLS, which must provide its own recovery mechanism.

This is an important reason for supporting protection mechanisms directly at the MPLS layer: the packet network should not have to assume that every underlying transport infrastructure provides the required protection service.

Fast rerouting through link protection.

A first MPLS protection strategy is to protect a specific link of an existing LSP.

Consider a primary LSP crossing a sequence of Label Switching Routers. If one of the links belonging to that path fails, it is not necessary to replace the entire end-to-end LSP immediately. Traffic can instead be redirected locally around the failed resource and then returned to the original LSP.

Figure 3.19 illustrates this mechanism.

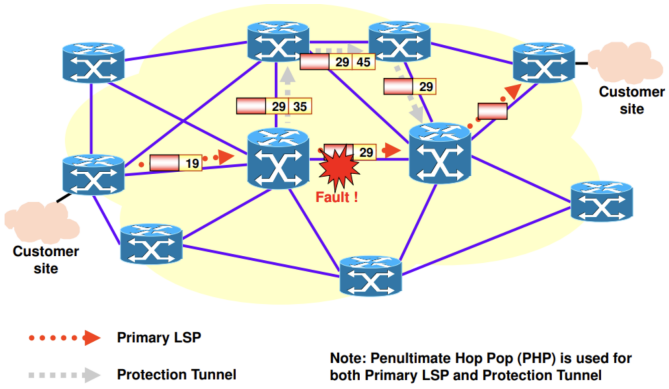


Figure 3.19: MPLS Fast Reroute through local link protection.

Before the failure, packets follow the primary LSP. When the protected link becomes unavailable, the router immediately upstream of the failure redirects the affected packets through a pre-established *protection tunnel*. The tunnel bypasses the failed resource and rejoins the original LSP at a downstream router.

The important point is that the recovery operation is *local*. The failure does not first

have to propagate to the ingress of the complete LSP. The node close to the failure can react as soon as the failure has been detected.

The forwarding path therefore changes from

$$R_1 \to \cdots \to R_i \to R_{i+1} \to \cdots \to R_n$$

to

$$R_1 \to \cdots \to R_i \to \boxed{\text{protection tunnel}} \to R_j \to \cdots \to R_n.$$

The protection tunnel can be implemented by exploiting MPLS label stacking. When the packet reaches the node performing the local repair, an additional label associated with the protection tunnel is pushed on top of the label already associated with the original LSP:

$$\boxed{\text{protection-tunnel label} \mid \text{original LSP label} \mid \text{packet}}.$$

Routers belonging to the protection tunnel process only the outer label. When the packet reaches the point at which the tunnel rejoins the original LSP, the protection label is removed and the original MPLS label becomes visible again. Forwarding can then continue along the original LSP without requiring the upstream portion of that LSP to be modified.

This mechanism provides a very fast reaction because the recovery decision is taken locally. The price is that the resulting route is not necessarily the optimal path in the network after the failure.

The local repair node only needs to bypass the failed resource and reconnect the packet to the existing LSP. It does not recompute the complete end-to-end route on the residual network topology. The resulting path may therefore contain a detour with respect to the shortest or otherwise optimal route that would be selected after full routing convergence.

Link protection therefore trades path optimality for reaction speed:

$$\boxed{\text{local information} \to \text{immediate reaction} \to \text{possibly sub-optimal path}}.$$

End-to-end recovery and path protection.

A different strategy is to recover the complete LSP rather than locally protecting an individual resource.

In this case, the information concerning the failure must reach the node responsible for selecting the end-to-end path, typically the ingress of the LSP. This takes longer than the purely local reaction used for link protection, since the failure information has to propagate through the network.

Once the ingress is aware of the changed topology, however, it can select an alternative complete path from the source to the destination.

Figure 3.20 illustrates this approach.

The alternative route now replaces the complete primary LSP:

$$\begin{aligned} \text{Primary LSP:} \quad & R_{\text{in}} \to \cdots \to R_{\text{out}}, \\ \text{Alternative LSP:} \quad & R_{\text{in}} \to \cdots \to R_{\text{out}}. \end{aligned}$$

Unlike local link protection, the protection mechanism itself does not require an additional stacked label to tunnel the packet around the failed resource. The packet is simply injected at the ingress onto the alternative end-to-end LSP and forwarded using the labels associated with that LSP.

The reaction is therefore slower than local repair because the failure must first become known at the ingress. On the other hand, the alternative path can be computed considering the complete residual topology after the failure.

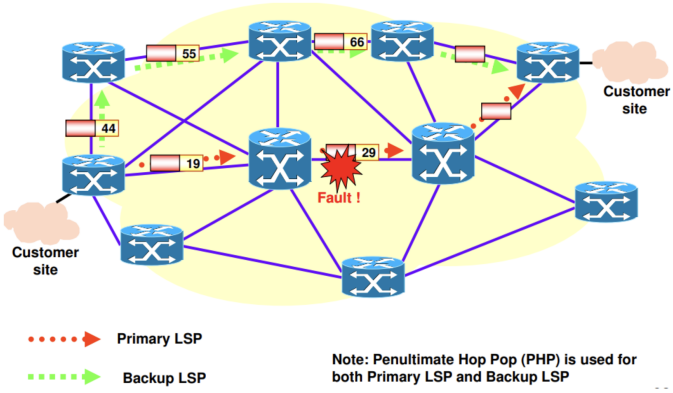


Figure 3.20: MPLS path protection through a backup Label Switched Path.

If the normal path-selection algorithm is applied to the updated topology, the new LSP can therefore follow the optimal route according to the selected routing metric or path-computation criterion.

The two recovery strategies exhibit a clear trade-off:

Link protection: local failure detection and local repair
→ very fast reaction
→ possibly sub-optimal detour,

Path protection/recovery: failure information propagated toward the ingress
→ slower reaction
→ end-to-end path optimized on the residual topology.

In the first case, label stacking makes it possible to temporarily tunnel the existing LSP around the failed resource. In the second case, a complete alternative LSP is used, and no additional protection-tunnel label is required by the recovery mechanism.

Fast local protection and routing convergence should therefore be considered complementary. Local repair can restore forwarding immediately after the failure, while the control plane subsequently disseminates the new topology and can establish a better end-to-end path.

MPLS protection thus extends the path-control capabilities already introduced for Traffic Engineering. Under normal conditions, LSPs determine where selected traffic should travel. After a failure, the same connection-oriented forwarding abstraction can be used either to bypass the failure immediately or to move the traffic onto a newly selected end-to-end path.

3.7 Chapter Summary

This chapter introduced Multiprotocol Label Switching (MPLS) as a technology that extends conventional IP forwarding with a connection-oriented forwarding abstraction. The main motivation is to provide network operators with greater control over how packets are transported through a shared backbone, enabling services and mechanisms that are difficult to implement using destination-based IP forwarding alone.

The fundamental concept is the *Forwarding Equivalence Class* (FEC), which identifies packets that should receive the same forwarding treatment. At the ingress of an MPLS domain, packets are classified into FECs and associated with a *Label Switched Path* (LSP). Intermediate routers forward packets using short, locally significant labels rather than repeatedly performing conventional IP destination lookups. The three basic data-plane

operations are PUSH, SWAP, and POP, while label stacking makes it possible to transport one LSP inside another.

Establishing an LSP requires the control plane to distribute label bindings and configure consistent forwarding state along the selected route. An important distinction is between *hop-by-hop LSPs*, which follow paths determined by conventional IP routing, and *explicitly routed LSPs*, whose paths can be selected independently of the default routing decisions.

The chapter then examined three major applications of MPLS in provider networks.

The first is the provision of *Layer-3 Virtual Private Networks*. An MPLS backbone can support multiple logically isolated customer networks over the same physical infrastructure. Provider Edge routers maintain customer-specific routing information through VRFs, while MP-BGP distributes VPN reachability information among the PEs. MPLS label stacking separates transport across the provider core from the customer-specific forwarding operations performed at the network edges.

The second application is *Traffic Engineering*. Conventional shortest-path routing does not necessarily distribute traffic efficiently over the available network capacity. By associating selected traffic with explicitly routed LSPs, MPLS allows a provider to control traffic placement and exploit alternative network resources without relying exclusively on the modification of IGP metrics.

Finally, the chapter discussed *fast recovery and protection*. MPLS can provide mechanisms that restore forwarding without waiting for complete routing convergence. Local link protection uses pre-established protection tunnels and label stacking to bypass failures rapidly, potentially at the cost of a sub-optimal route. End-to-end recovery instead replaces the affected LSP with an alternative path, potentially selected according to the updated network topology. These mechanisms illustrate the trade-off between reaction speed and path optimization.

Overall, MPLS provides a common forwarding abstraction that can be exploited for different purposes: isolating customer networks, controlling traffic distribution, and recovering from failures. Its main contribution is not simply replacing IP destination lookups with label lookups, but enabling the provider to manage forwarding paths and services with greater flexibility and control.

3.8 Further Reading

The following documents provide additional details on the MPLS architecture, its control-plane protocols, and the service applications discussed in this chapter.

- E. Rosen, A. Viswanathan, and R. Callon, *Multiprotocol Label Switching Architecture*, RFC 3031, IETF, January 2001.
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- L. Andersson, I. Minei, and B. Thomas, *LDP Specification*, RFC 5036, IETF, October 2007.
- D. Awduche et al., *RSVP-TE: Extensions to RSVP for LSP Tunnels*, RFC 3209, IETF, December 2001.
- E. Rosen and Y. Rekhter, *BGP/MPLS IP Virtual Private Networks (VPNs)*, RFC 4364, IETF, February 2006.
- P. Pan, G. Swallow, and A. Atlas, *Fast Reroute Extensions to RSVP-TE for LSP Tunnels*, RFC 4090, IETF, May 2005.
- D. Katz and D. Ward, *Bidirectional Forwarding Detection (BFD)*, RFC 5880, IETF, June 2010.